Recommendations

- Huawei Learning Website
  - http://learning.huawei.com/en
- Huawei e-Learning
  - https://ilearningx.huawei.com/portal/#/portal/ebg/51
- Huawei Certification
  - http://support.huawei.com/learning/NavigationAction!createNavi?navId=_31&lang=en
- Find Training
  - http://support.huawei.com/learning/NavigationAction!createNavi?navId=_trainingsearch&lang=en

More Information

- Huawei learning APP

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A Brief Introduction to Cloud Computing

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Foreword

- IT is a fast-changing industry. Cloud computing has been developing rapidly in recent years and has become the foundation of a wide range of major applications. So, what is cloud computing? How has it evolved to what it is today? This chapter will offer you a brief introduction to the history and present of cloud computing.

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Objectives

- Upon completion of this course, you will be able to:
  - Describe what cloud computing is.
  - Describe the history of cloud computing.
  - List a few use cases of cloud computing.
  - Describe the characteristics of cloud computing.

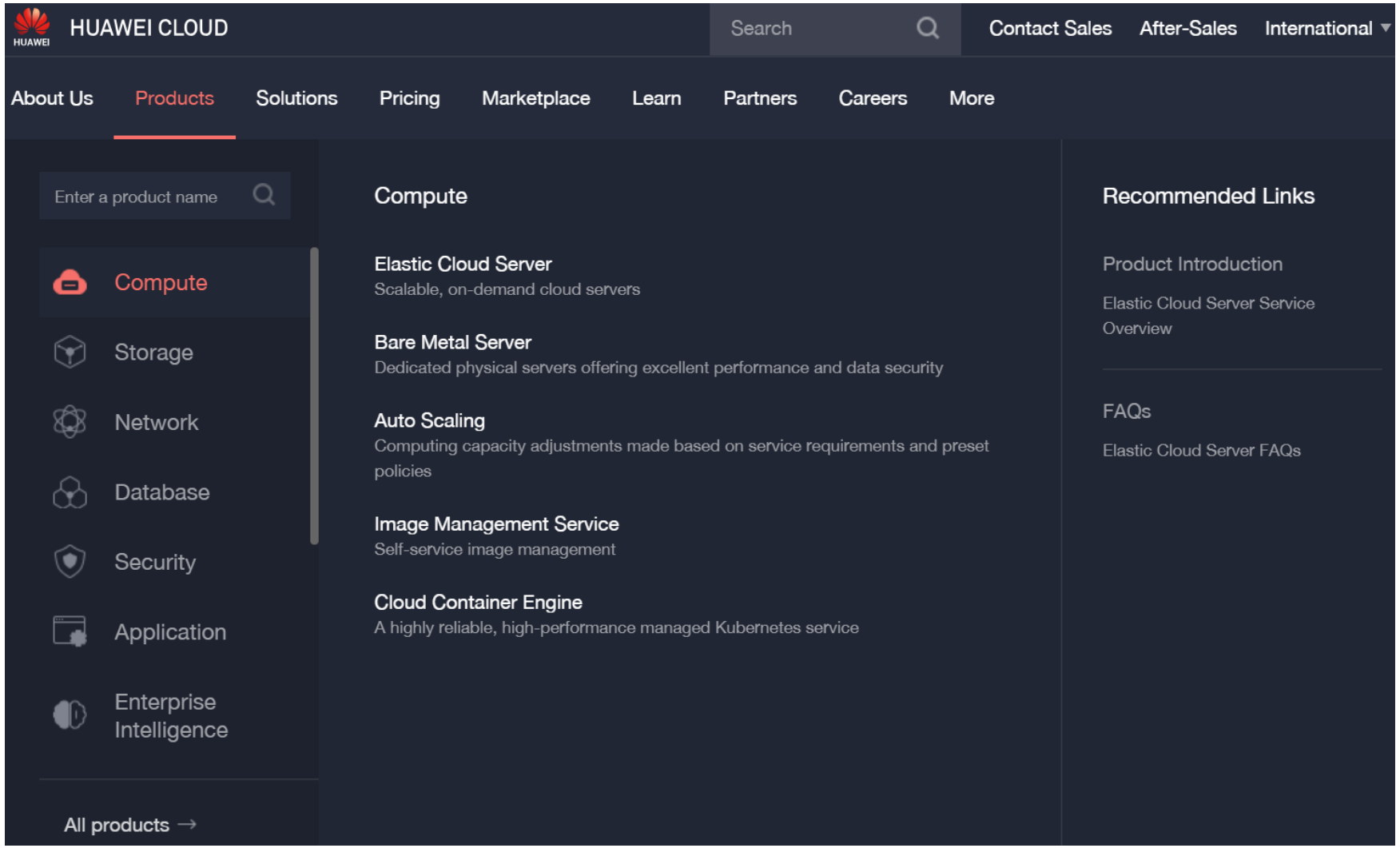
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2. Cloud Computing Advantages
3. Cloud Computing Definition
4. Origin and Development of Cloud Computing
5. Cloud Computing Models

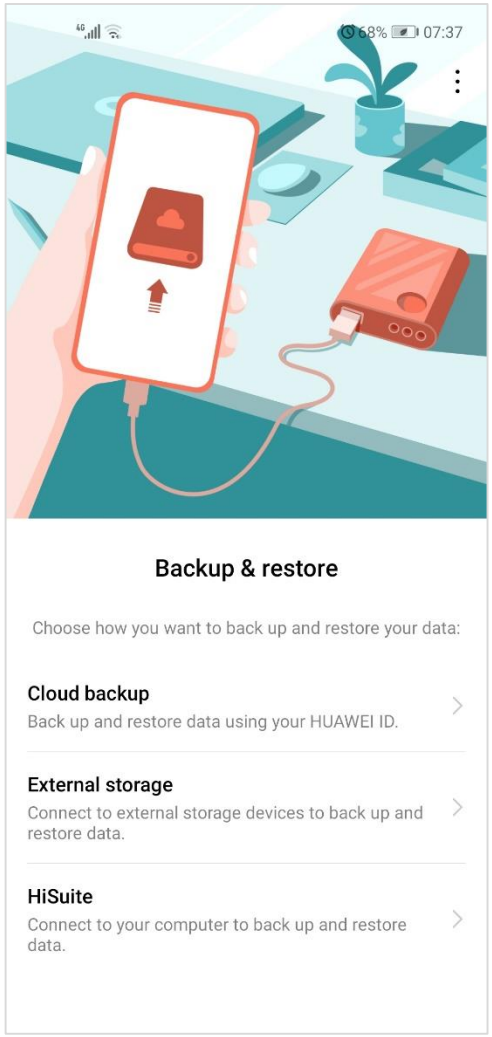
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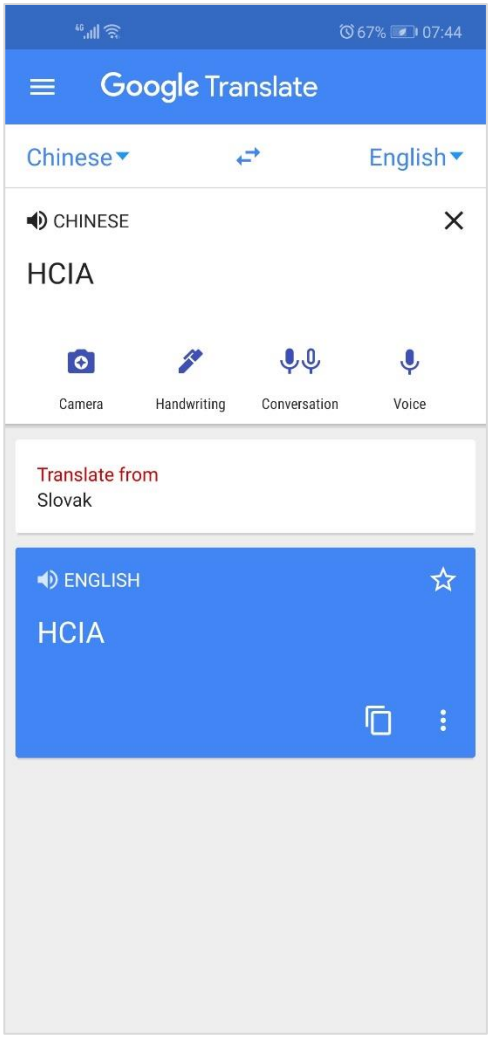
What Can Cloud Computing Do?

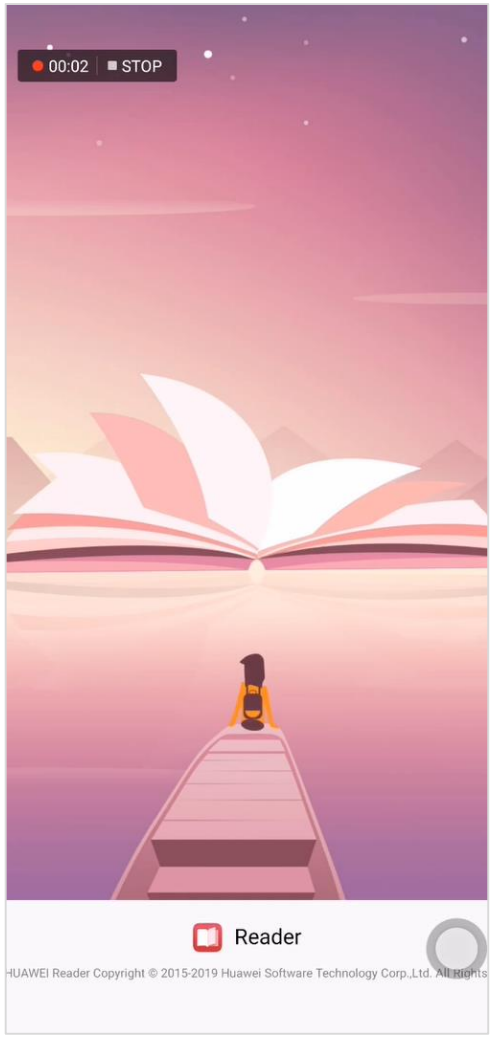


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You May Be Using the Cloud Already







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5. Cloud Computing Models

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Cloud Computing Advantages



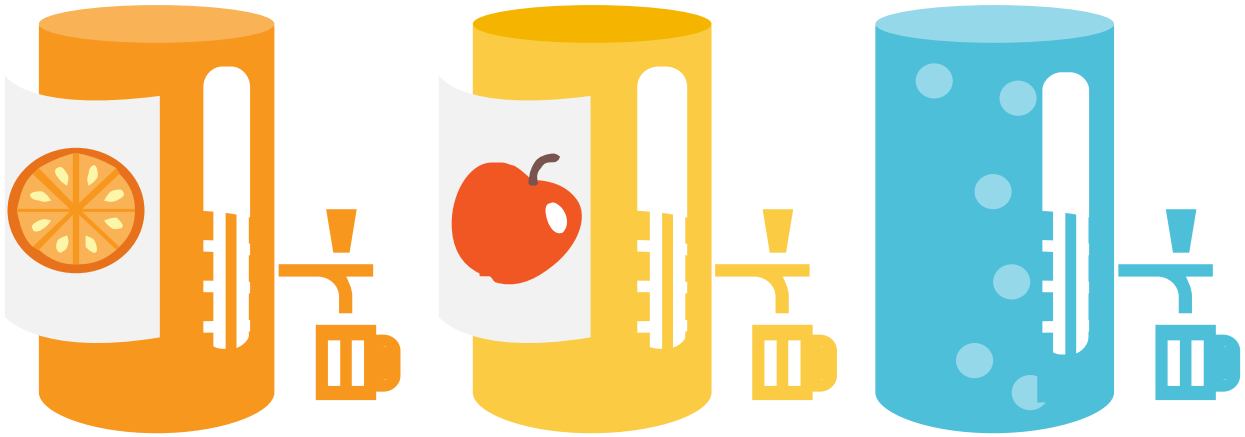
1. On-demand self-service



2. Broad network access

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Cloud Computing Advantages



3. Resource pooling

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Cloud Computing Advantages



L: 1.3 m, Φ: 1 m



L: 1 m, Φ: 0.7 m



L: 0.71 m, Φ: 0.5 m



4. Rapid elasticity

5. Measured service

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Cloud Computing Definition

- The National Institute of Standards and Technology (NIST) defines cloud computing as follows:

  Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

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Why the Term “Cloud Computing”?

- The cloud is a metaphor for the Internet. It is an abstraction of the Internet and the infrastructure that underpins it. Computing refers to computing services provided by a sufficiently powerful computer capable of providing a range of functionalities, resources, and storage. Put together, cloud computing can be understood as the delivery of on-demand, measurable computing services over the Internet.

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A Brief History of the Internet

1969

ARPANET was born.

1981

The complete specifications of the TCP/IP protocol suite were established.

1983

ARPANET, PRNET, and SATNET communicated through TCP/IP.

1984

DNS technology was introduced.

1986

The modern email routing system MERS was developed.

1989

The first commercial network operator PSINet was founded.

1990

The first network search engine Archie was launched.

1991

WWW was officially open to the public.

1995

E-commerce platforms such as Amazon and eBay were created.

2000

The dotcom bubble burst.

2004

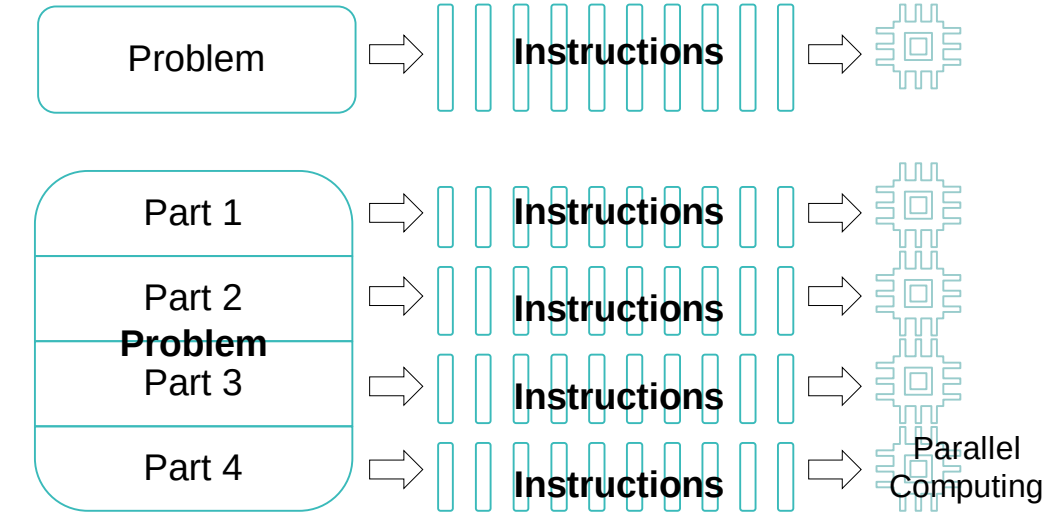
Facebook was founded. With it came the phenomenon of social networking.

2014

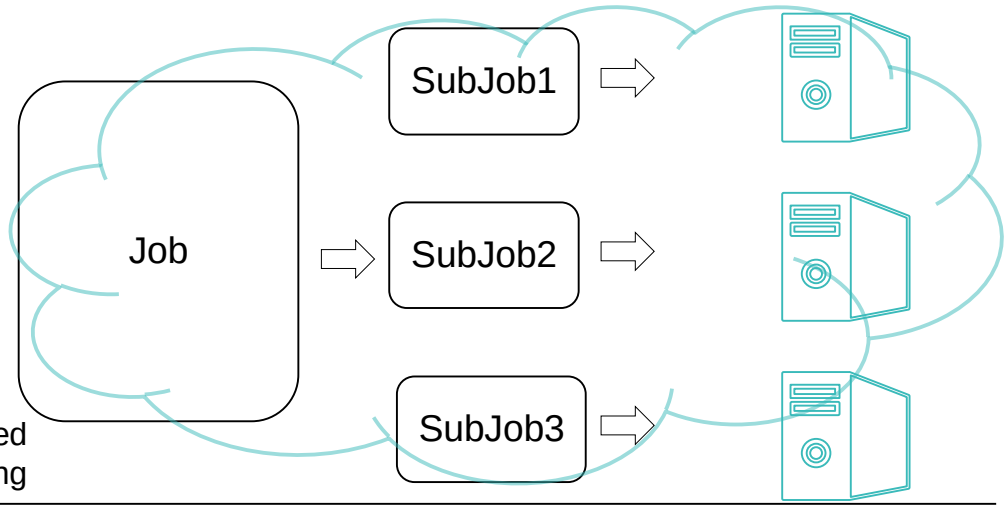
The browser alliance led by Google reinforced support for HTTPS.

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A Brief History of Computing

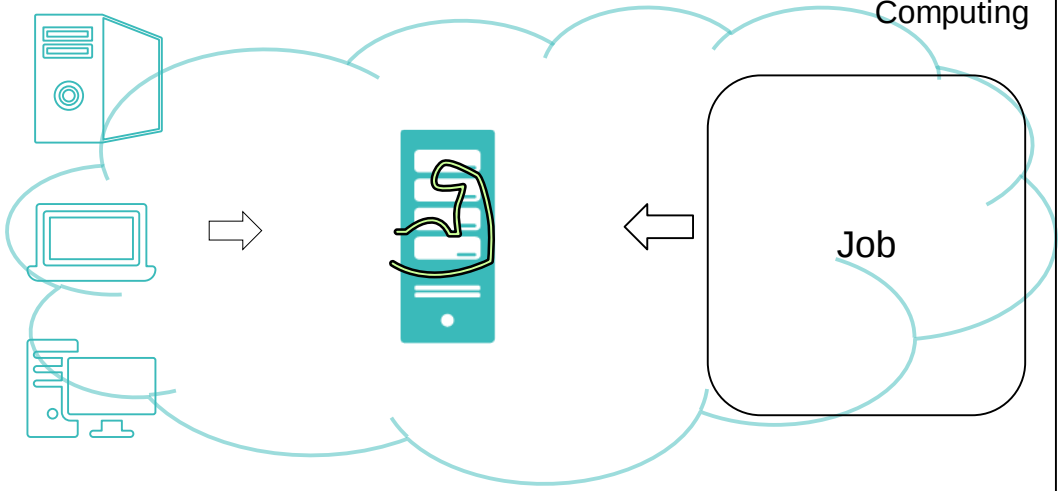


Parallel Computing

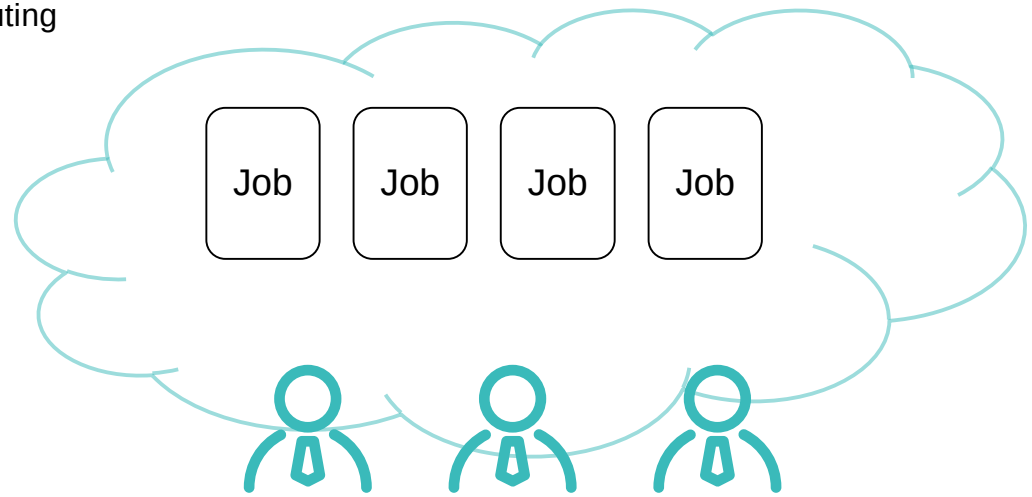


Distributed Computing

Grid Computing



Cloud computing



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Cloud Computing Evolution

Cloud computing 1.0



Computing virtualization

Hyper-V

Xen KVM

VMware ESX

Virtualization

Higher resource utilization

Cloud computing 2.0



Software-defined, integration

OpenStack

VMware AWS

Cloud-based infrastructure

Standardization and automation of resource services

Cloud computing 3.0



Cloud native, app redesign

Docker

CoreOS

Cloud Foundry

Cloud-native applications

Agile application development and lifecycle management

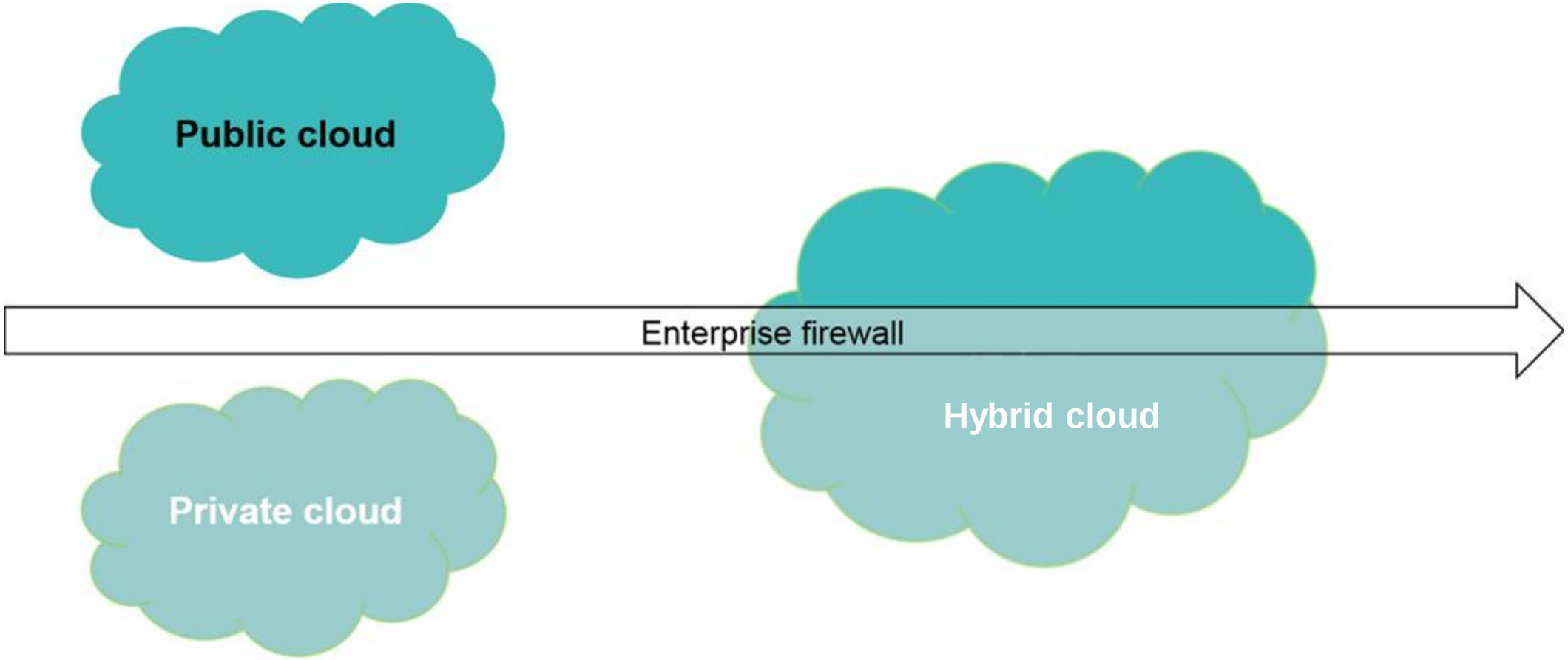
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Cloud Deployment Models



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Cloud Deployment Models



Transportation cloud



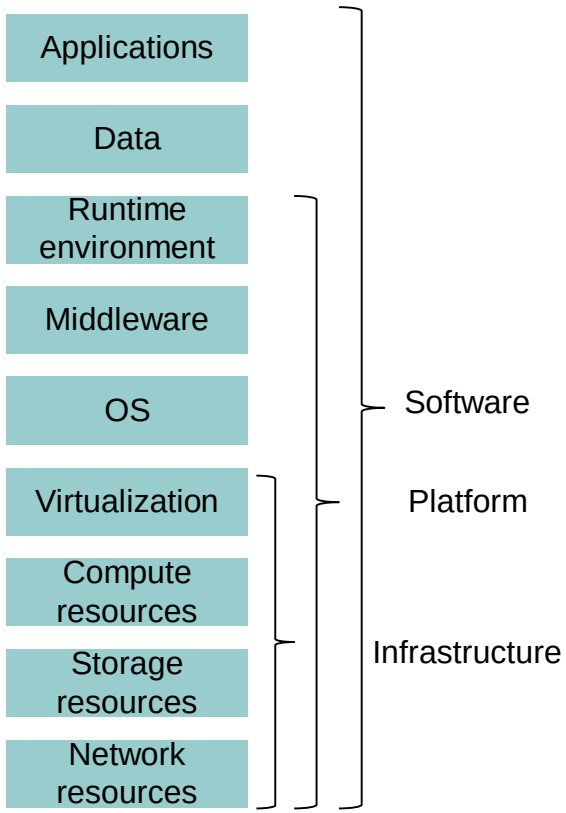
Financial cloud

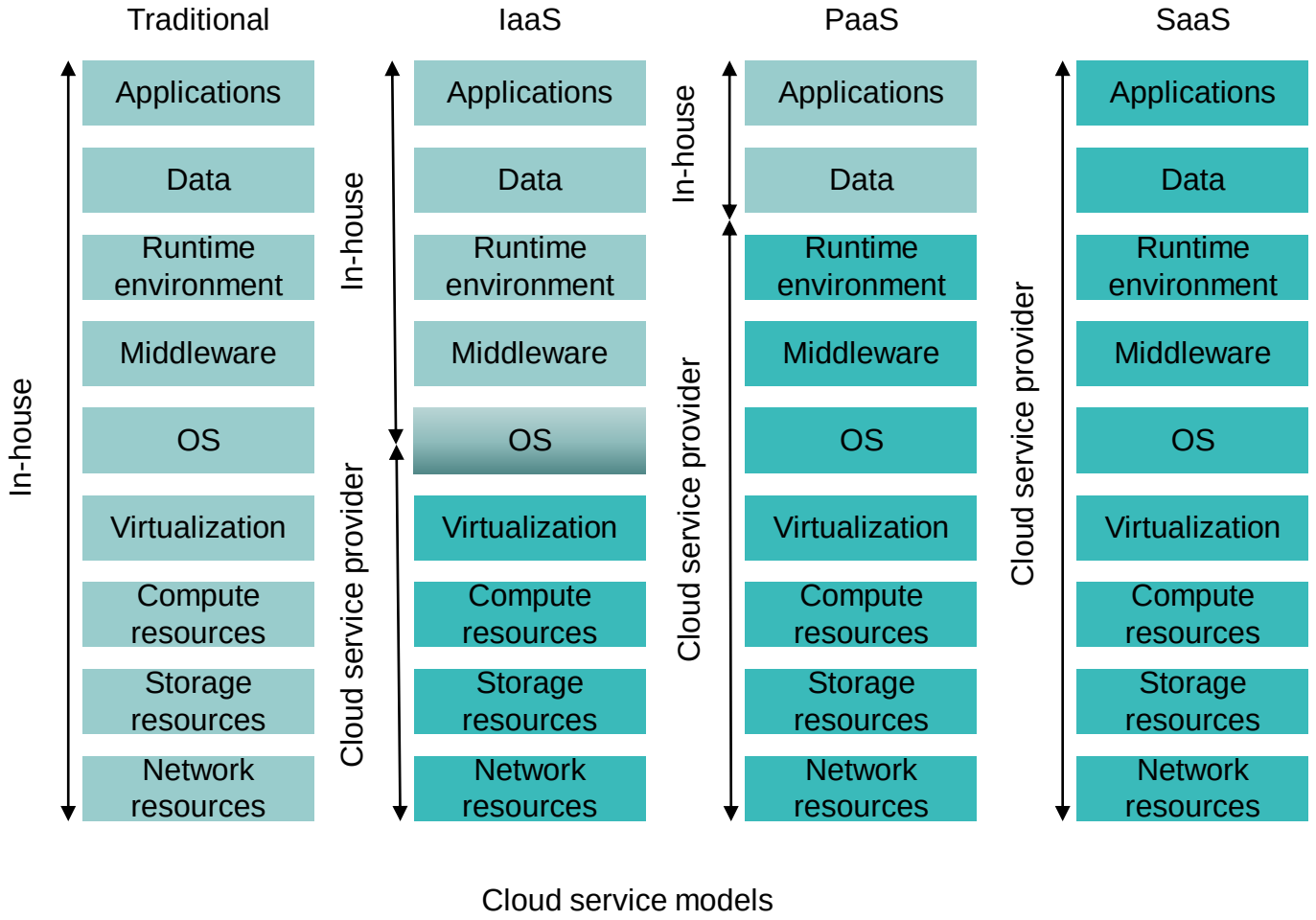


Healthcare cloud

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Cloud Service Models





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Quiz

1. Which of the following services offered by HUAWEI CLOUD (public cloud) falls into the category of SaaS services?

   A. Elastic Cloud Server (ECS)

   B. Facial recognition

   C. Elastic Volume Service (EVS)

   D. Object Storage Service (OBS)

2. The measurable service characteristic of cloud computing means that users pay for how long and how much they use the cloud service.

   A. TRUE

   B. FALSE

3.

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Summary

- Describe what cloud computing is.
- Describe the brief history of cloud computing.
- List a few use cases of cloud computing.
- Describe the characteristics of cloud computing.

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Recommendations

- Huawei e-Learning website:
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- Huawei Support case library:
  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Thank You
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Introduction to Compute Virtualization

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Foreword

- Cloud computing 1.0 focused on virtualization which today has become the foundation of cloud computing. This chapter focuses on the architecture and technologies involved in compute virtualization.

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Objectives

- Upon completion of this course, you will:
  - Be able to describe what virtualization is.
  - Understand the differences between virtualization and cloud computing.
  - Understand KVM technology.
  - Understand Huawei’s FusionSphere virtualization solution.

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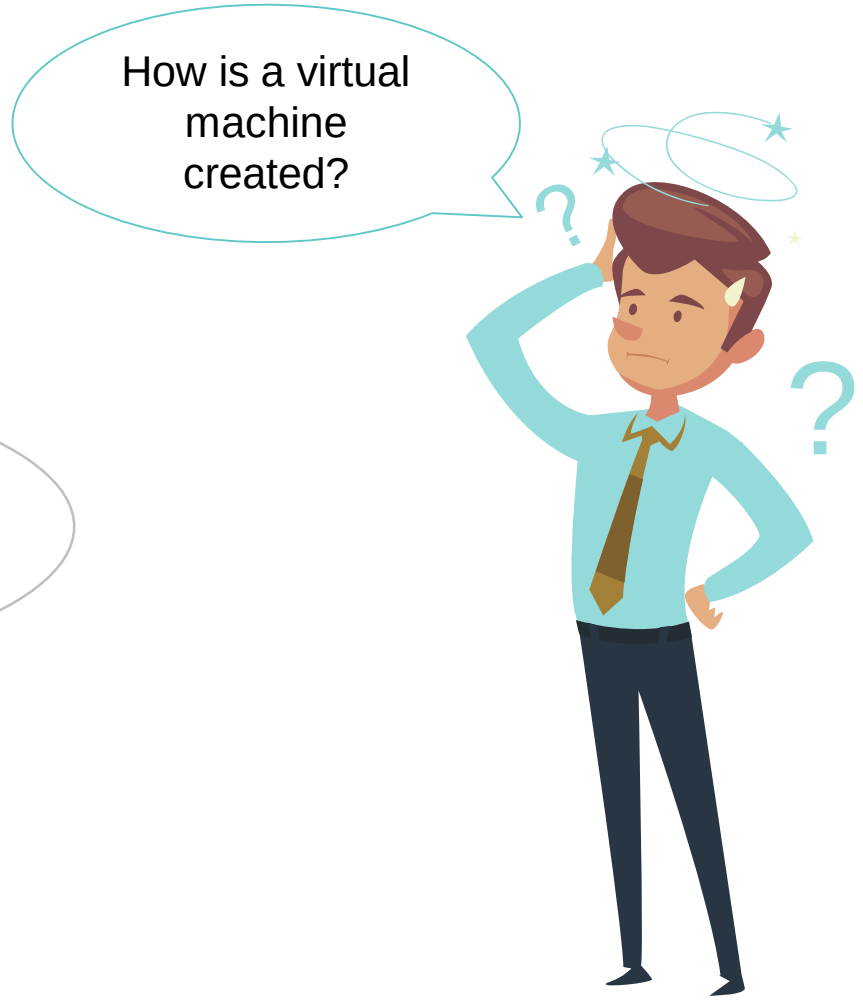
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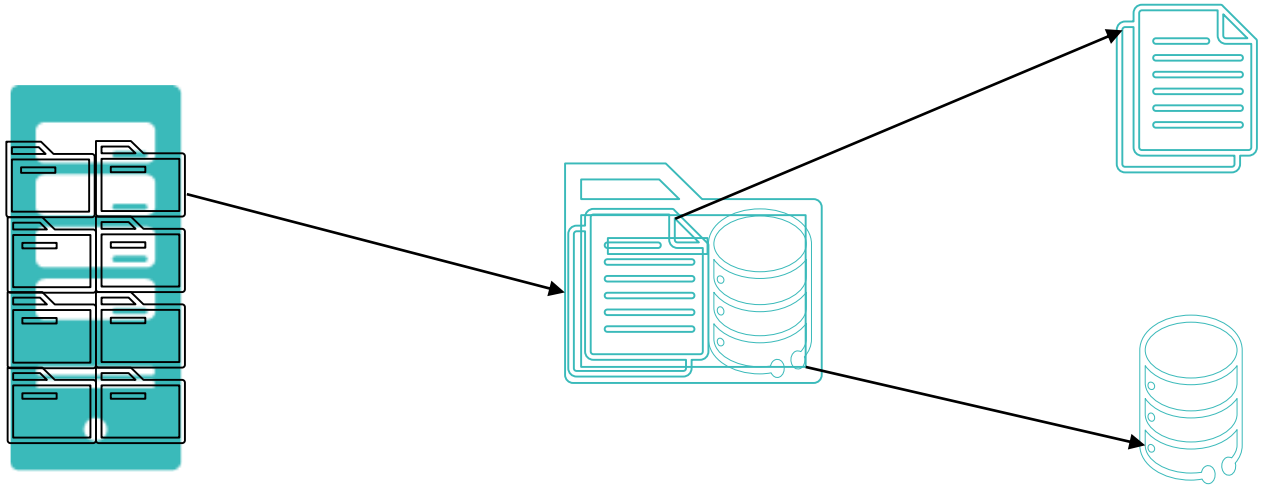
What’s a Virtual Machine (VM)?





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What’s Virtualization?



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A Brief History of Compute Virtualization

1964

IBM began to explore virtualization on mainframes.

1972

IBM launched VMs running on mainframes.

1999

VMware launched x86-based virtualization products.

2002

Xen was officially open sourced.

2006

Qumranet first announced KVM.

2007

HP launched HP-UX Integrity VMs.

2008

Microsoft added Hyper-V in Windows Server 2008 R2.

2008

Linux Container LXC was launched.

2010

By default, RHEL 6.0 supported KVM as the only virtualization option.

2011

IBM, Red Hat, HP, and Intel formed the Open Virtualization Alliance to accelerate KVM adoption.

2013

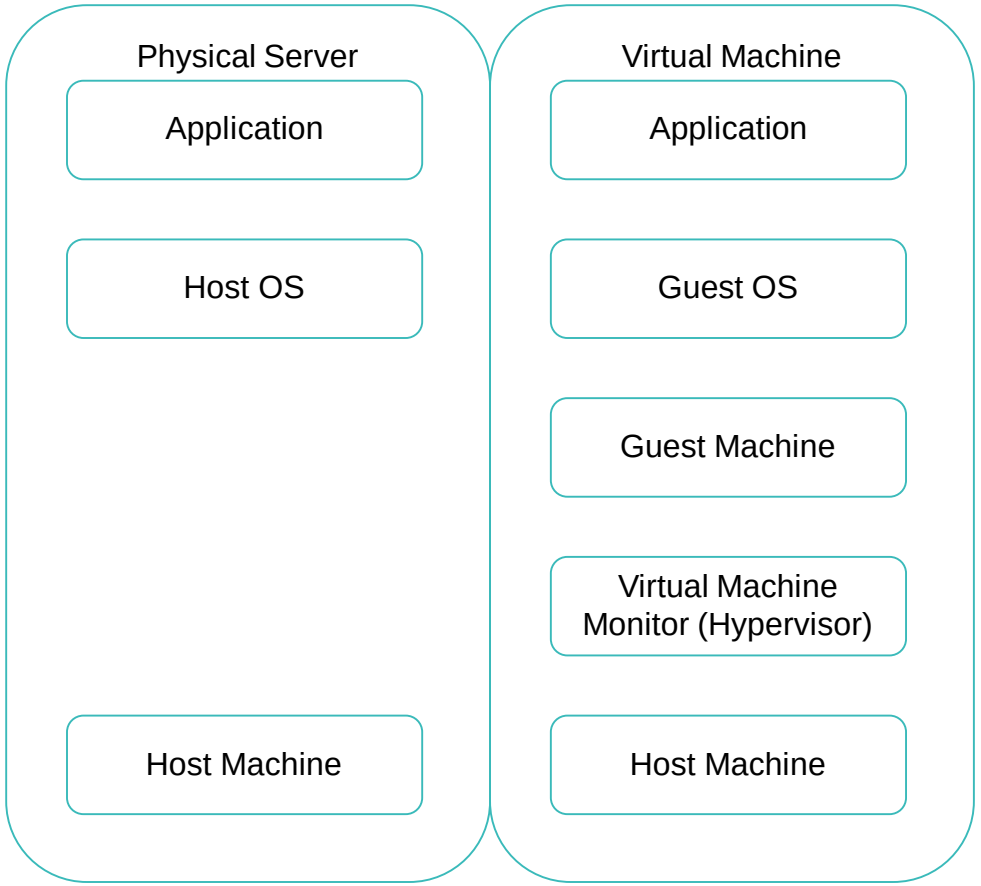
Docker was launched.

2014

Rocket was launched.

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Important Concepts in Compute Virtualization



Guest OS:

Operating system running in a virtual machine (VM)

Guest Machine:

Virtual machine created through virtualization

Hypervisor:

Virtualization software layer, or

Virtual Machine Monitor (VMM)

Host OS:

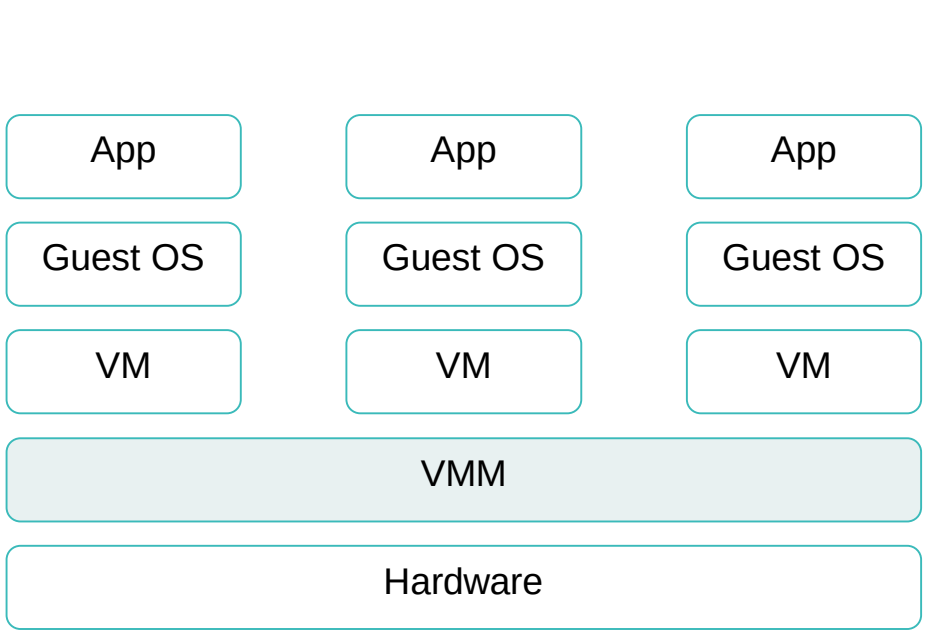
Operating system running in a physical machine

Host Machine:

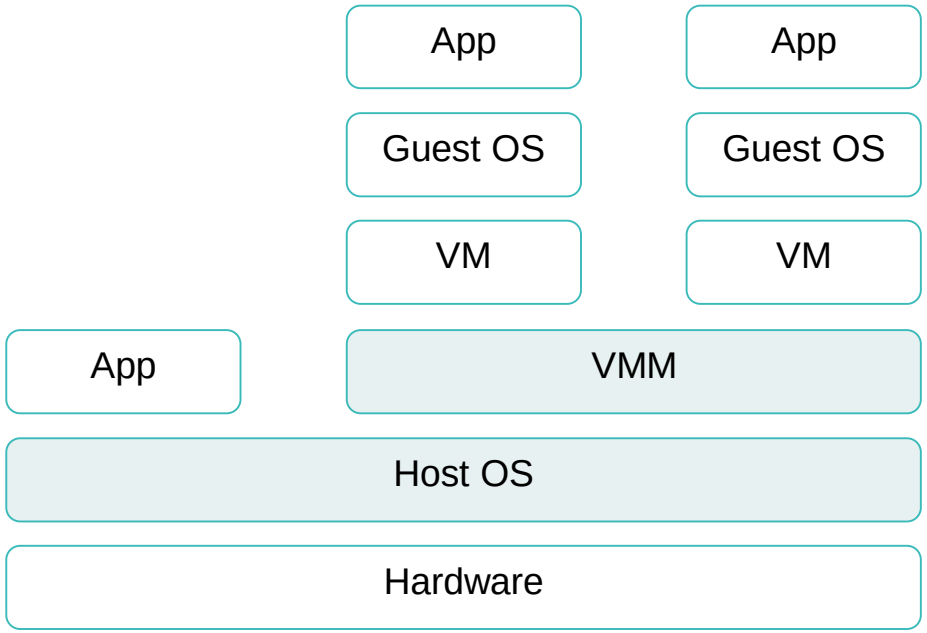
Physical machine

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Types of Compute Virtualization (Hypervisors)



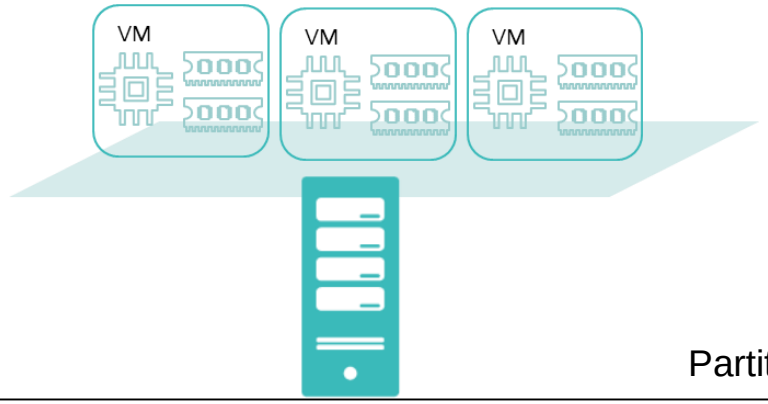
Bare-Metal Virtualization
(Type 1)



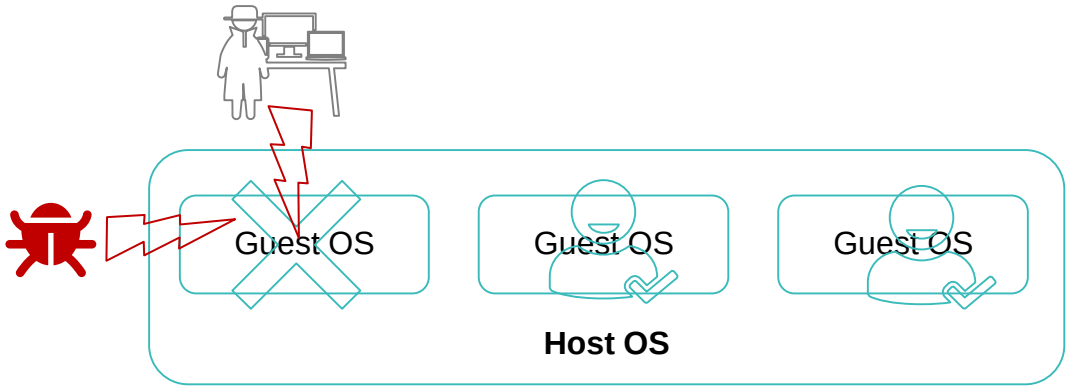
Hosted Virtualization
(Type 2)

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Characteristics of Virtualization



Partitioned

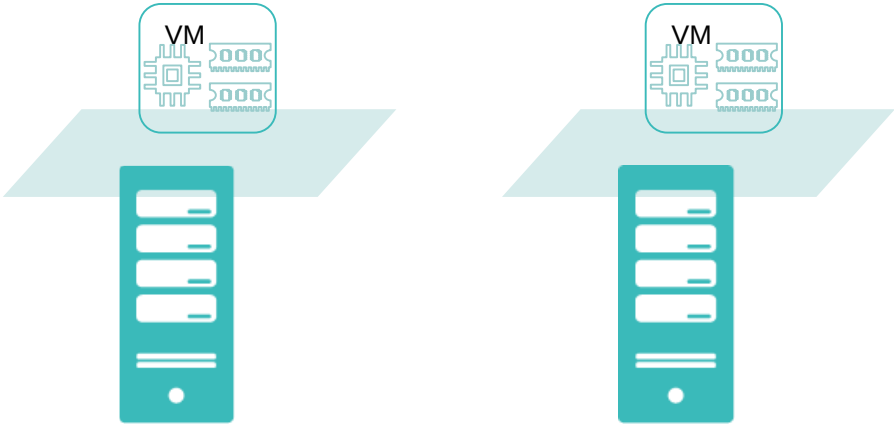


Isolated

Encapsulated



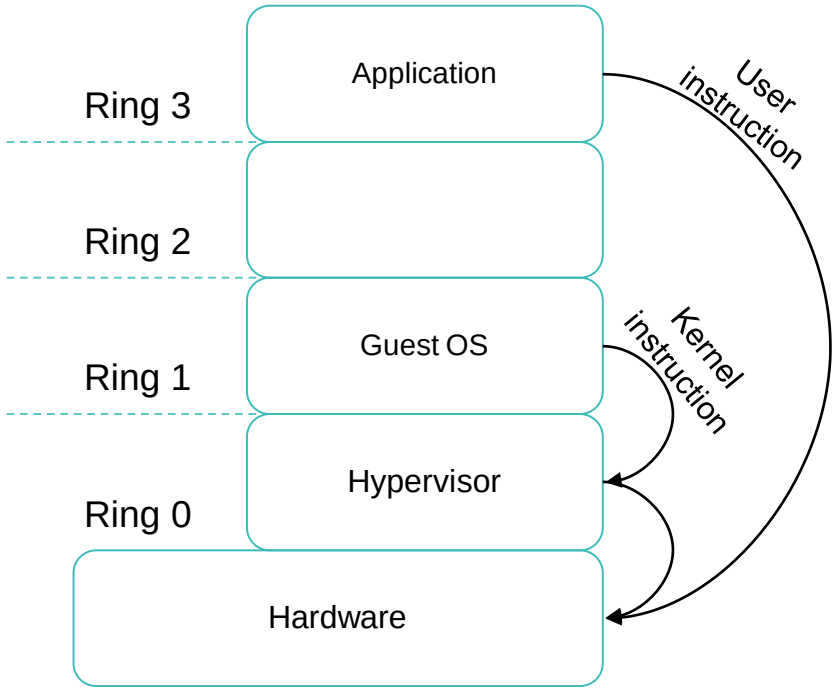
Independent



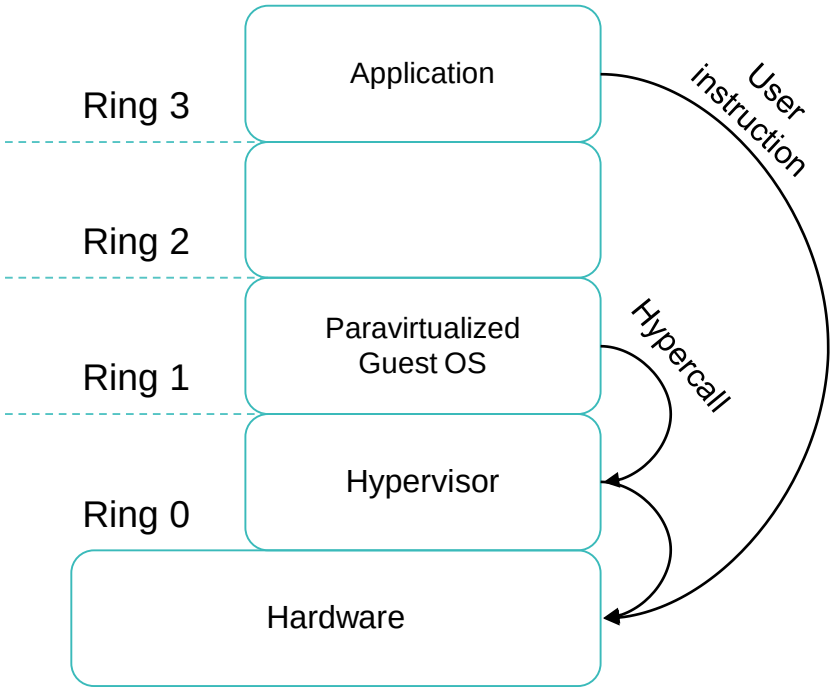
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CPU Virtualization

Full virtualization



Paravirtualization

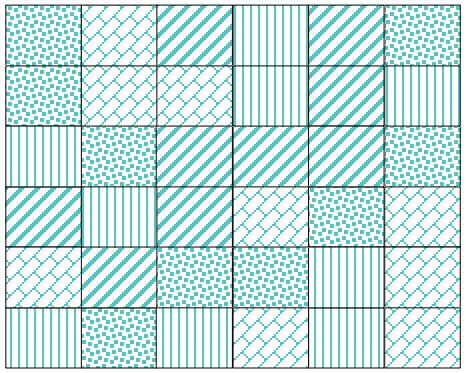


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Memory Virtualization



Host memory



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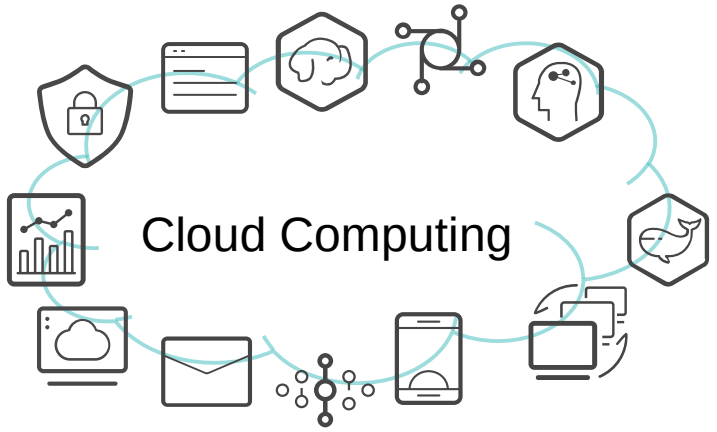
I/O Virtualization

- Emulation [Full Virtualization]: Complete simulation of the hardware, for example, keyboard and mouse. Access to such hardware depends on the capture of focus by the host, and leads to poor performance in some cases.
- Paravirtualization: Access to hardware drivers is transferred from the I/O frontend to the I/O backend. This mode is usually only used for hard disks and NICs, and delivers high performance.
- IO-through: Hardware devices capable of I/O passthrough, such as hard disks and NICs, are directly allocated to VMs. In Xen, Dom0 allocates hardware devices but does not take part in the access of the hardware. Hardware support is required.

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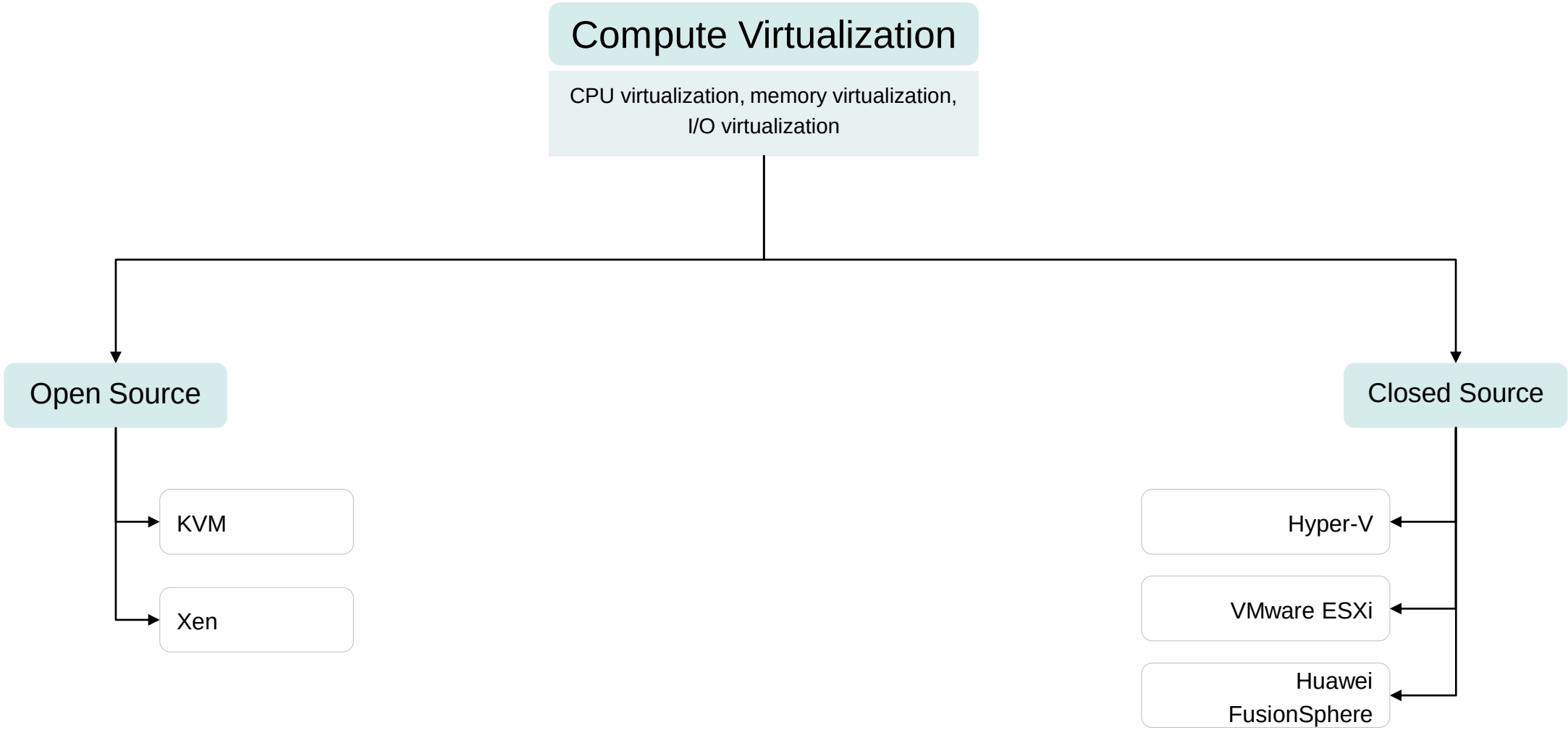
Cloud Computing and Virtualization





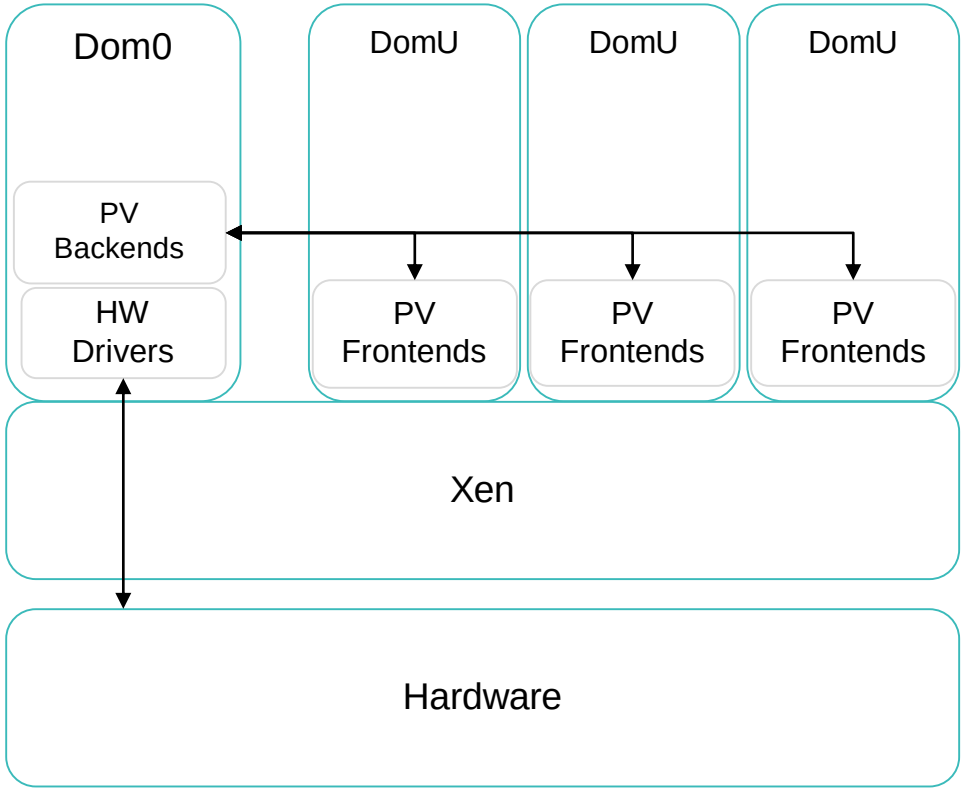
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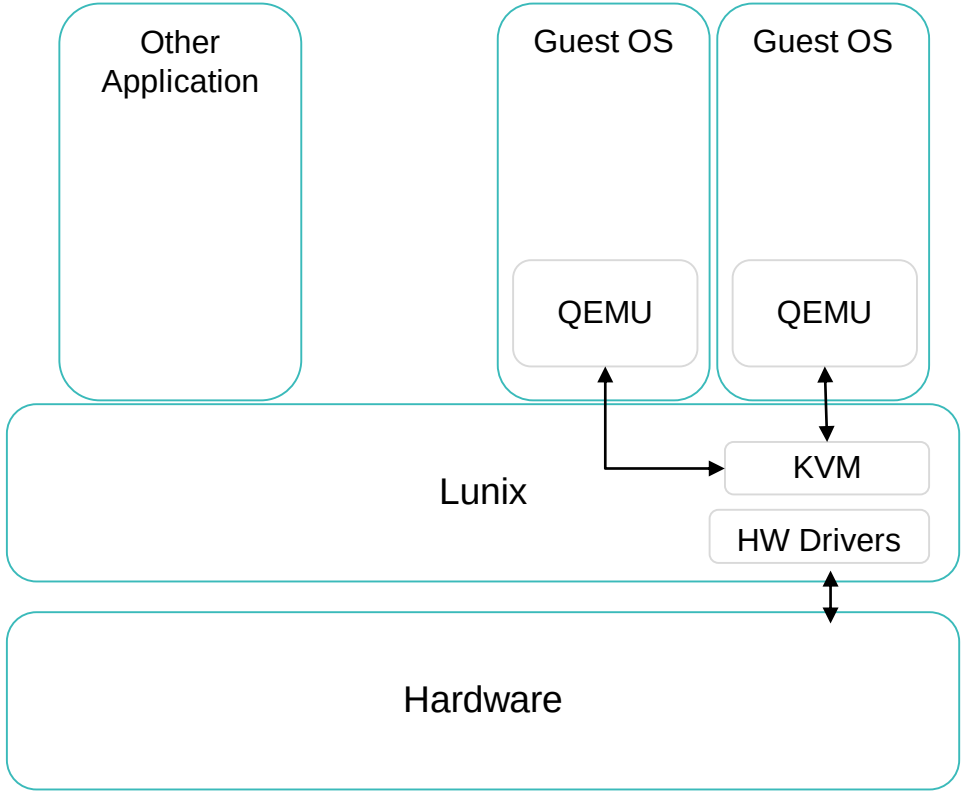
Mainstream Compute Virtualization Technologies



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Xen vs. KVM





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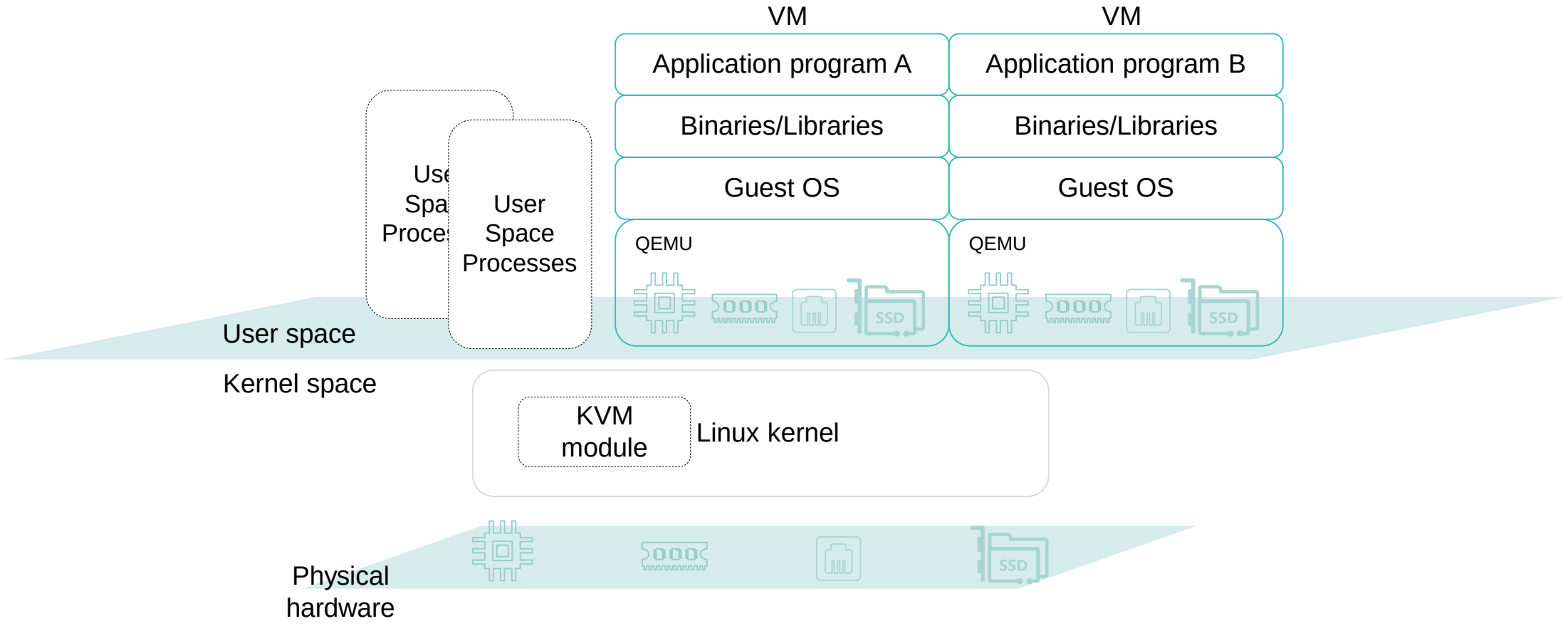


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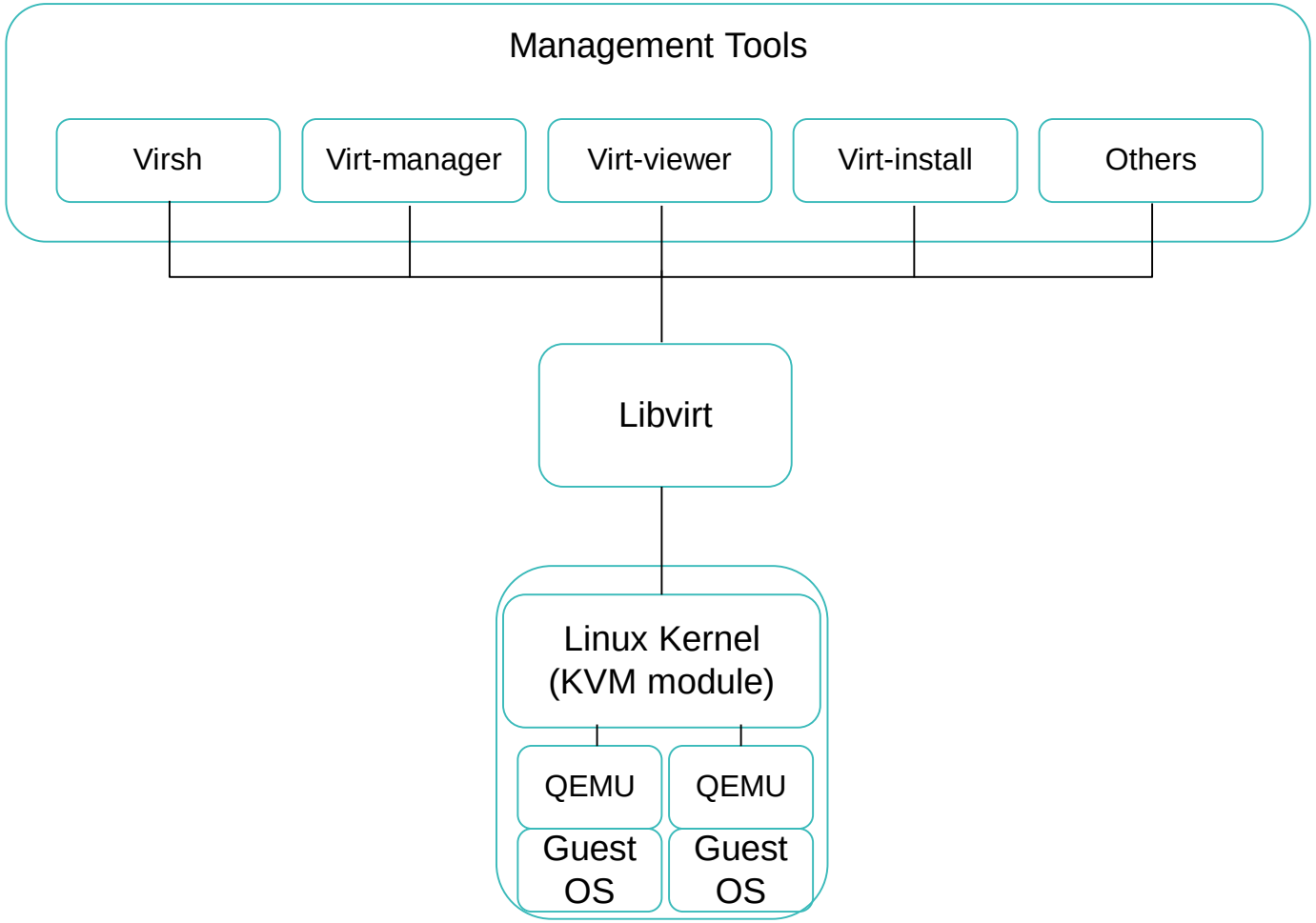
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What's KVM?



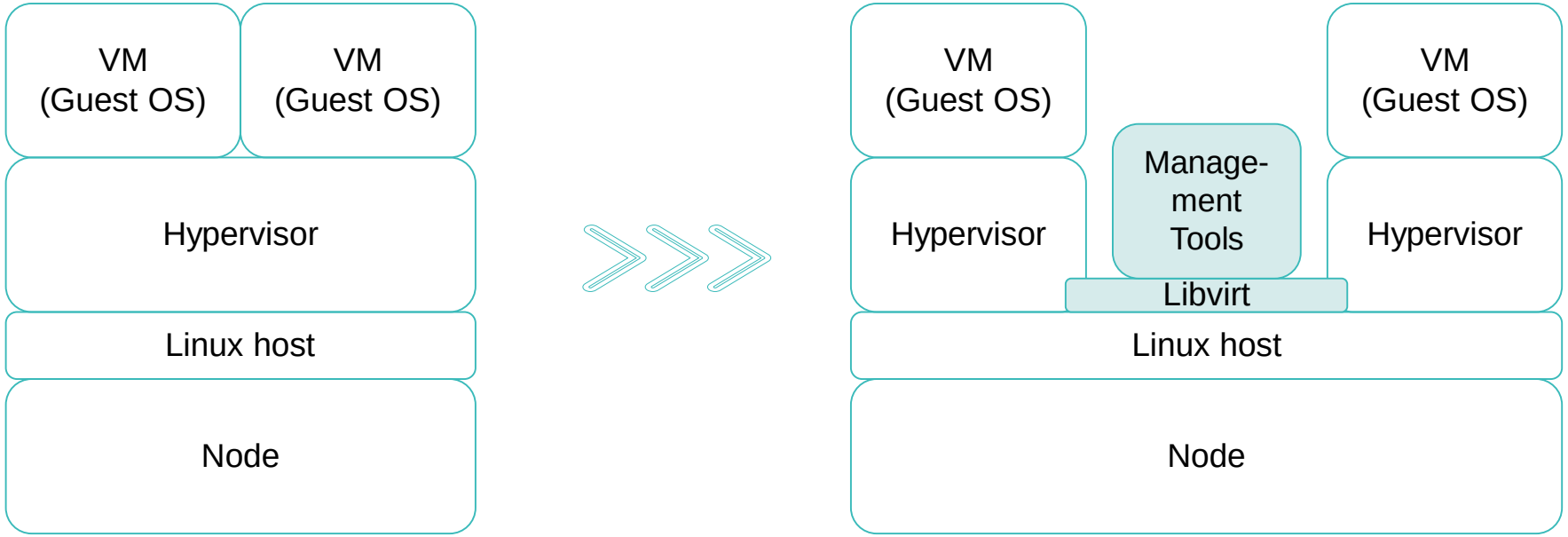
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KVM Architecture



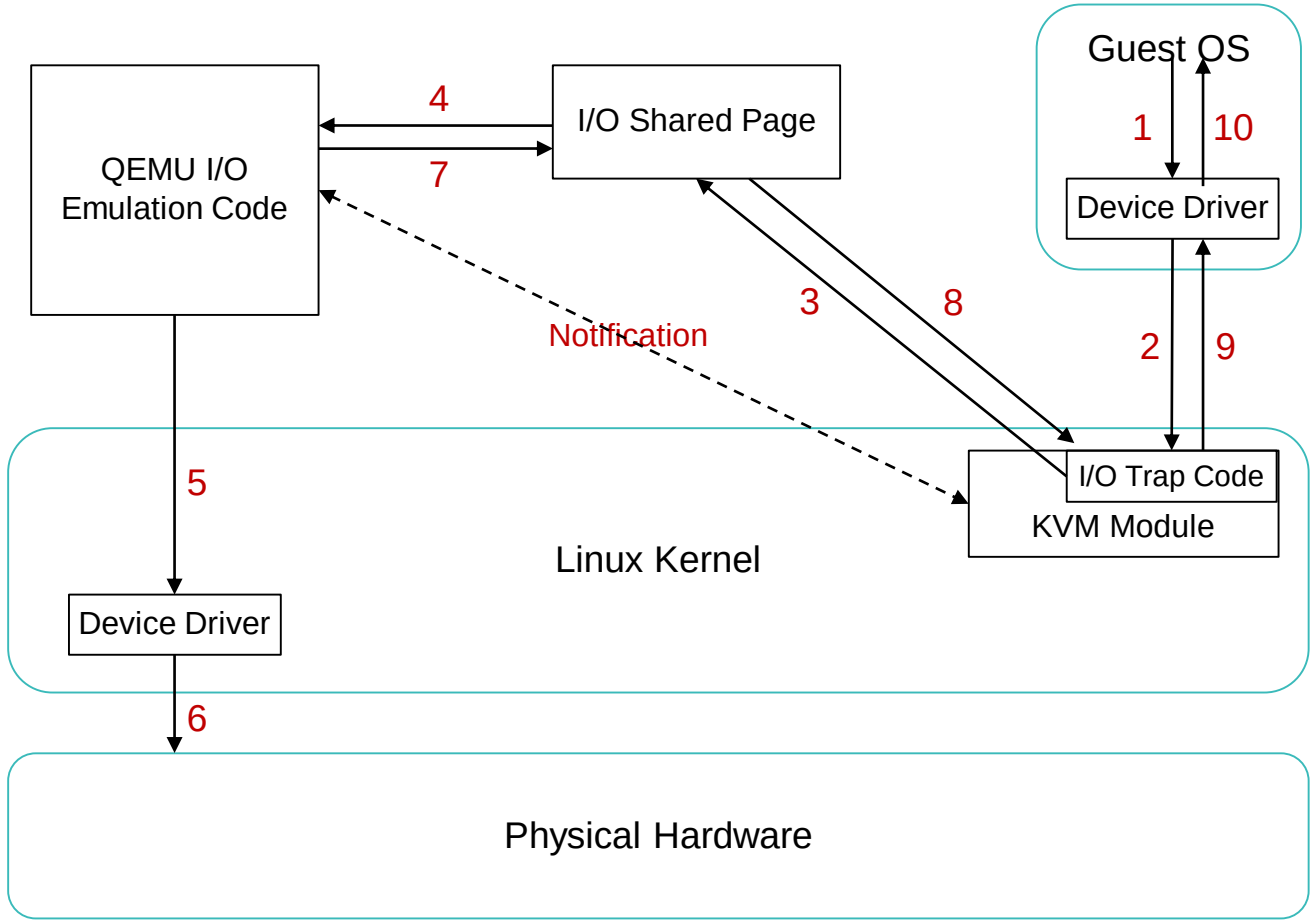
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Libvirt



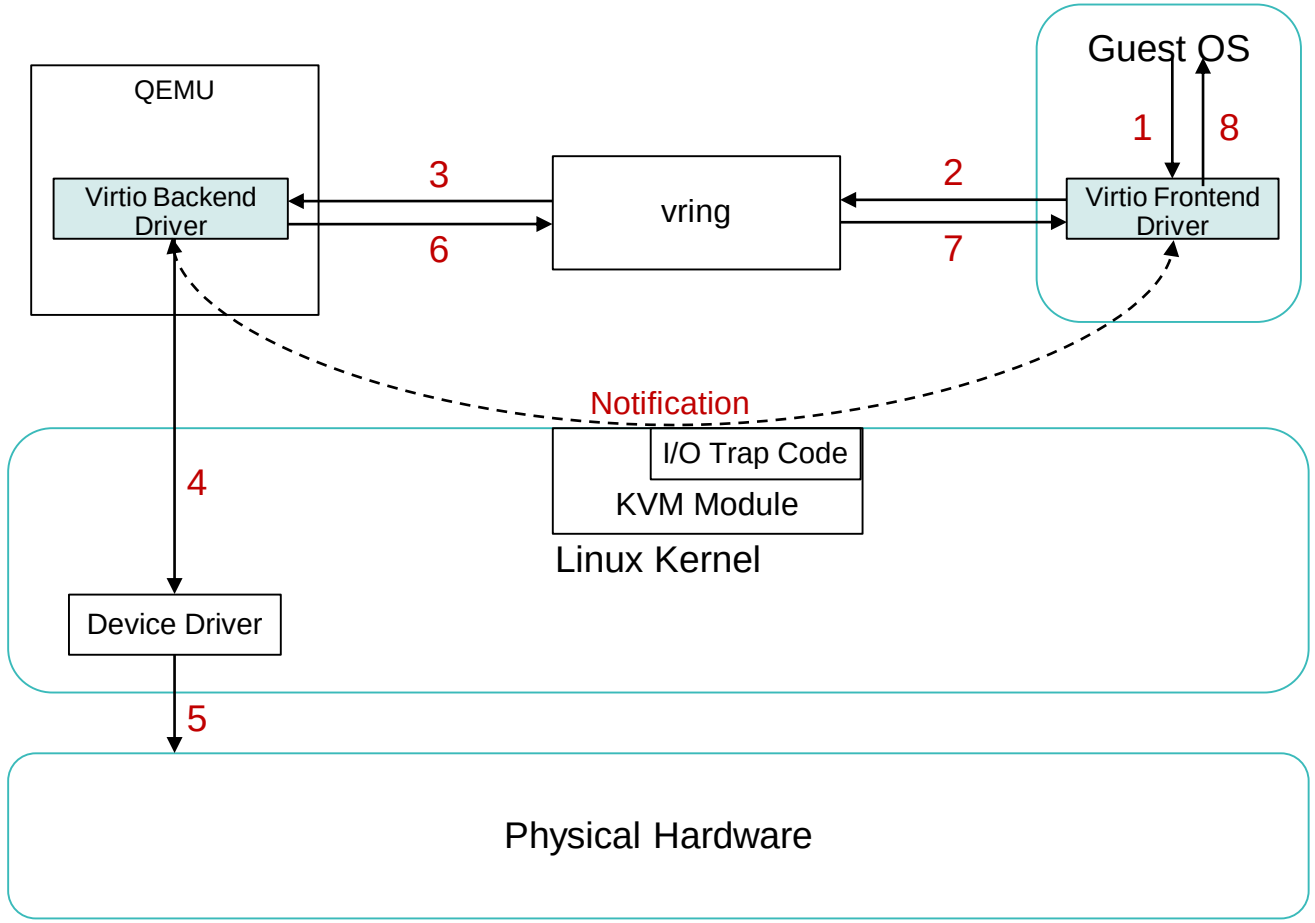
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KVM I/O Process - Default



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KVM I/O Process - Virtio



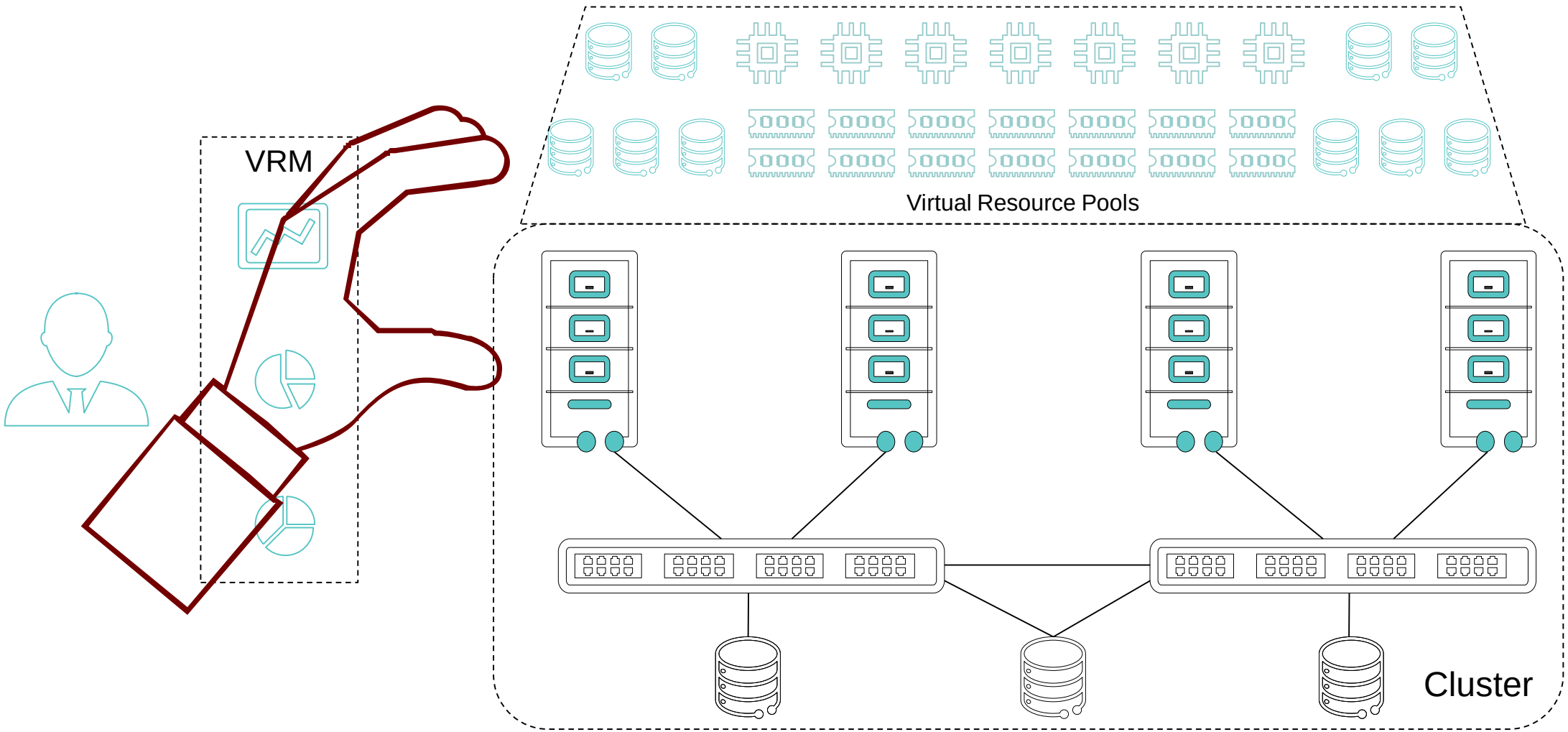
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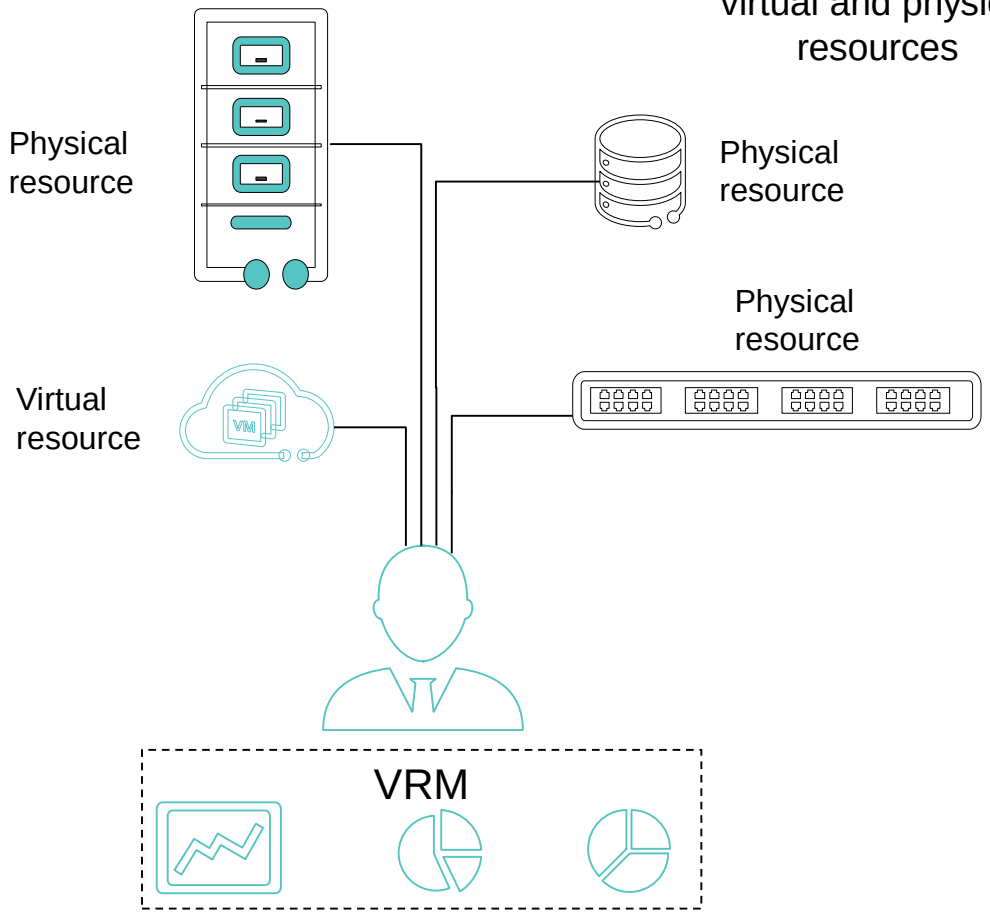
FusionCompute Architecture



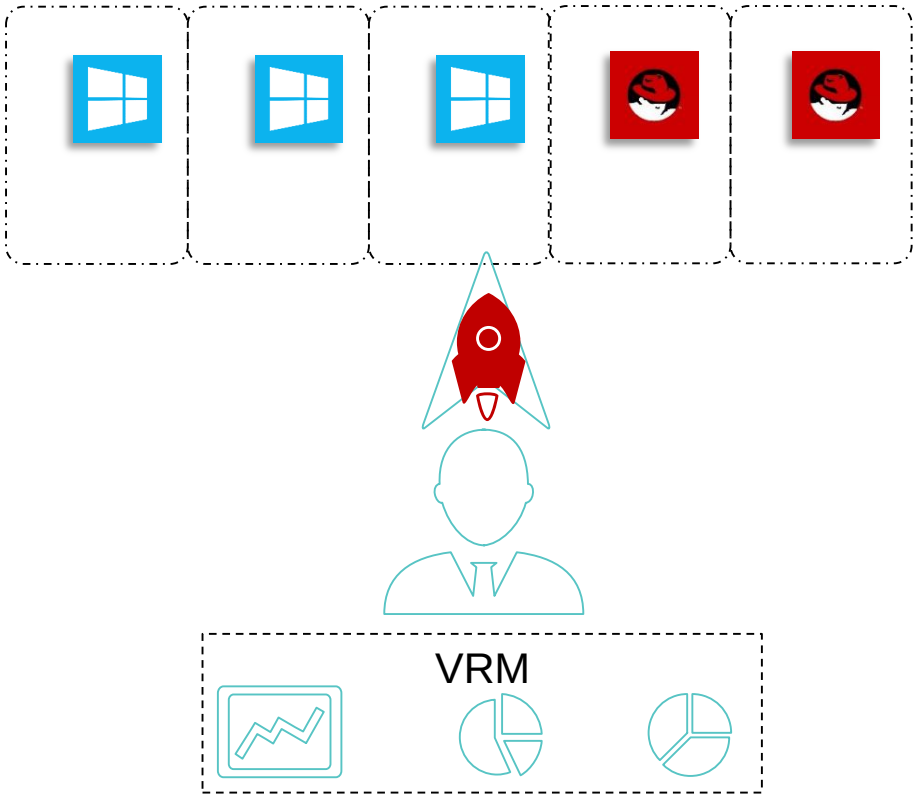
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FusionCompute Advantages

Unified management of virtual and physical resources



Quick VM provisioning



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Functions of FusionCompute Components

Component	Function
CNA	CNA provides the following functions: • Provides virtual computing functions. • Manages VMs on compute nodes. • Manages compute, storage, and network resources on compute nodes.
VRM	VRM provides the following functions: • Manages block storage resources in clusters. • Manages network resources, such as IP addresses and VLANs in clusters, and assigns IP addresses to VMs. • Manages the lifecycle of VMs in clusters, and allocates and migrates VMs across compute nodes. • Dynamically schedules resources in clusters. • Manages virtual resources and user data in a unified manner and provides elastic computing, storage, and IP address services. • Provides a unified Web UI portal, allowing O&M engineers to remotely access the FusionCompute system to monitor and manage resources, and create and view resource reports.

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Quiz

1. In compute virtualization, which of the following allocates CPU and memory resources to VMs? ( )

A. Guest OS

B. VMM

C. Hypervisor

D. Host OS

2. All open-source virtualization technologies are type 1 virtualization, while all closed-source ones are type 2 virtualization.

A. TRUE

B. FALSE

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Summary

- This chapter provides basic knowledge about compute virtualization, including CPU, memory, and I/O virtualization. It introduces you to open-source virtualization technology by using KVM as an example, and also to commercial virtualization products by using FusionCompute as an example.

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  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Network Basics for Cloud Computing

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Foreword

- The network consists of a variety of networking devices. In addition to the traditional physical network, the networks for a virtualized environment also include invisible networks running inside servers. This chapter will cover basic knowledge about both types of networks.

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Objectives

- Upon completion of this course, you will:
  - Understand how physical and virtual switches work.
  - Understand the network architecture used for virtualized environments.
  - Understand the traffic flows between VMs.
  - Understand concepts related to VLAN.

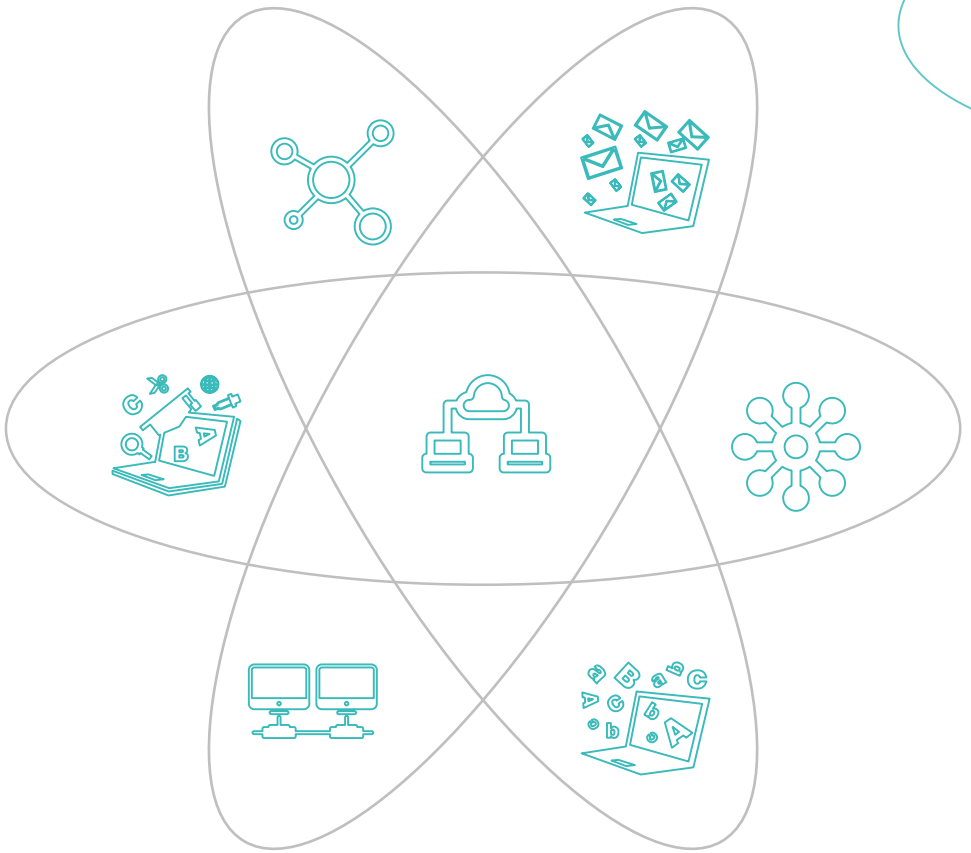
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2. Physical Network for Virtualization
3. Virtual Network for Virtualization
4. Network Features of Huawei Virtualization Product

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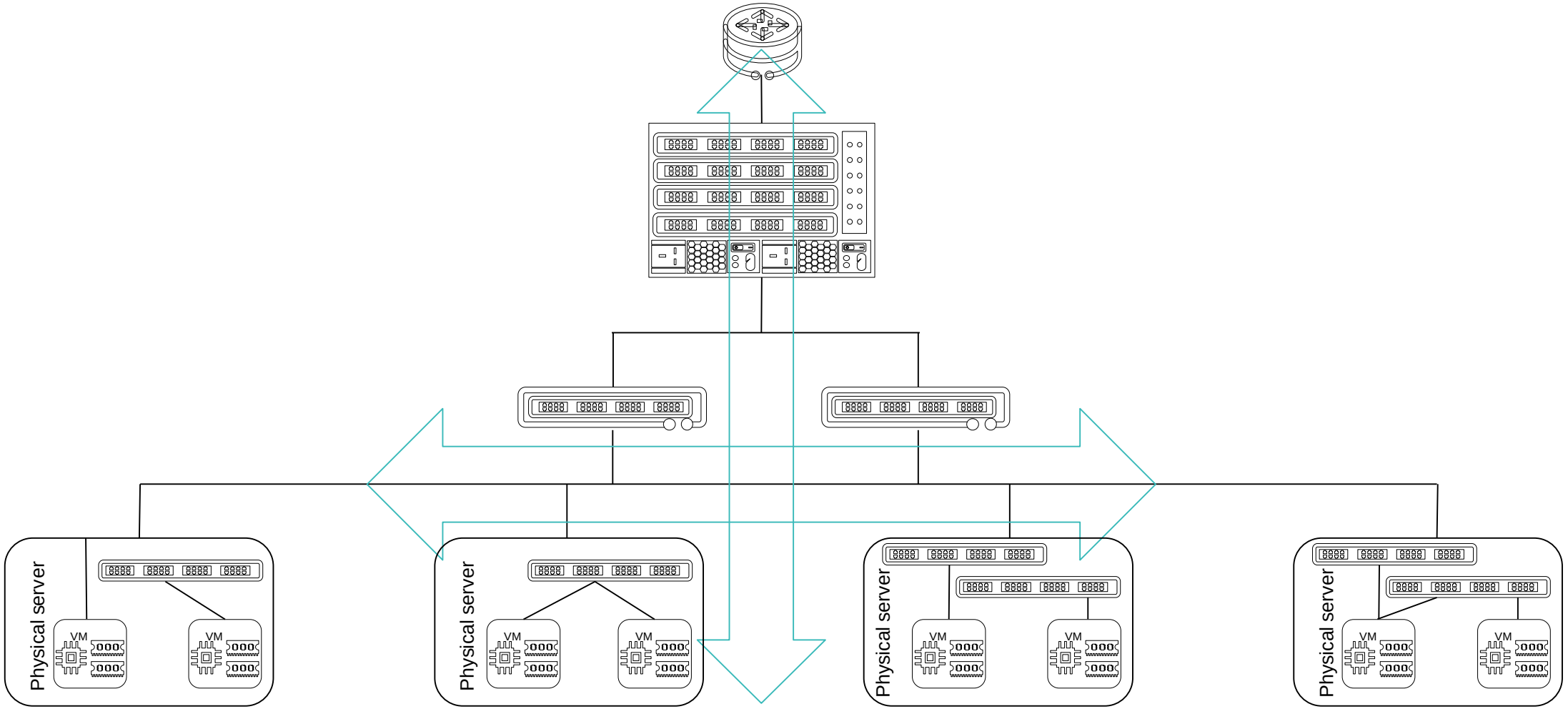
How Do VMs Communicate?





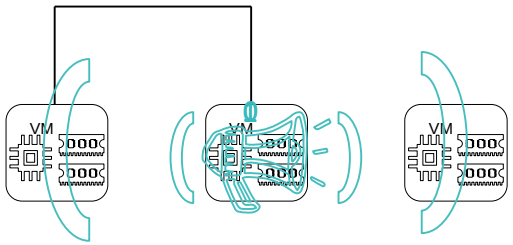
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Network Architecture for Virtualization

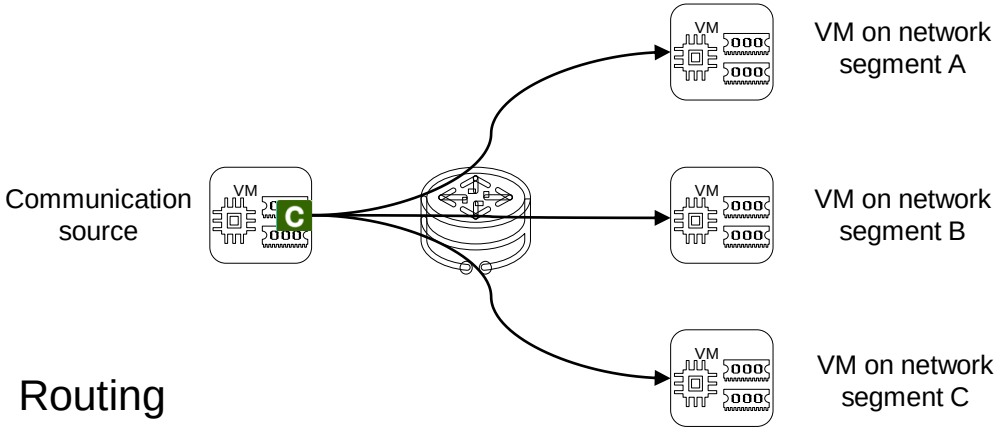


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Basic Network Concepts

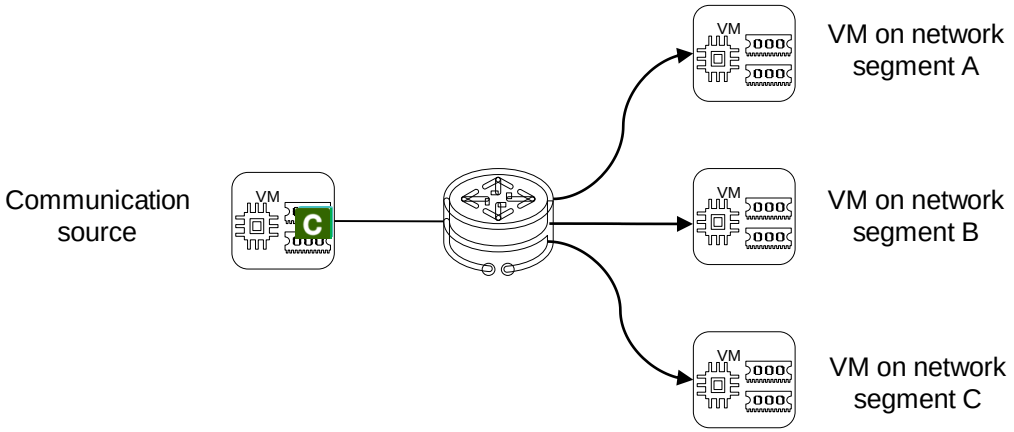


Broadcast & Unicast

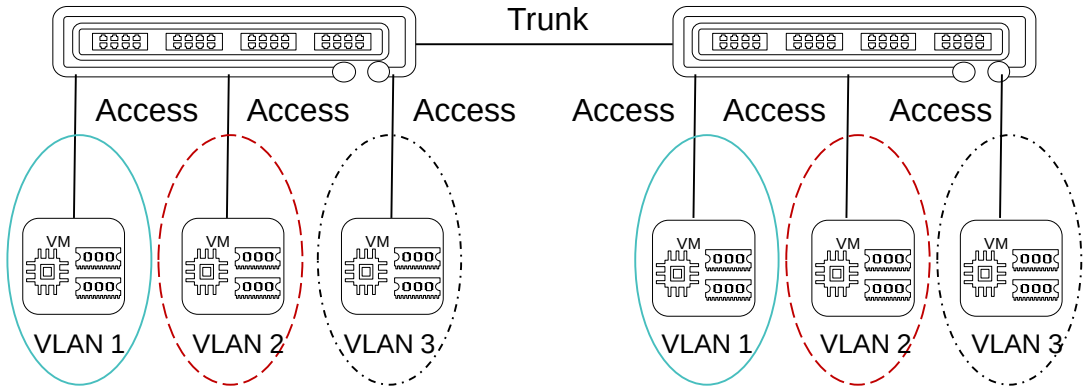


Routing

Default gateway



VLAN



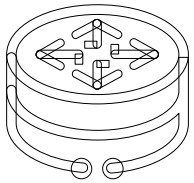
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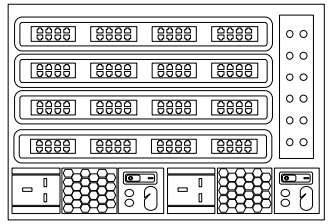
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Physical Network Devices

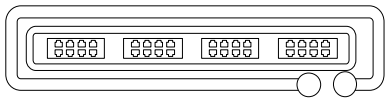


Router



Layer 3 switch

Layer 2 switch

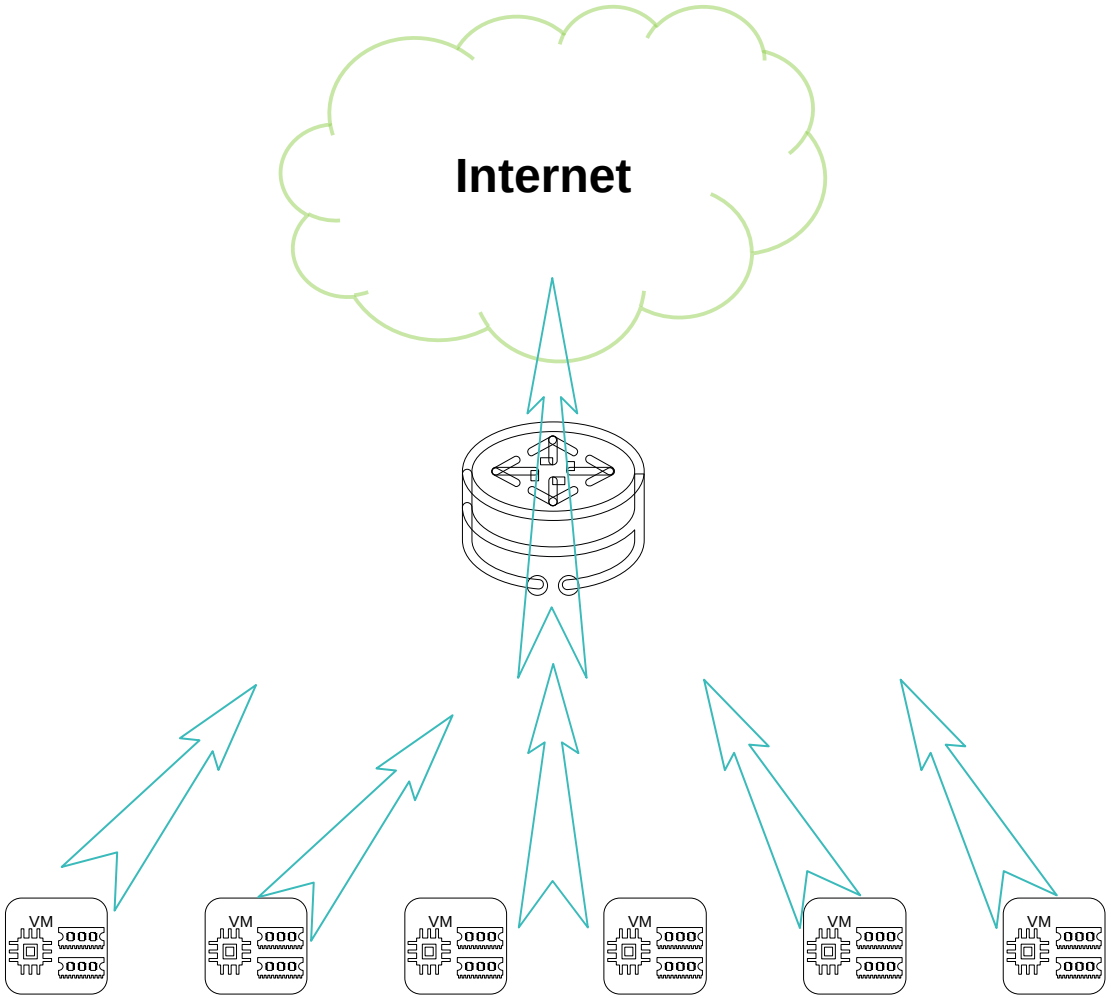


Server NIC



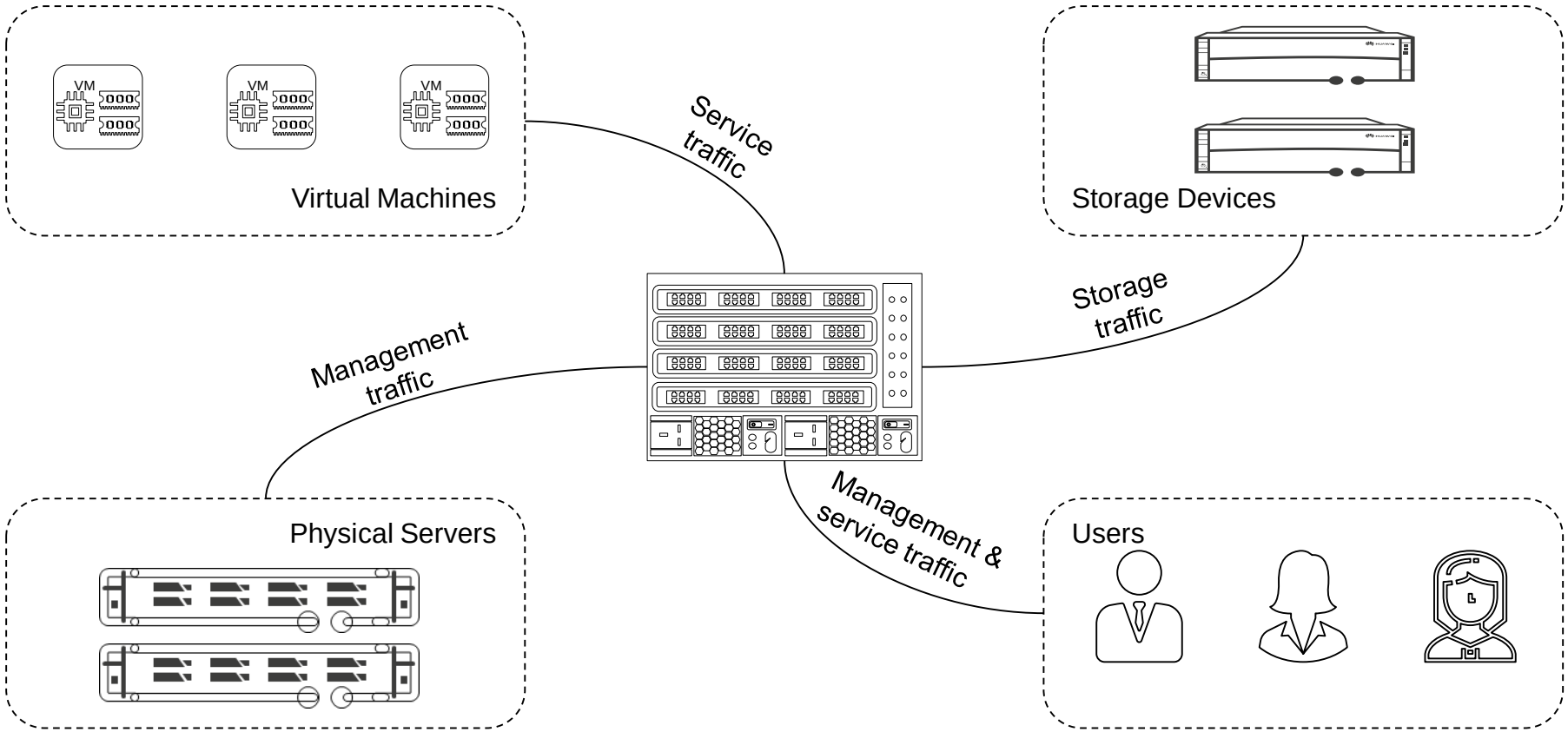
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Router in a Virtualized Environment



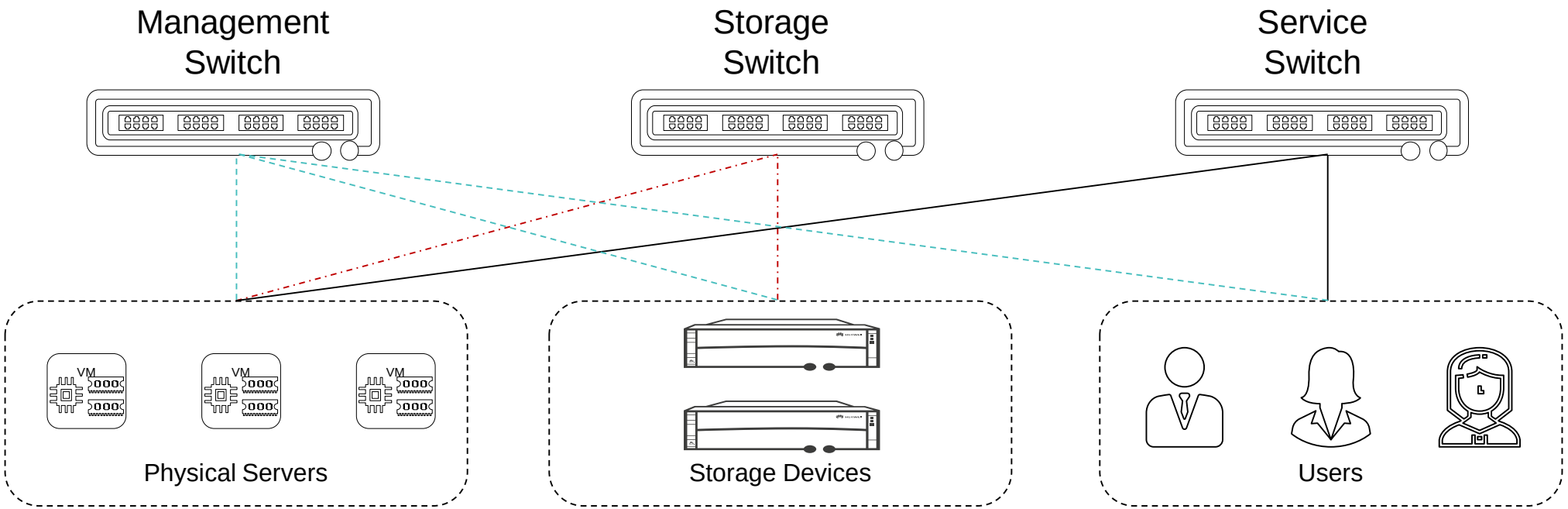
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Layer 3 Switch in a Virtualized Environment



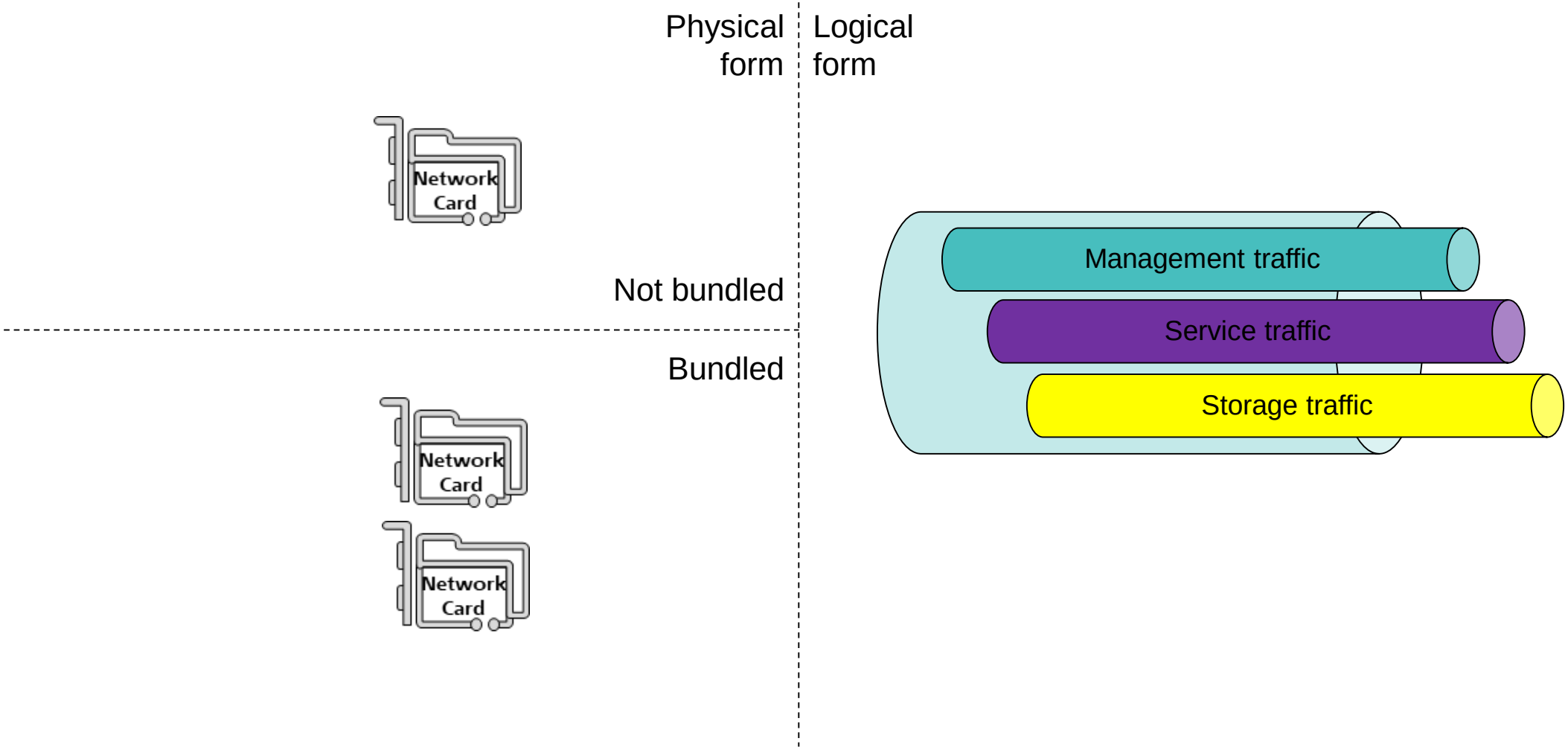
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Layer 2 Switches in a Virtualized Environment



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Physical NICs in a Virtualized Environment



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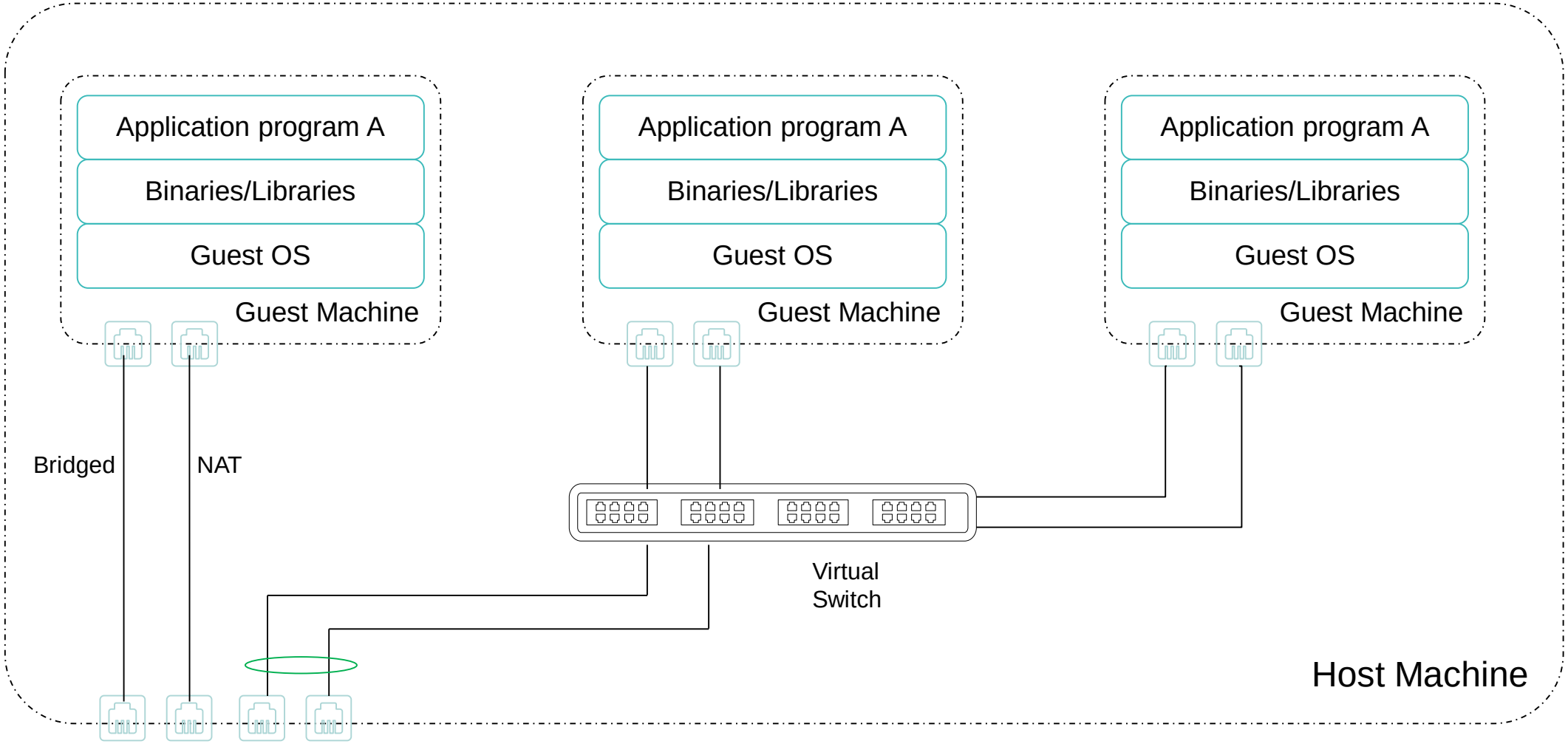
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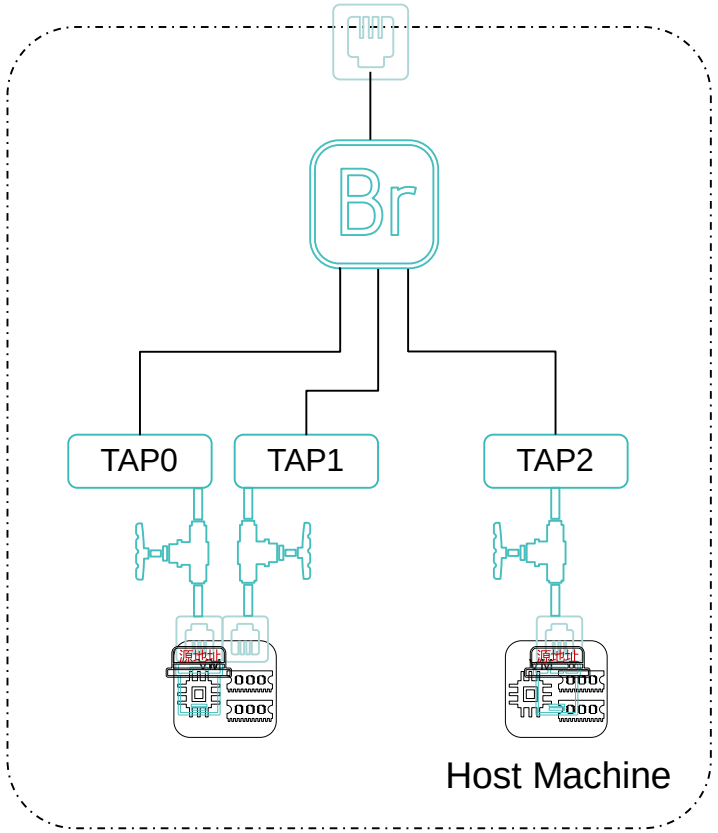
Mainstream Virtual Network Architecture



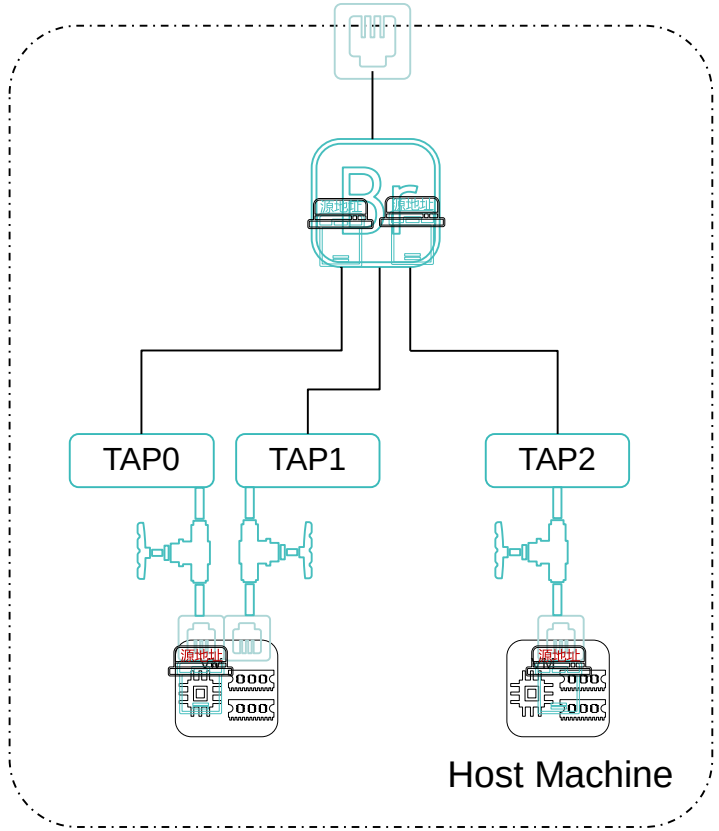
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Bridged vs. NAT

Bridged

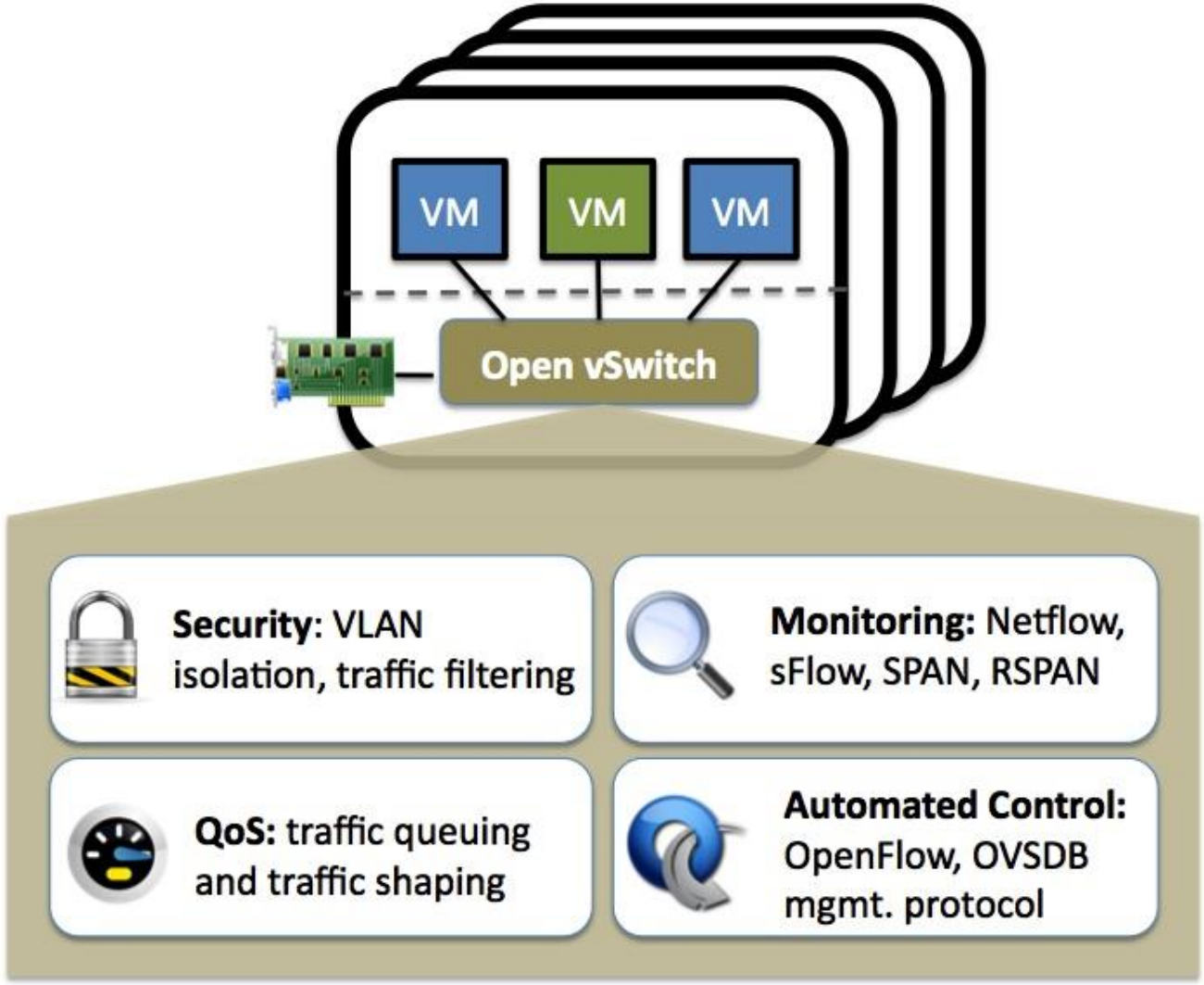


NAT



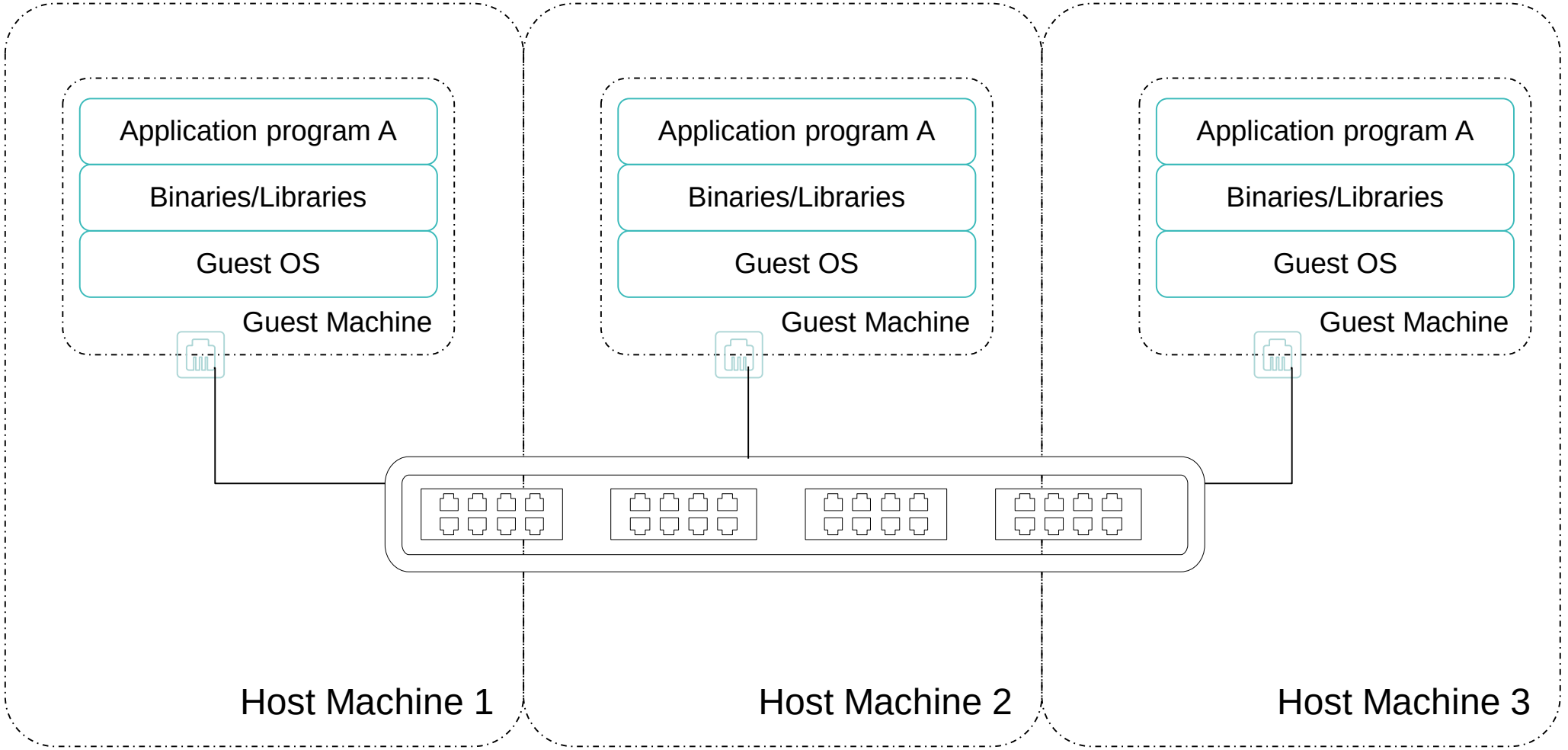
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Virtual Switch - Open vSwitch



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Distributed Virtual Switch



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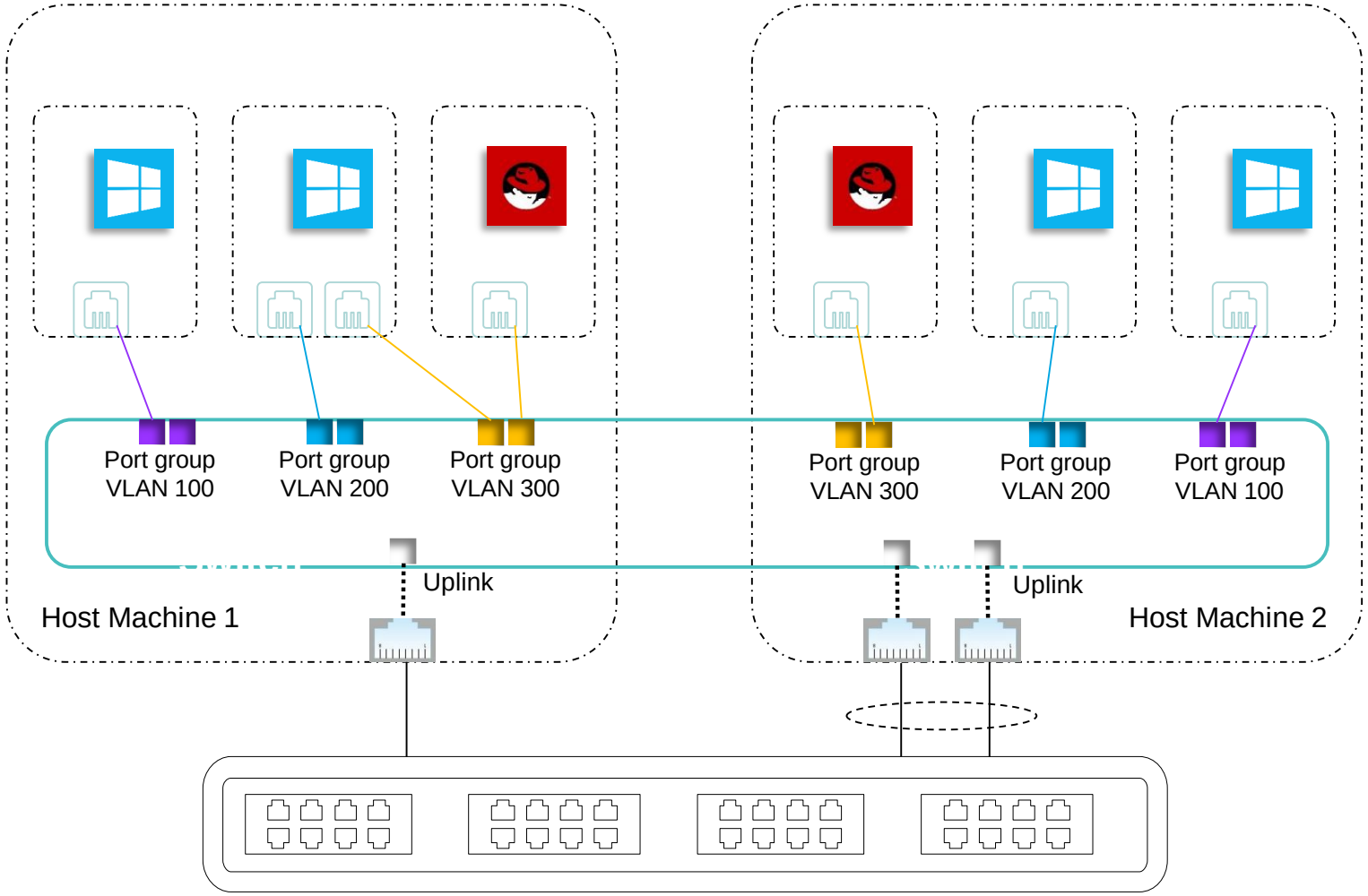
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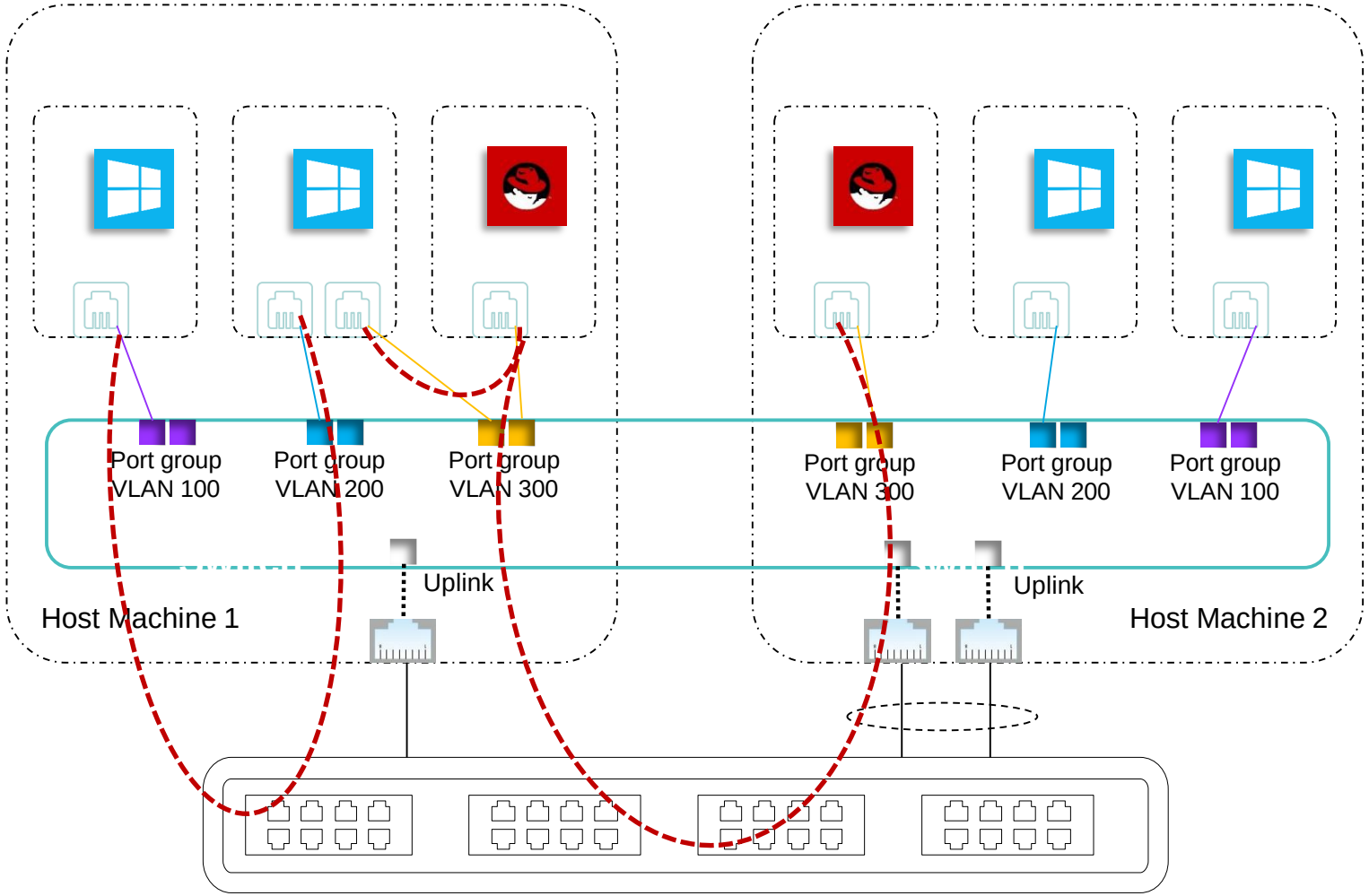
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FusionCompute Network Architecture



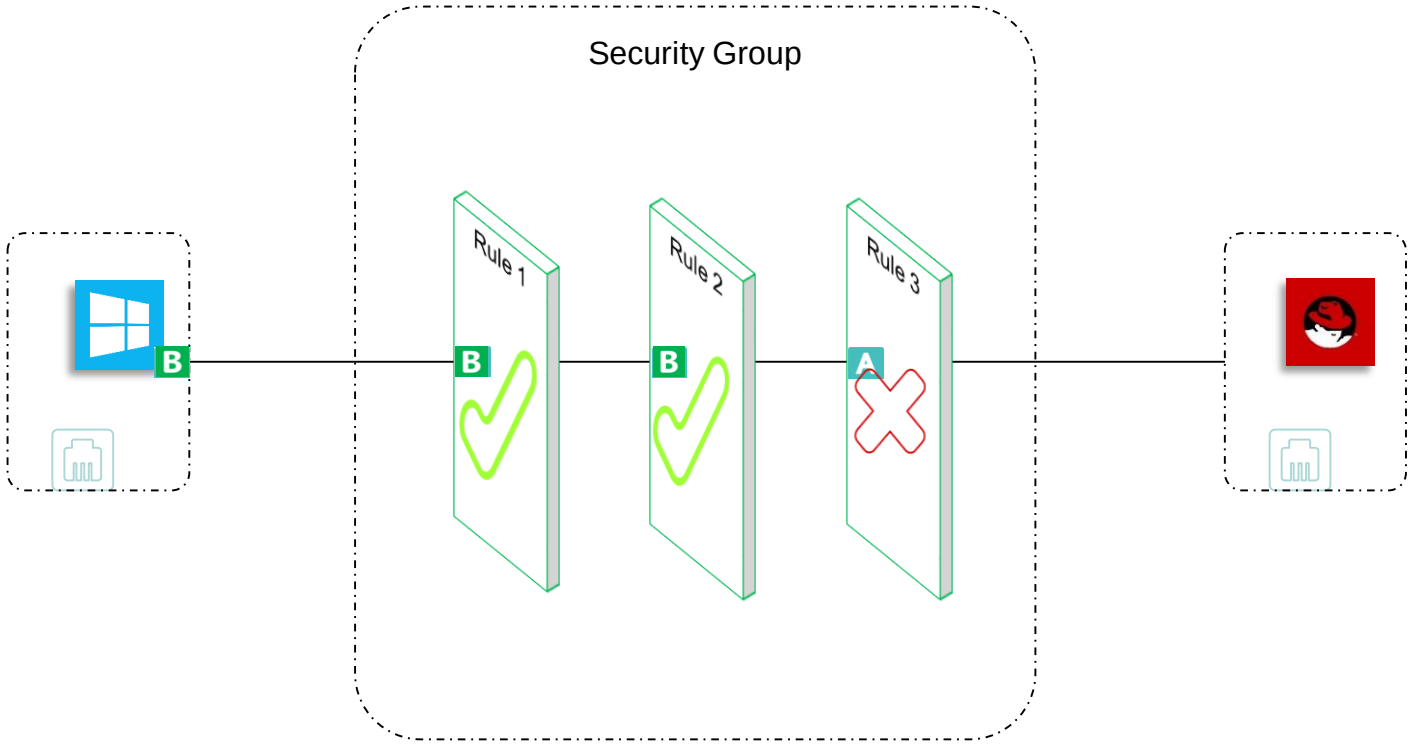
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Traffic Flow



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Security Group



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Quiz

1. Which of the following falls into the category of “north-south traffic”? ()

   A. Traffic between VMs in a data center

   B. Traffic generated when users access the data center through the Internet

   C. Traffic generated during VM live migration

   D. Traffic between VMs in different data centers

2. One VM can have multiple virtual NICs connecting it to different virtual switches.

   A. TRUE

   B. FALSE

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Summary

- This chapter covers basic knowledge about the physical and virtual networks used in virtualized environments.

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Recommendations

- Huawei e-Learning website:
  - http://support.huawei.com/learning/Index!toTrainIndex
- Huawei Support case library:
  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Storage Virtualization Basics

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Foreword

- Data has become an enterprise’s most important asset. How is data stored on the cloud? How is this different from the way it is stored with traditional IT? This chapter will answer these questions.

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Objectives

- Upon completion of this course, you will:
  - Understand Huawei's FusionSphere virtualization solution.
  - Understand the storage architecture used for virtualized environments.
  - Understand different types of disks.
  - Understand the differences between centralized and distributed storage.
  - Understand the differences between virtualized and non-virtualized storage.
  - Understand different types of VM disks.
  - Understand the storage features of Huawei's virtualization product.

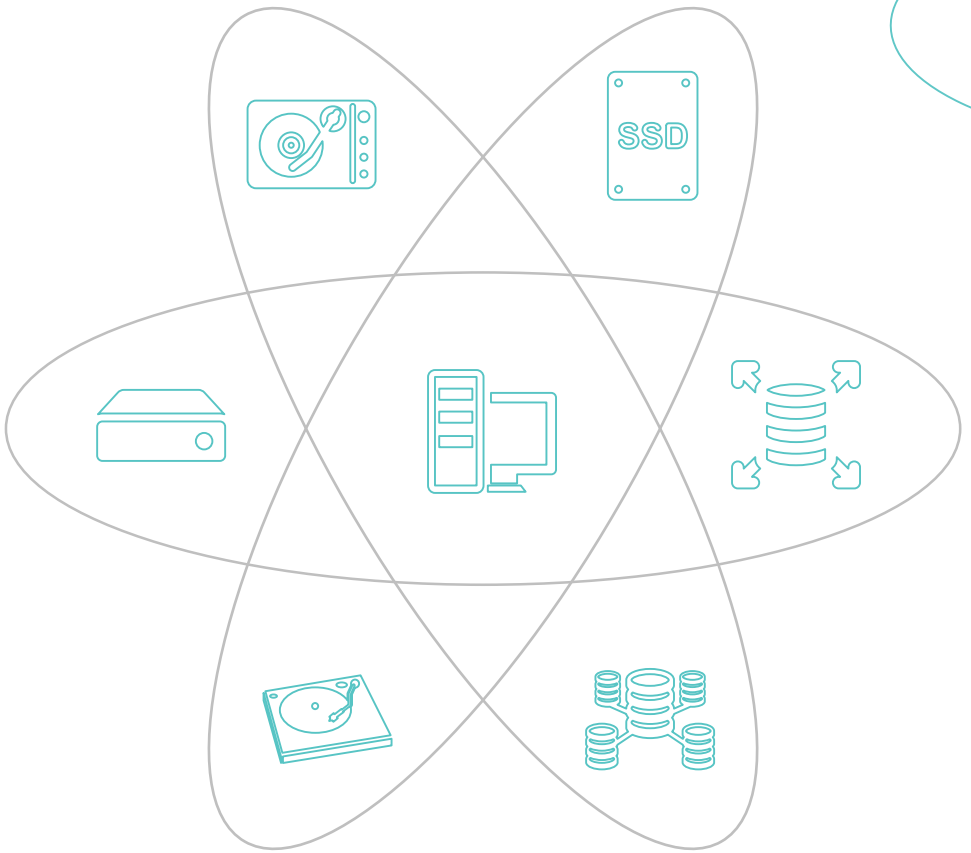
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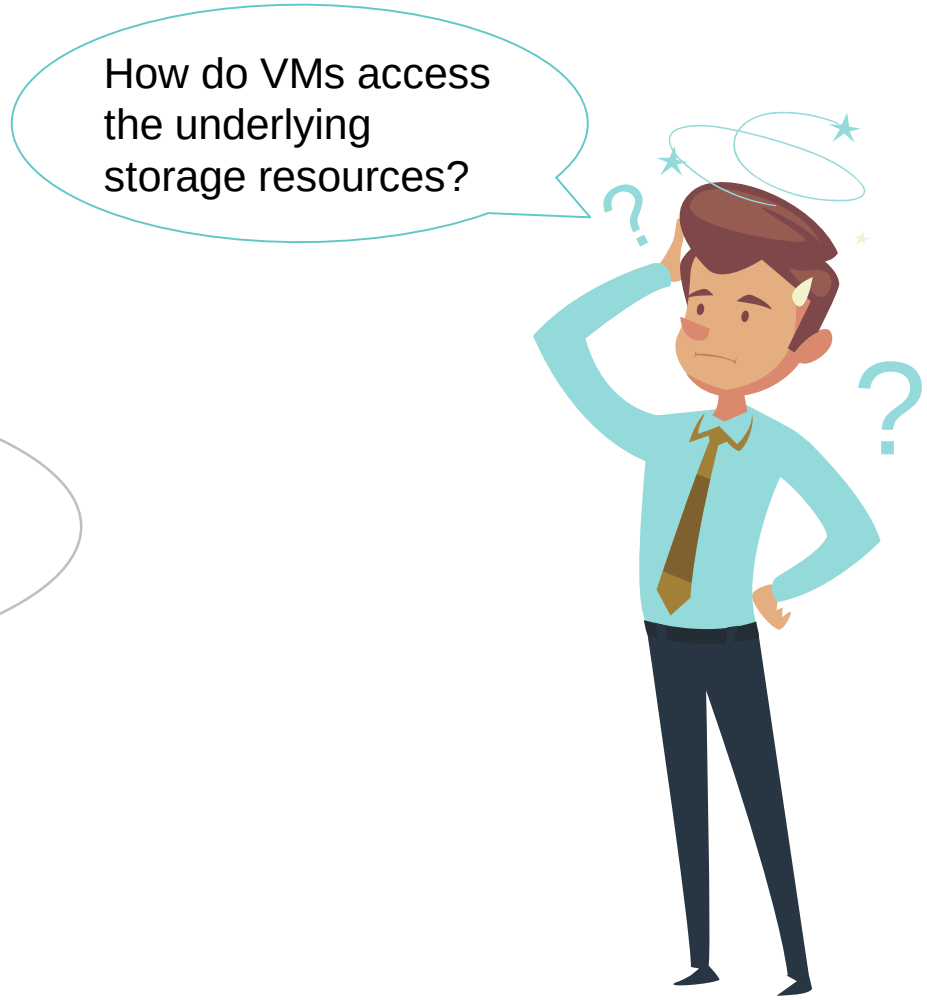
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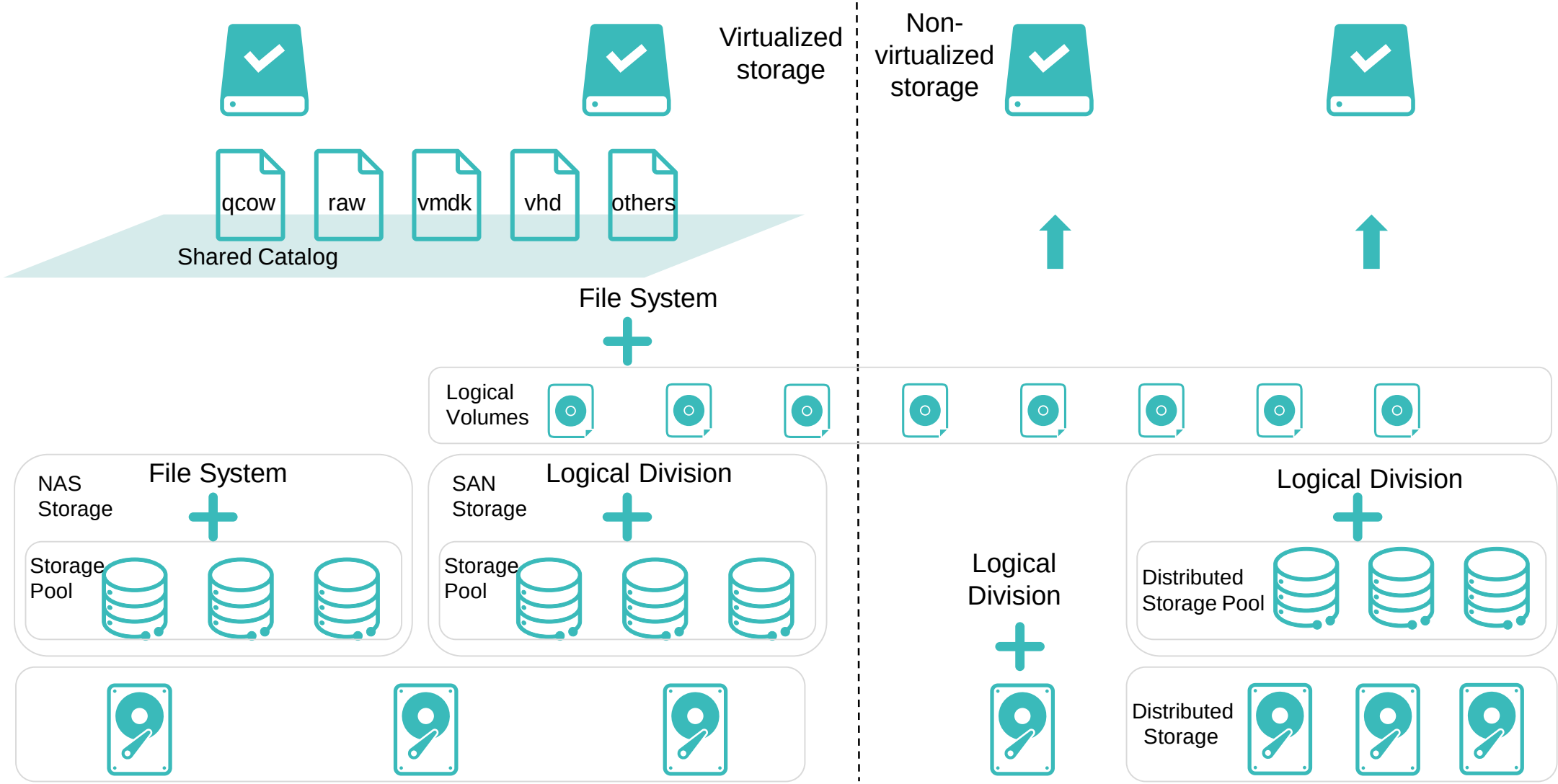
Where Do the Storage Resources Used by VMs Come From?





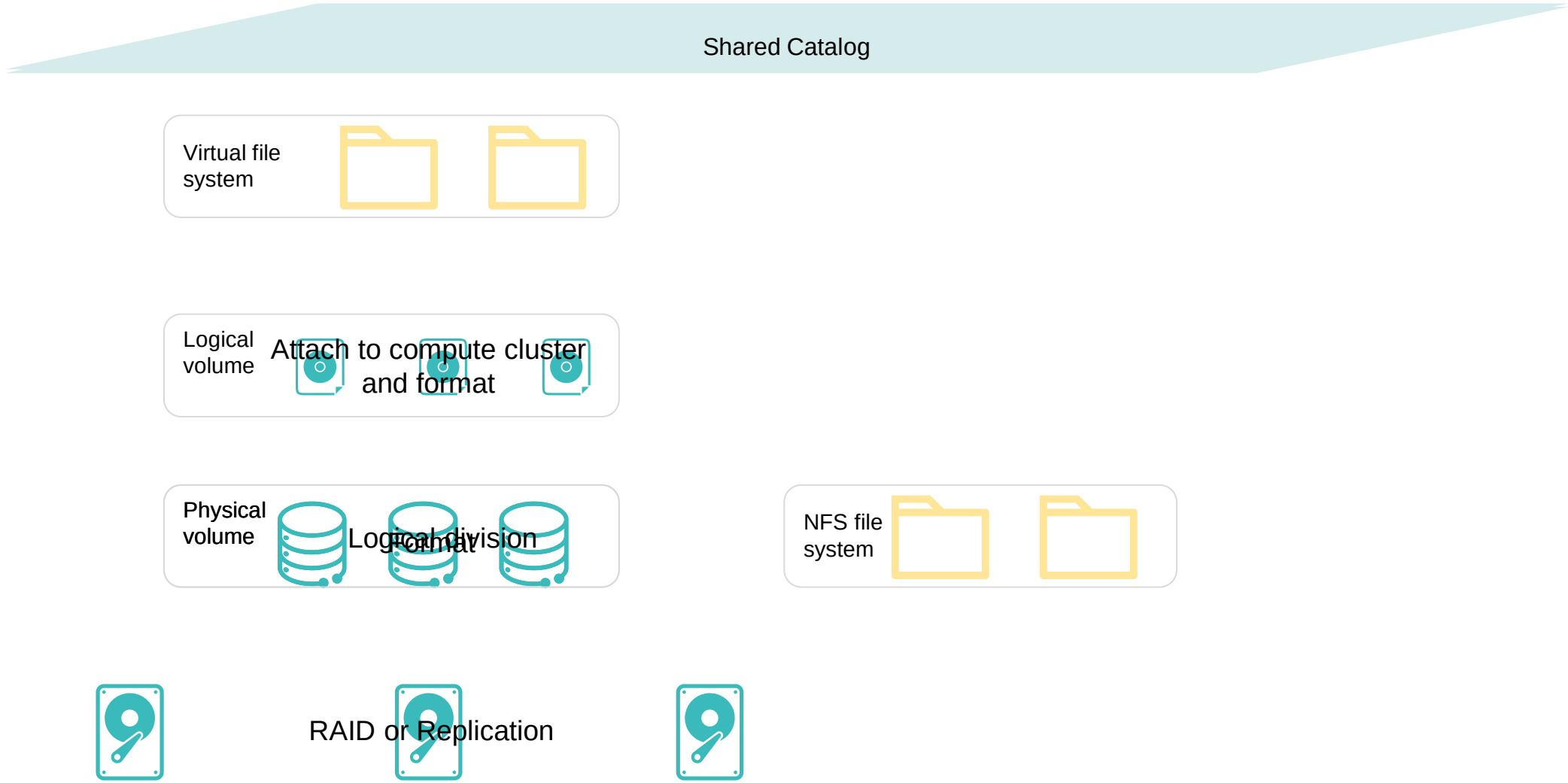
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Storage Architecture for Virtualization



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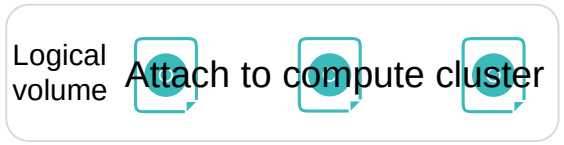
Virtualized Storage Conversion Path in Cloud Computing



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Non-Virtualized Storage Conversion Path in Cloud Computing









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Physical Disk Type - SATA

- SATA (Serial Advanced Technology Attachment), also called serial ATA, is a computer bus interface that connects host bus adapters to mass storage devices. Using serial communication, the serial ATA bus uses embedded clock signals to check transmission commands (not only data) and automatically corrects detected errors, ensuring reliable data transmission for SATA disks. The SATA interface is simple and the disks are hot-swappable.



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Physical Disk Type - SAS

- Serial Attached SCSI (SAS) is a new-generation SCSI technology, which, like the popular Serial ATA (SATA), also uses serial communication to achieve a higher transmission speed. Compared with the older parallel SCSI, SAS improves the efficiency, availability, and scalability of storage systems and offers optional compatibility with SATA.



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Physical Disk Type - NL-SAS

- Nearline SAS (NL-SAS) disks are a combination of a SAS interface with SATA disks. The rotational speed of NL-SAS disks is only 7200 rpm, so performance is lower than that of SAS disks. However, with the SAS interface, addressing and speed are improved.



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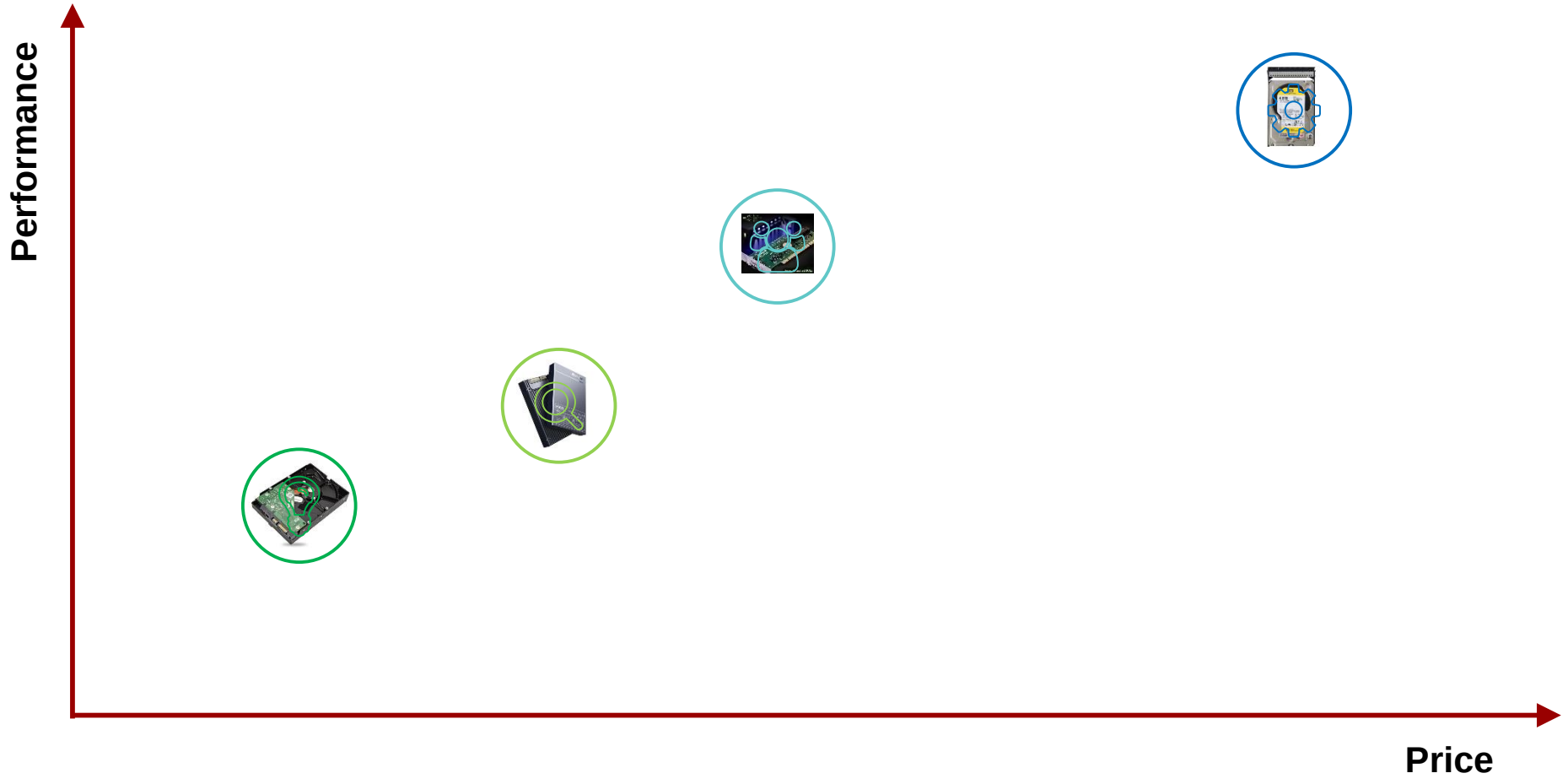
Physical Disk Type - SSD

- A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies as non-volatile memory. An SSD consists of a control unit and a storage unit (flash or DRAM chip). An SSD is the same as a common hard disk drive (HDD) in terms of interface specifications and definition, functionality, usage, and product shape and size. Although SSDs deliver fast read/write performance, light weight, low power consumption, and compact sizes which traditional HDDs cannot rival, SSDs have a shorter lifespan and higher price.



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Comparing Different Types of Physical Disks



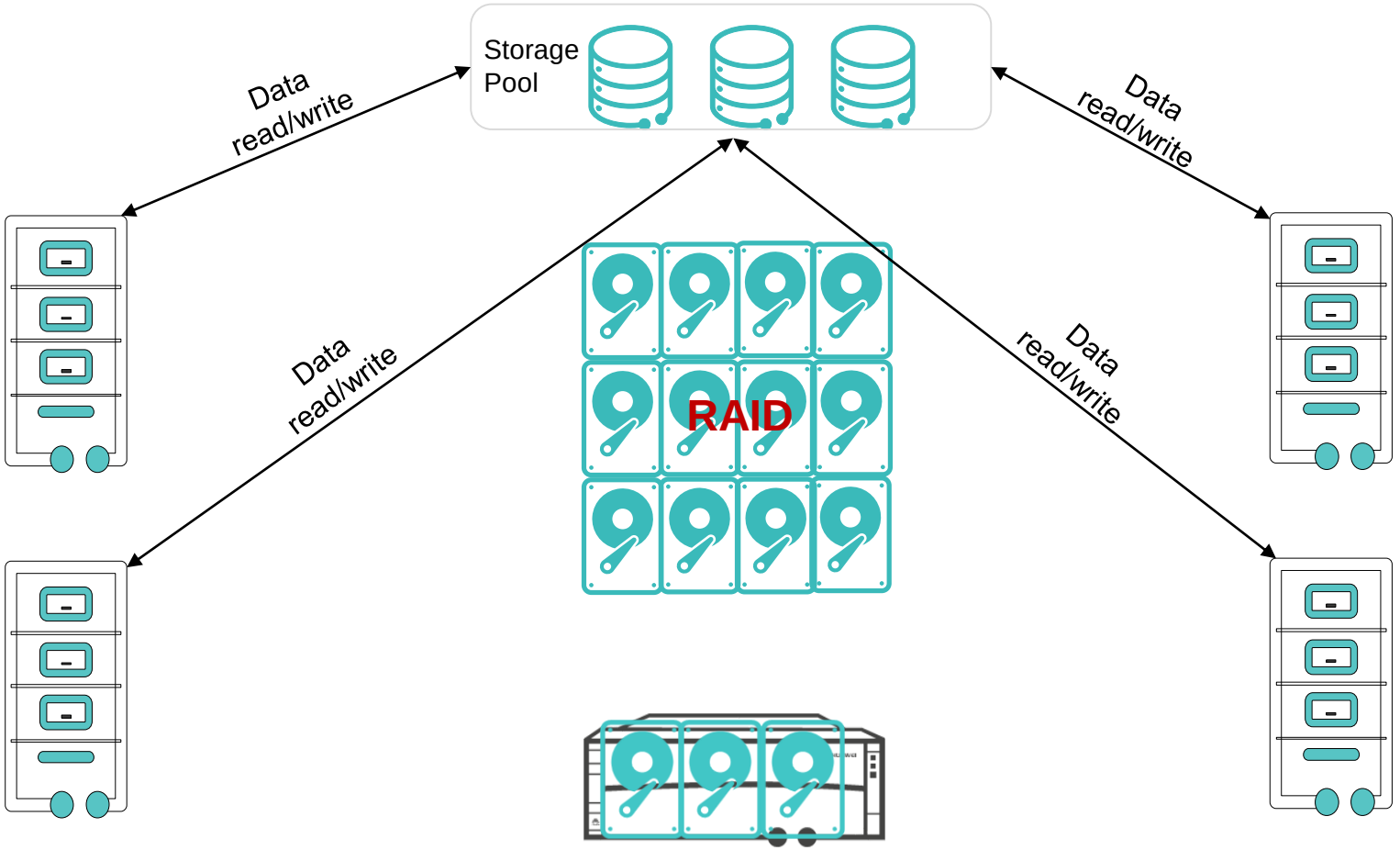
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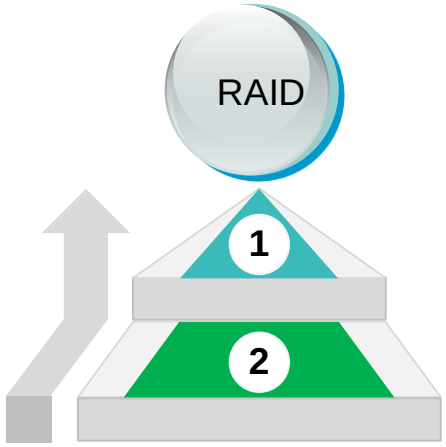
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Centralized Storage



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RAID Technology



Efficient

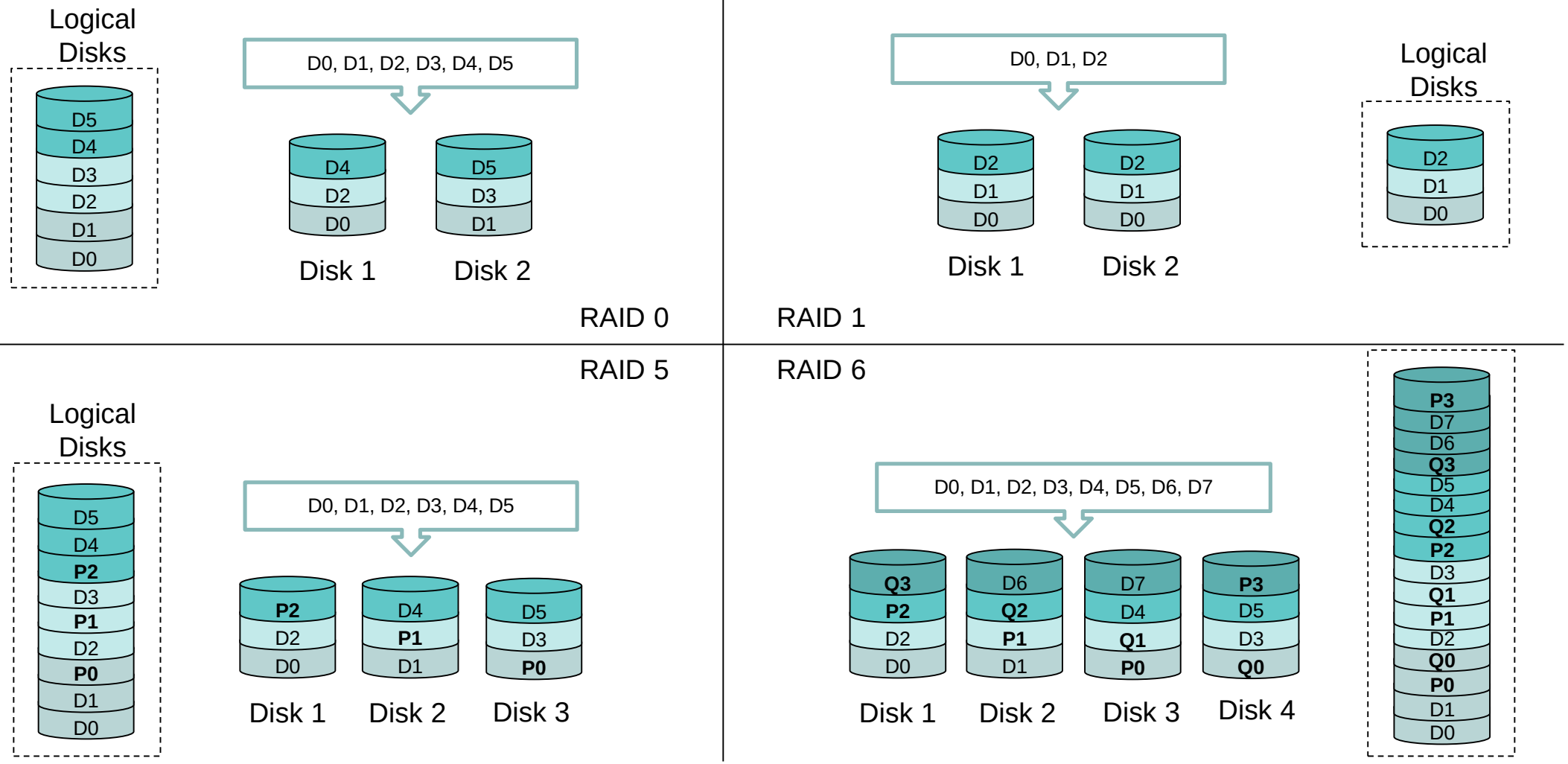
Parallel read and write on multiple disks

Safe

Parity check and hot backup

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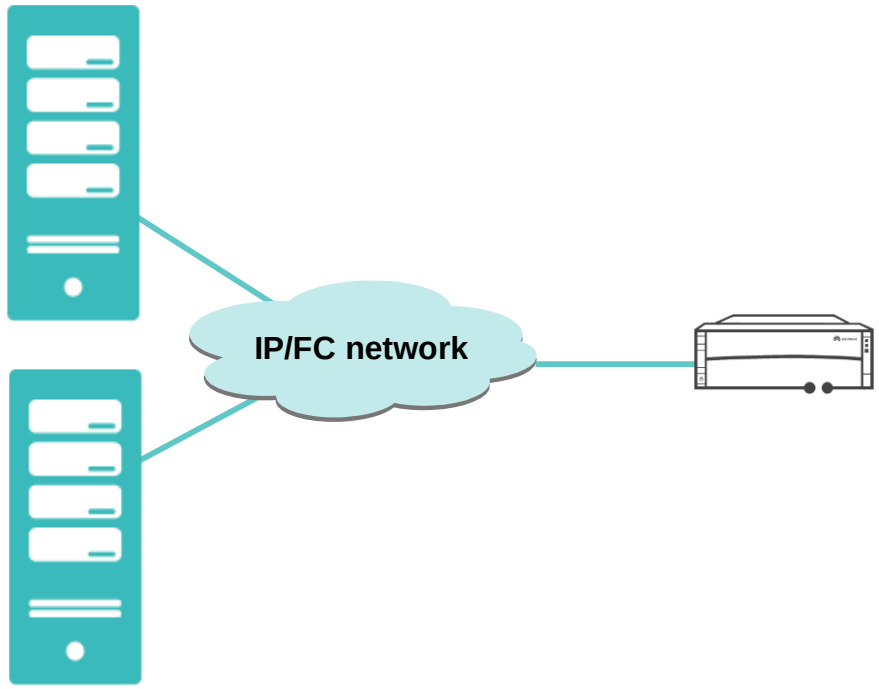
Common RAID Types



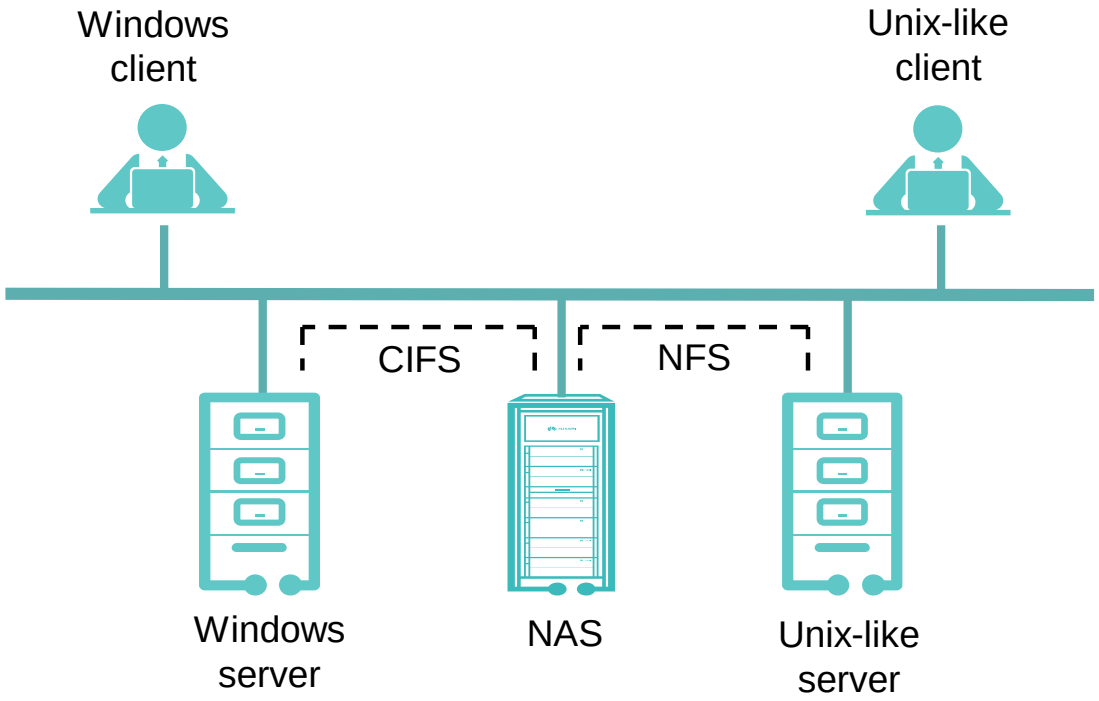
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Centralized Storage Types

SAN

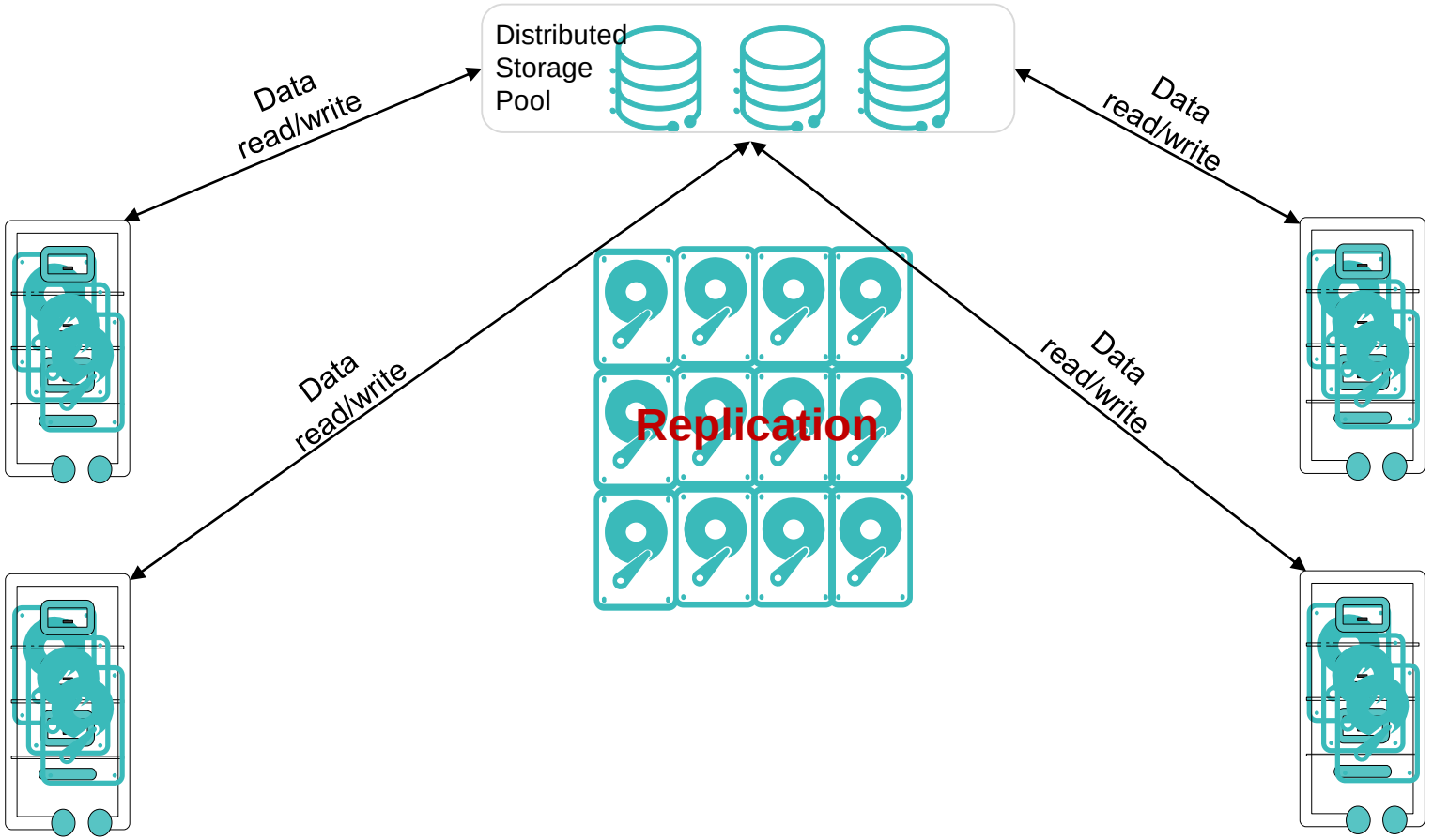


NAS



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Distributed Storage



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Replication

Data write



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Replication

Data write







Data read





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Popular Distributed Storage Products

ceph

hadoop HDFS



Powered By vmware vSAN™

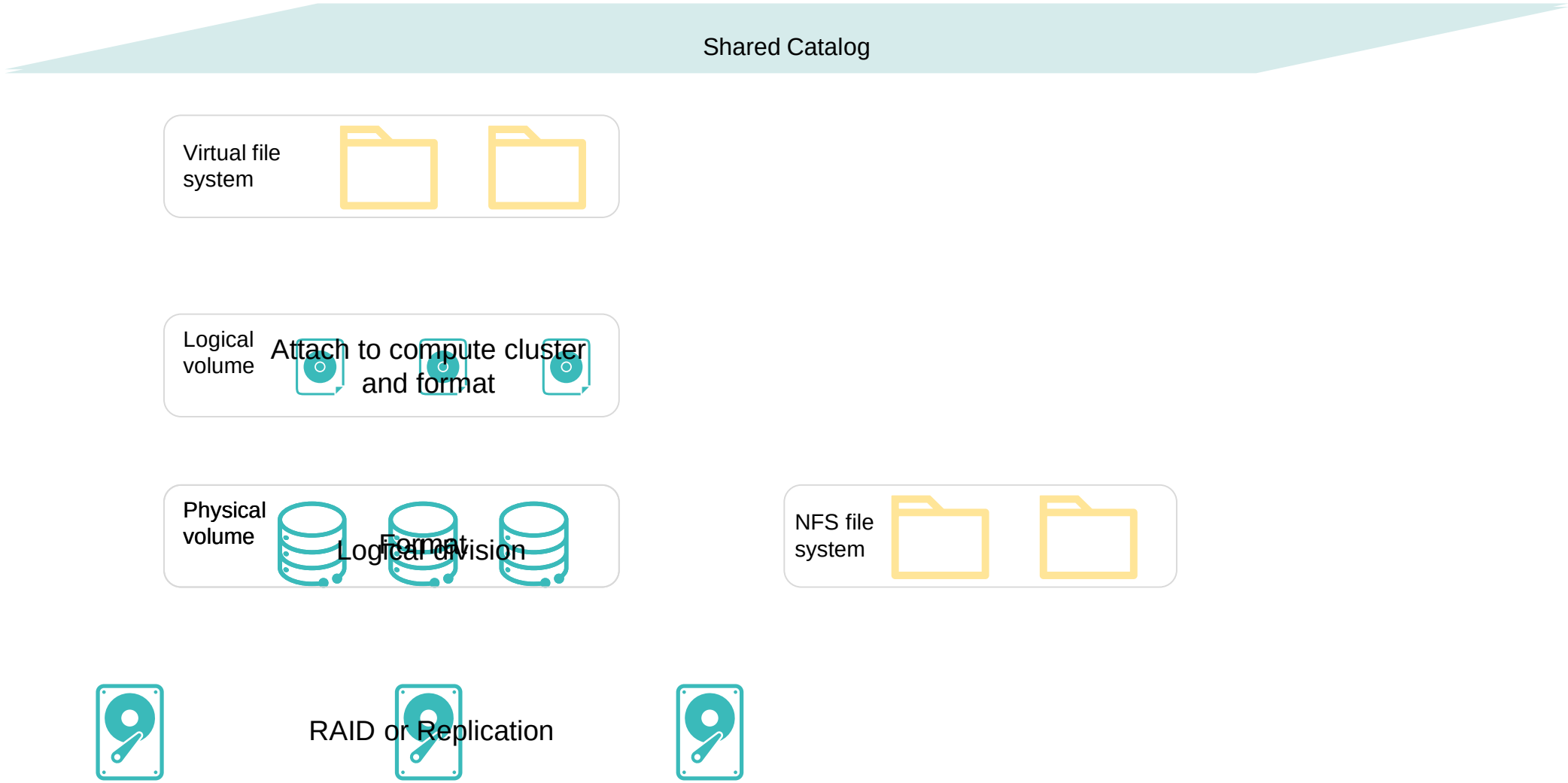
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Virtualized Storage Conversion Path in Cloud Computing



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Non-Virtualized Storage Conversion Path in Cloud Computing





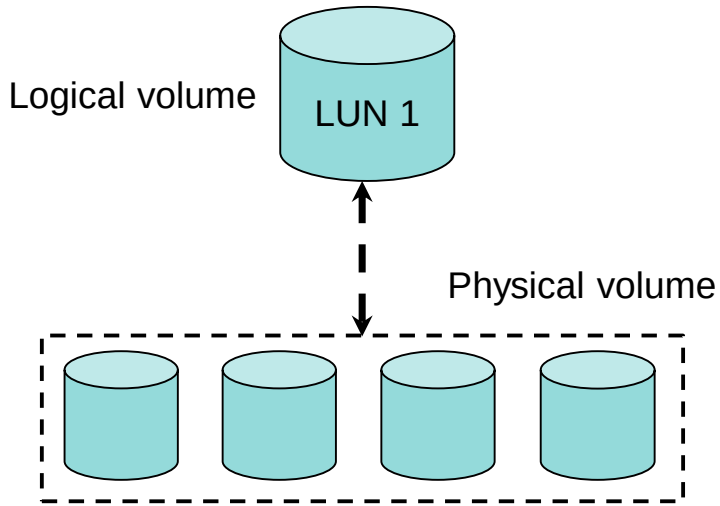




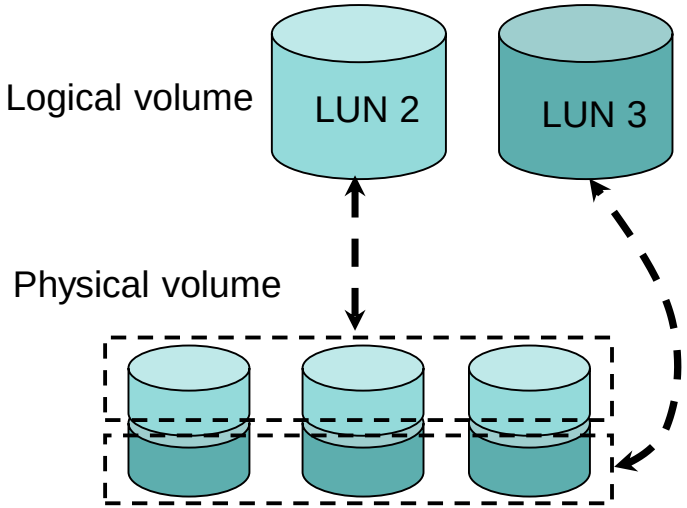
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Relationship Between RAIDs and LUNs

- A RAID array can be seen as a large physical volume made up of a number of hard disks.
- On top of a physical volume, you can create one or more logical units of specified capacities. These logical units are called Logical Unit Numbers (LUNs) and can be mapped to hosts as basic block devices.



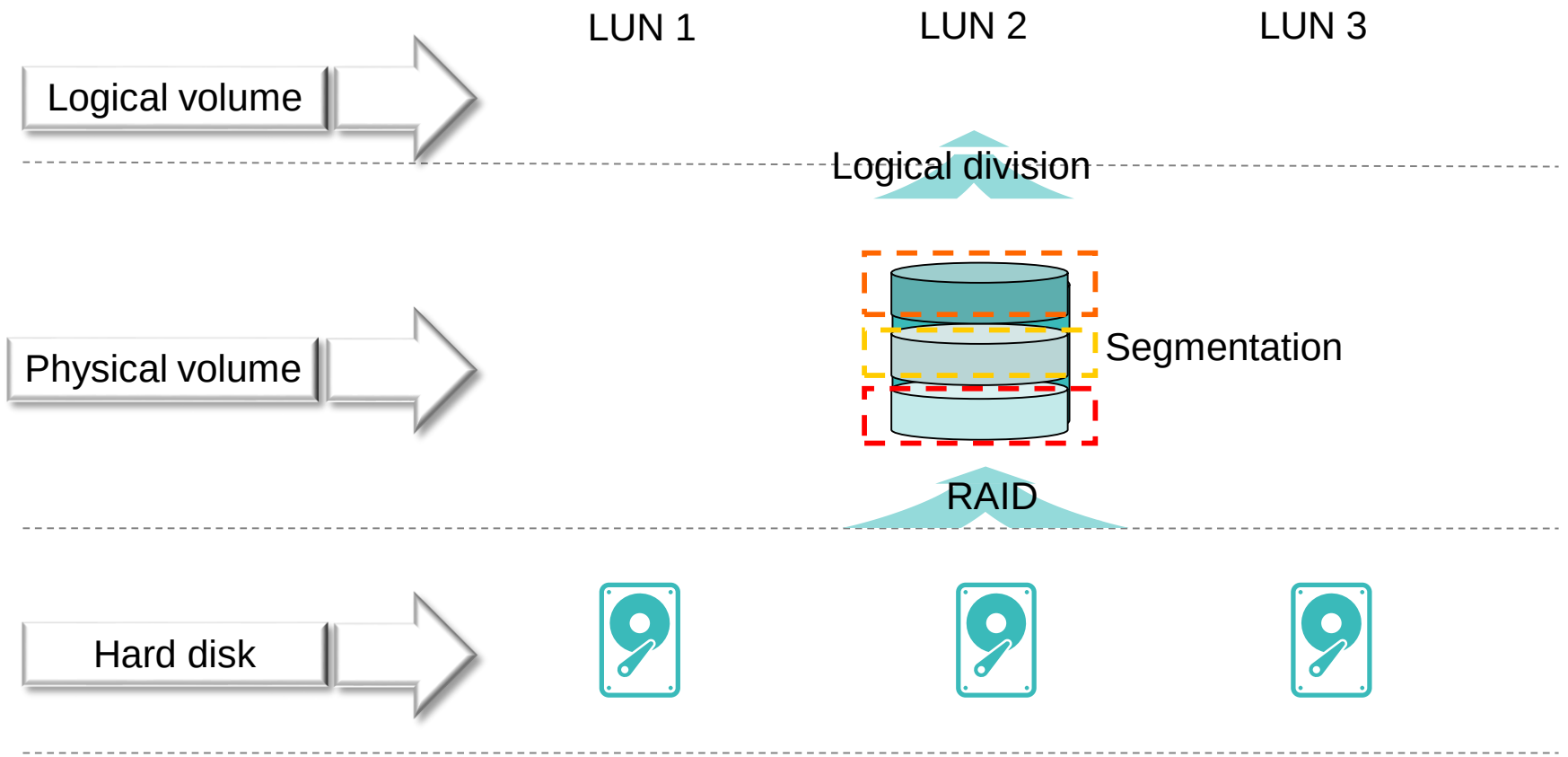
Create one logical volume on a single physical volume



Create multiple logical volumes on a single physical volume

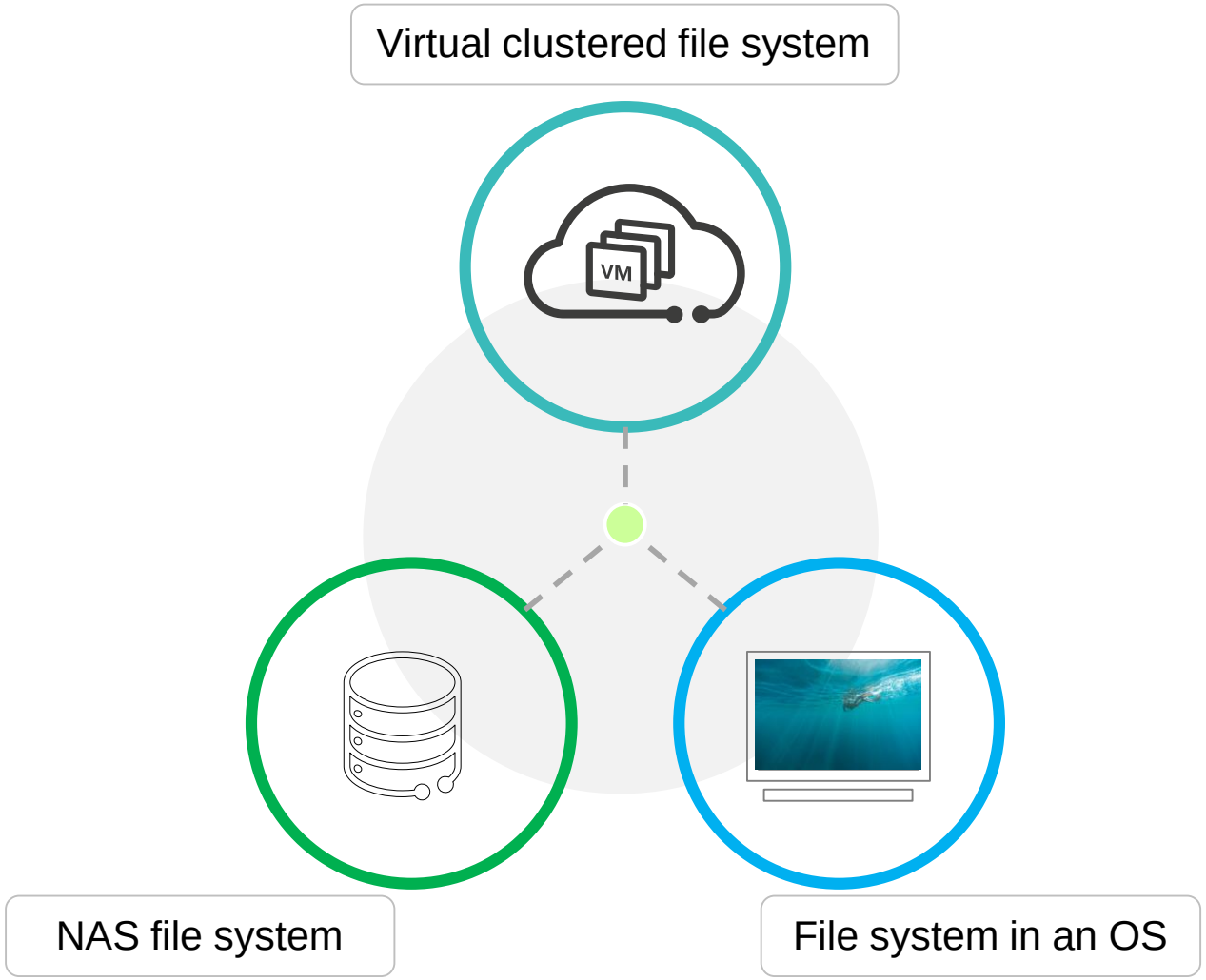
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Process of Creating a LUN



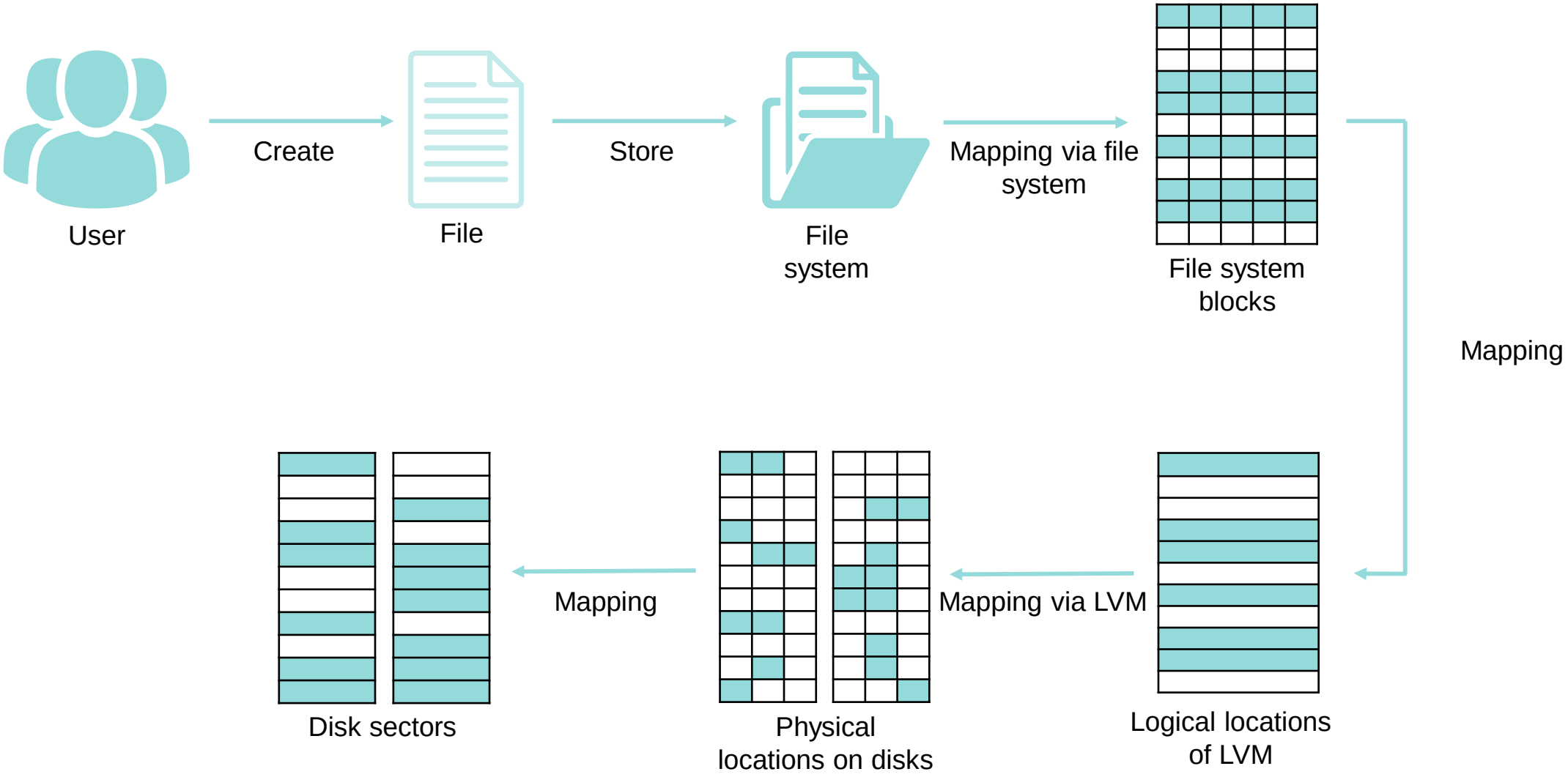
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Common File Systems



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Mapping a File to a Disk



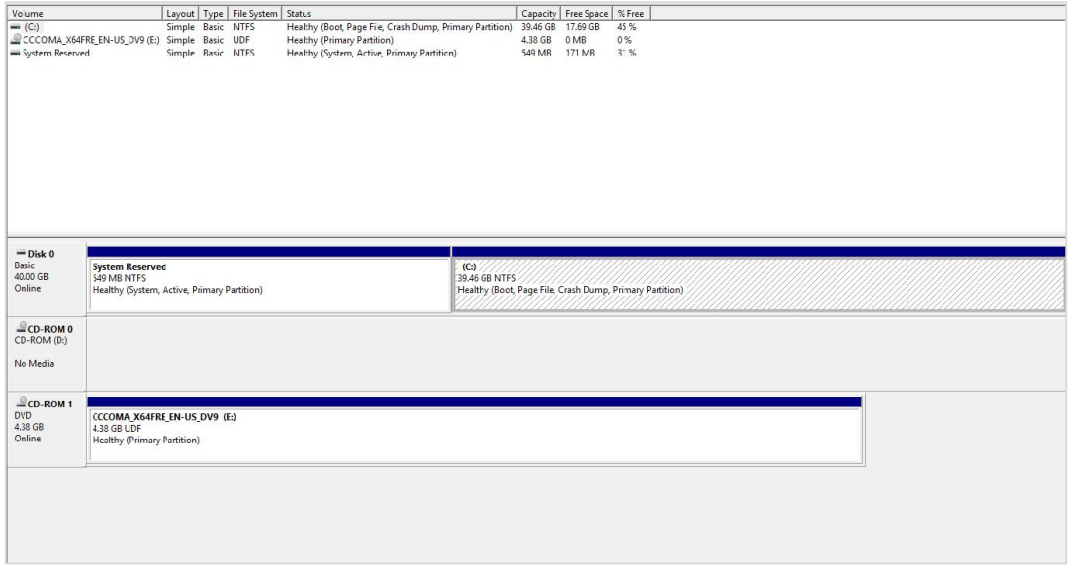
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VM Disks



User view

Administrator view

```
</disk>
<disk type='file' device='disk'>
  <driver name='qemu' type='raw' cache='none' io='native' iothread='1'/>
  <source file='/POME/datastore_3/vol/vol_f6db8fbd-c0c8-4cc4-9252-a862767219d0/vol_f6db8fbd-c0c8-4cc4-9252-a862767219d0.img'/>
  <target dev='vda' bus='virtio'/>
  <boot order='1'/>
  <address type='pci' domain='0x0000' bus='0x02' slot='0x01' function='0x0'/>
```


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Common VM Disk Formats

VM Disk File Format	Vendor
RAW	All
VMDK	VMware
VHD	Microsoft Hyper-V, Huawei FusionCompute
QCOW	QEMU or KVM virtualization platforms only
QED	
VDI	Oracle

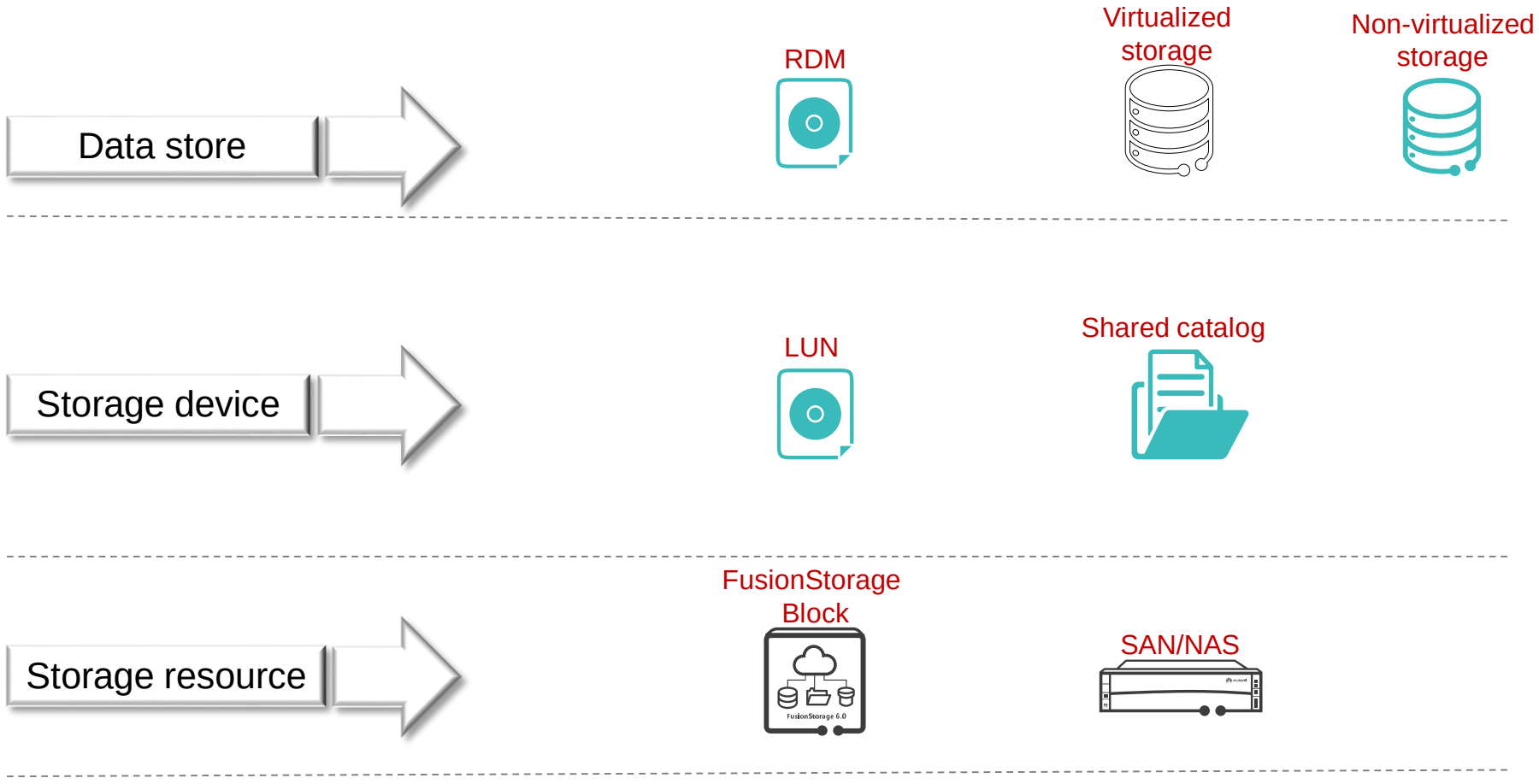
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2. Physical Disk Types and Related Techniques
3. Centralized Storage vs. Distributed Storage
4. Virtualized Storage vs. Non-Virtualized Storage
5. Introduction to VM Disks
6. **Storage Features of Huawei’s Virtualization Product**

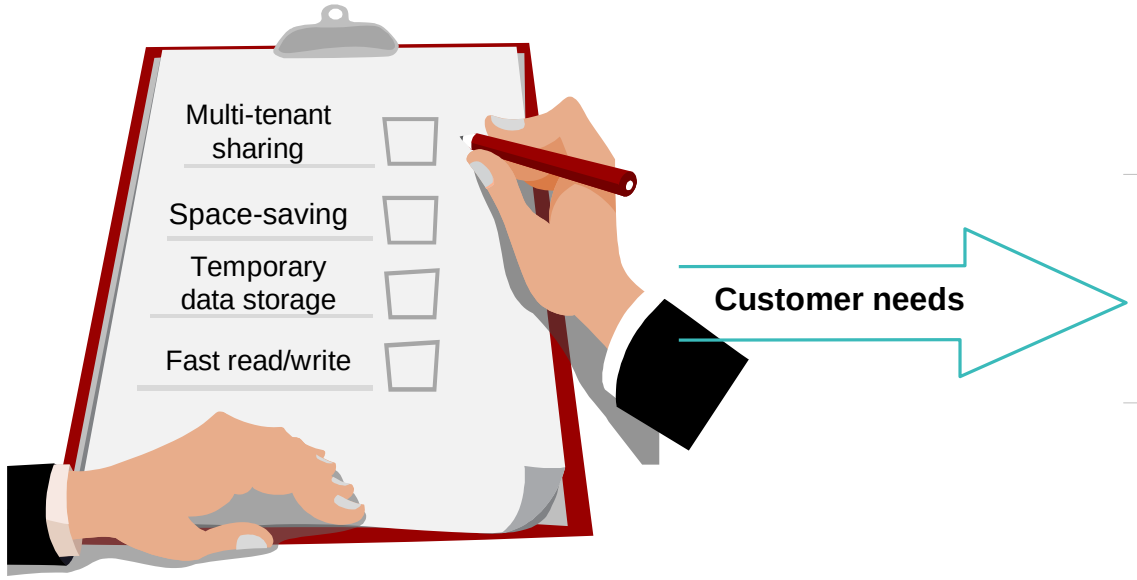
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Storage Architecture for Huawei Virtualization Product



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Huawei Virtual Disk Features



Type

- Common
- Shared

Configuration Mode

- Common
- Thin-provision

Disk Mode

- Dependent
- Independent & persistent
- Independent & nonpersistent

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Summary

- At this point you should understand how data is written to storage devices in virtualization scenarios.

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Quiz

1. Which of the following disk types is recommended for an archive room that needs to store large amounts of data that will be accessed infrequently over long periods of time? ( )

   A. SSD

   B. SAS disk

   C. SATA disk

   D. NL-SAS disk

2. SATA disks use a parallel interface while SAS disks use a serial interface.

   A. TRUE

   B. FALSE

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Recommendations

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- Huawei Support case library
  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Introduction to Virtualization Features

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Foreword

- Having attended previous courses about cloud computing, you may already have a good understanding of virtual machines (VMs) and the resources they use. This chapter focuses on when and where we would need to use VMs and what virtualization features VMs provide.

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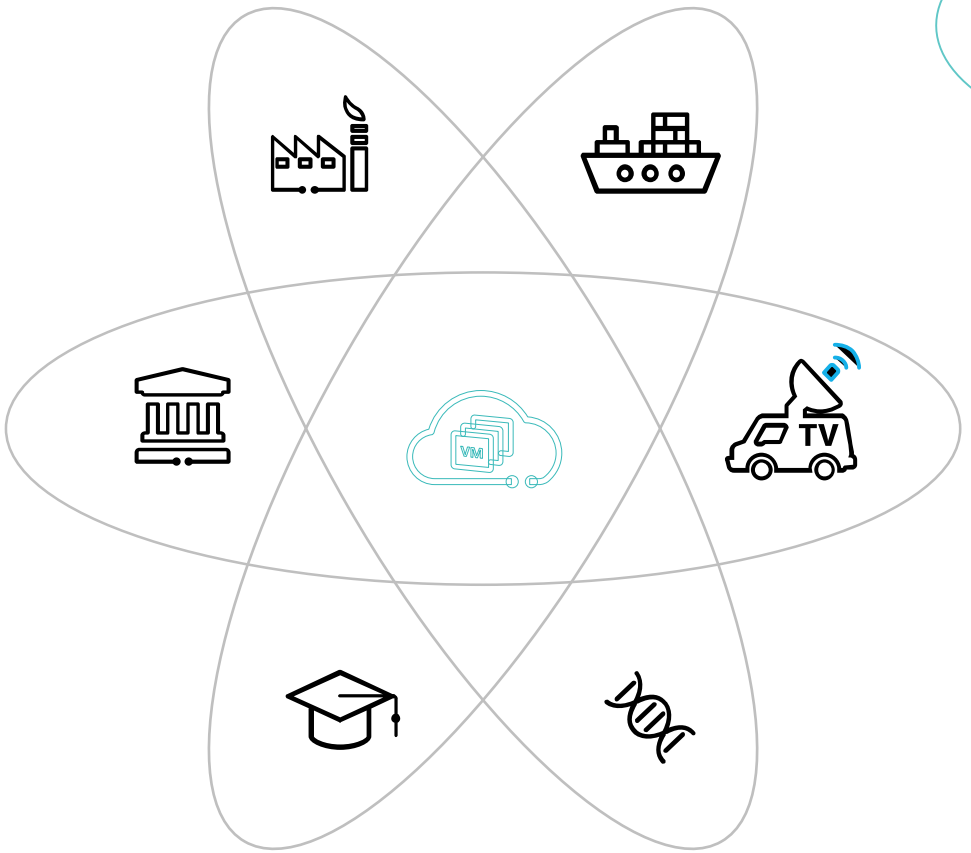


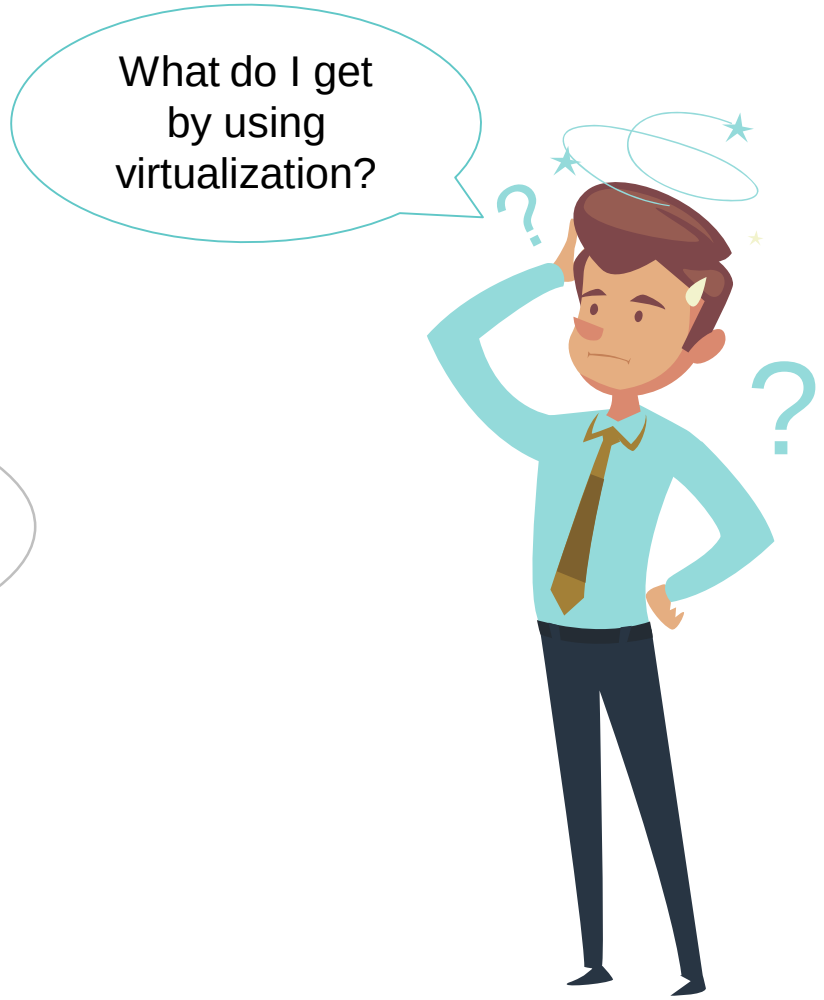
Objectives

- Upon completion of this course, you will:
  - Understand virtualization features.
  - Understand where and when virtualization features may be used.

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What Are the Benefits of Virtualization?





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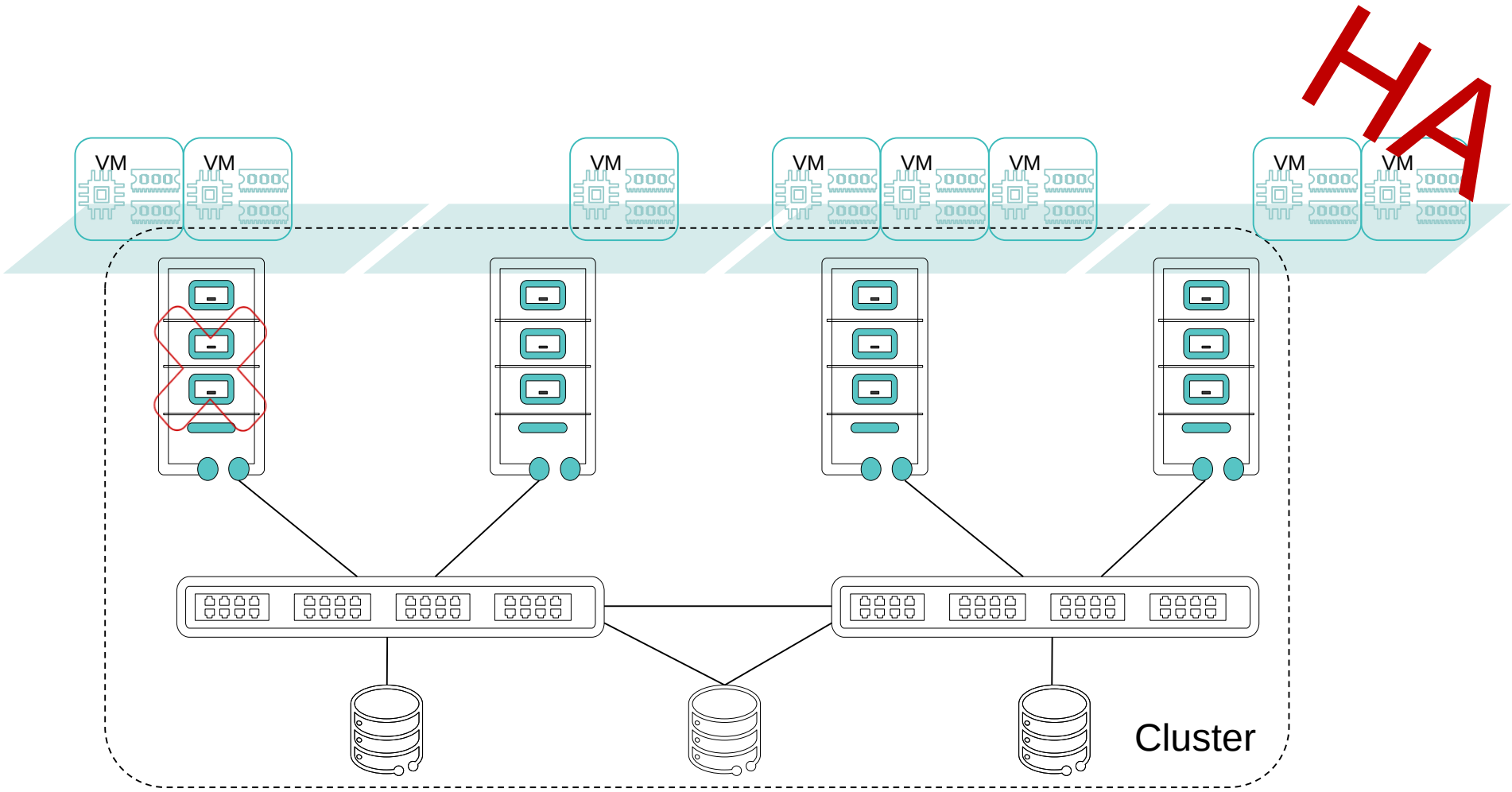
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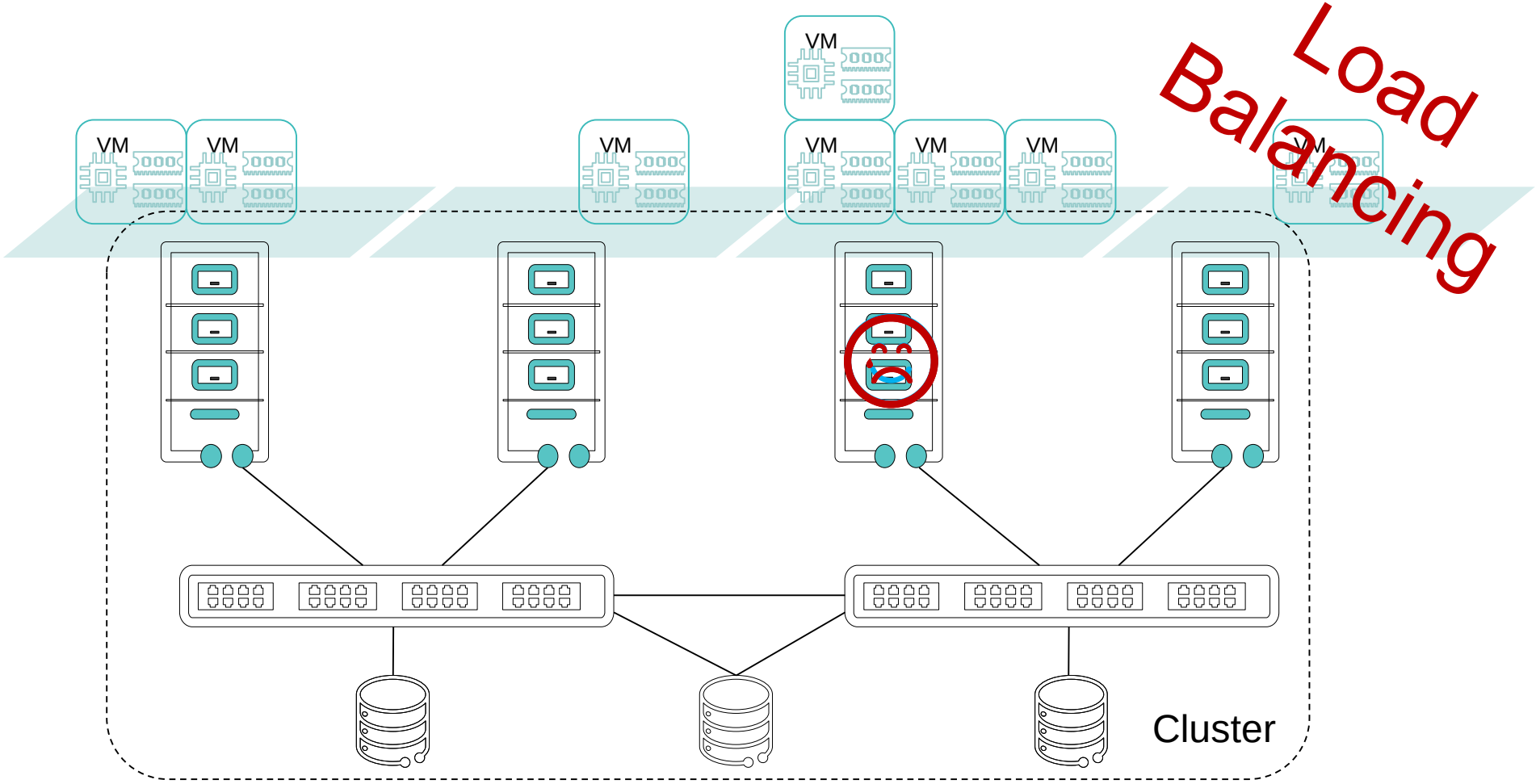
Virtual Cluster Features (1)





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Virtual Cluster Features (2)



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Virtual Cluster Features (3)

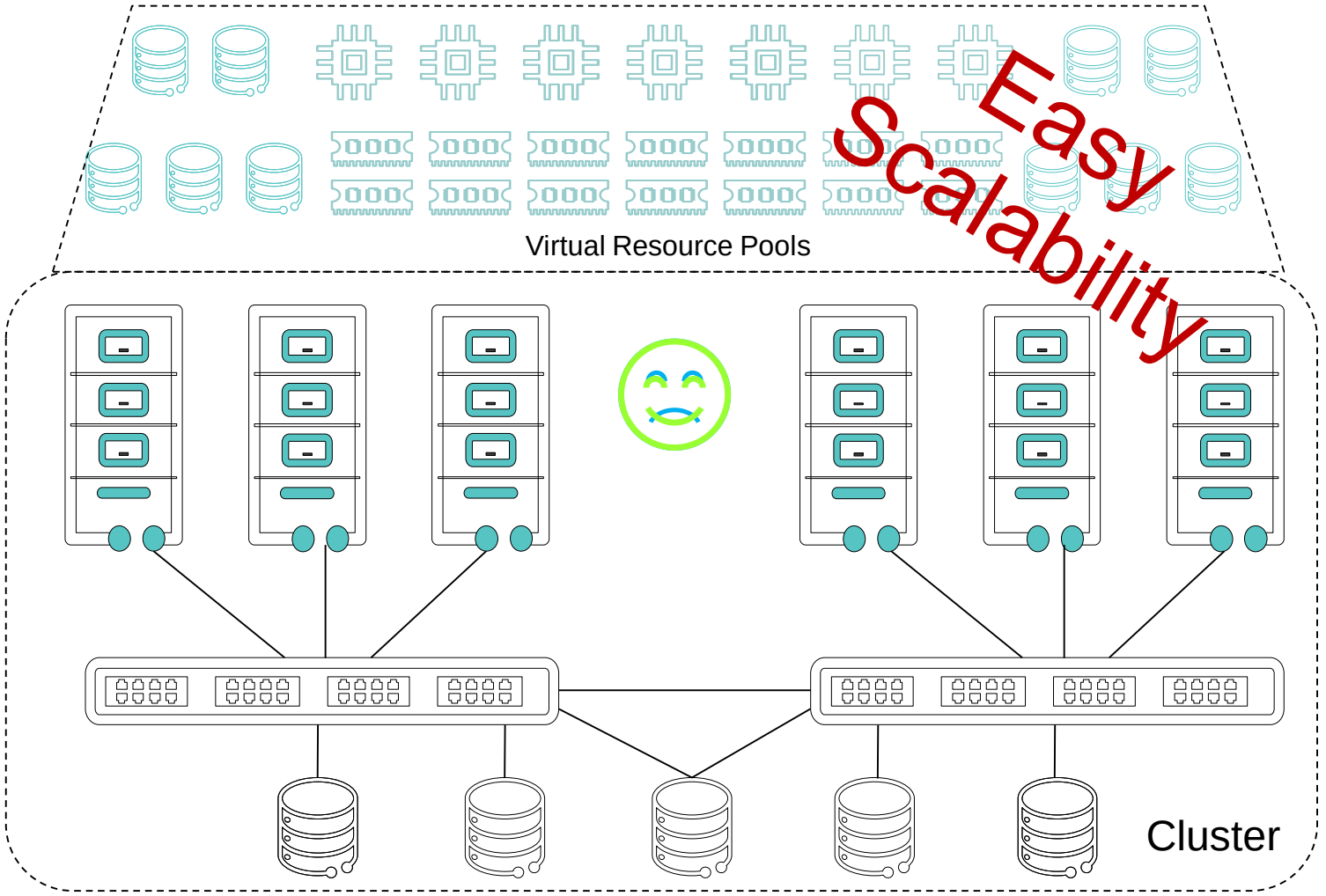
VM

VM

VM

VM

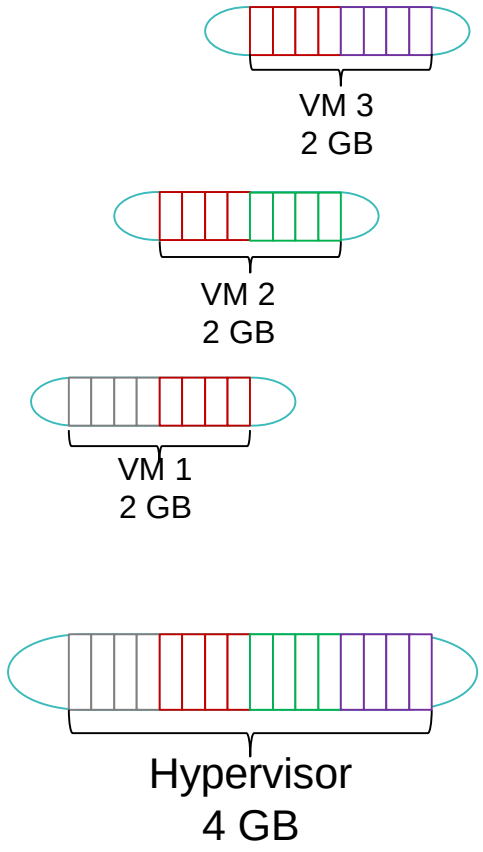
VM



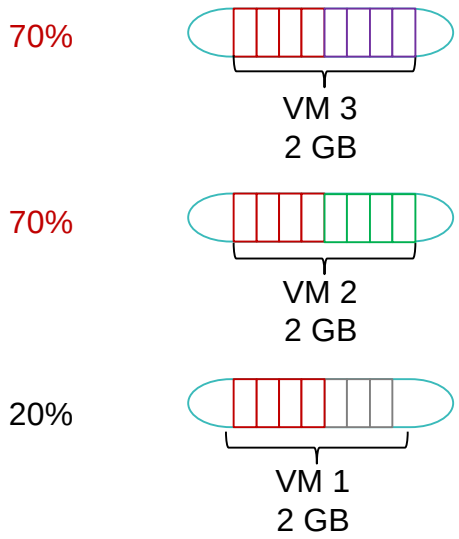
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Virtual Cluster Features (4)

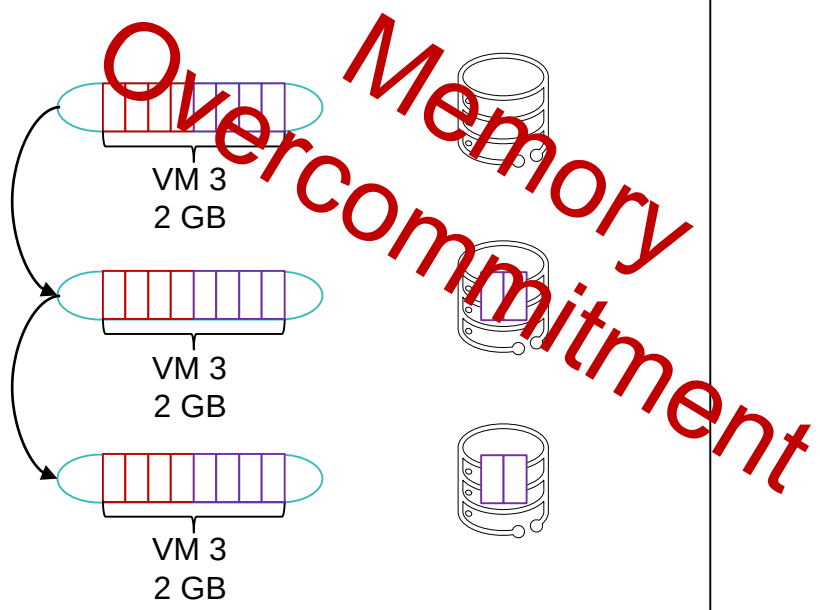
Memory Sharing



Memory Ballooning



Memory Swapping



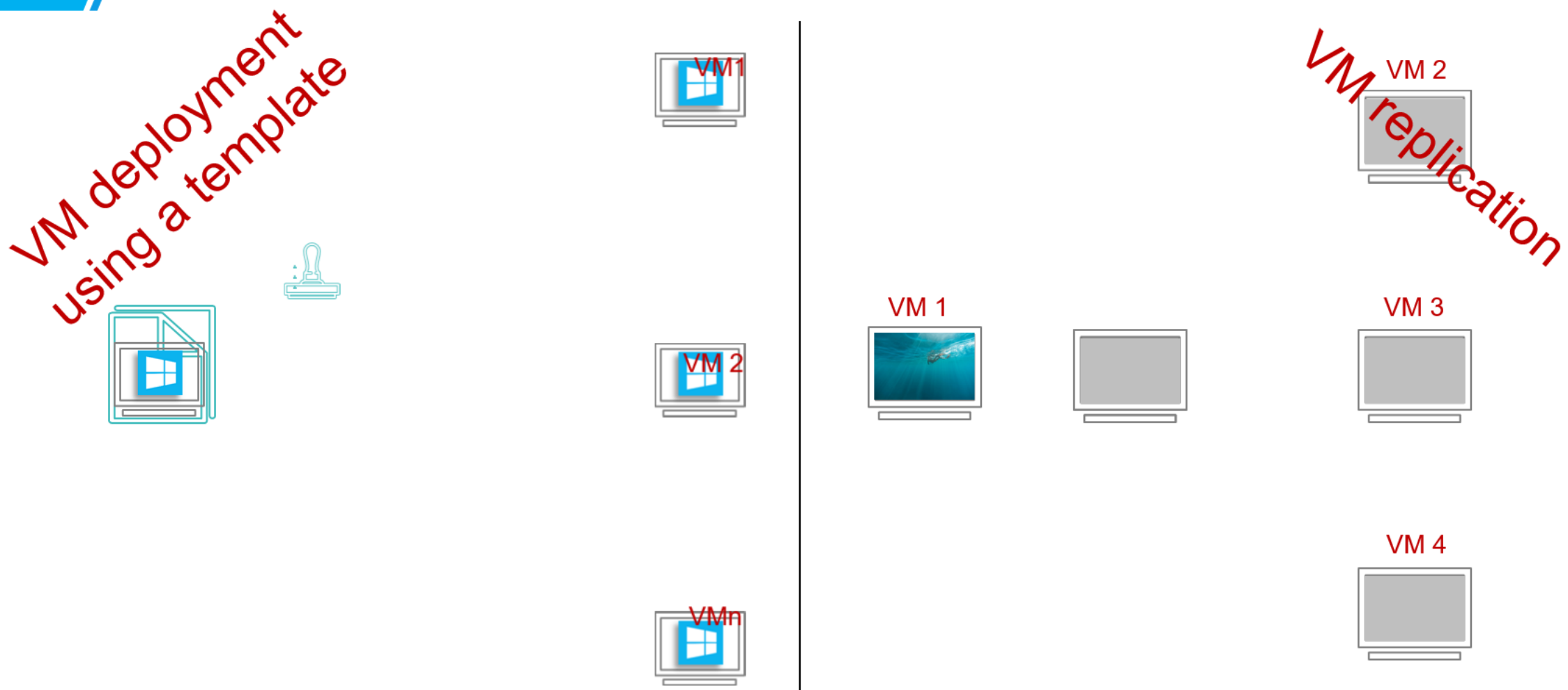
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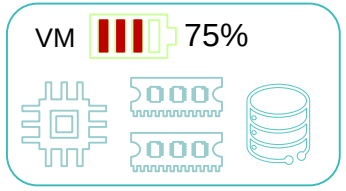


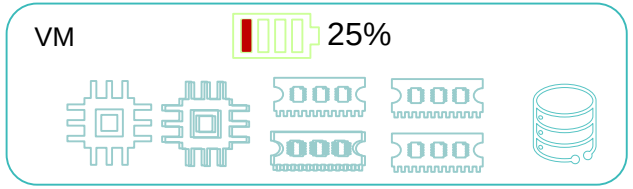
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Virtualization (VM) Features (2)

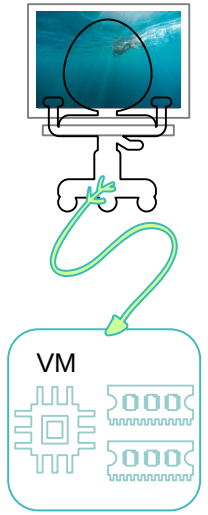
Hot-add resources





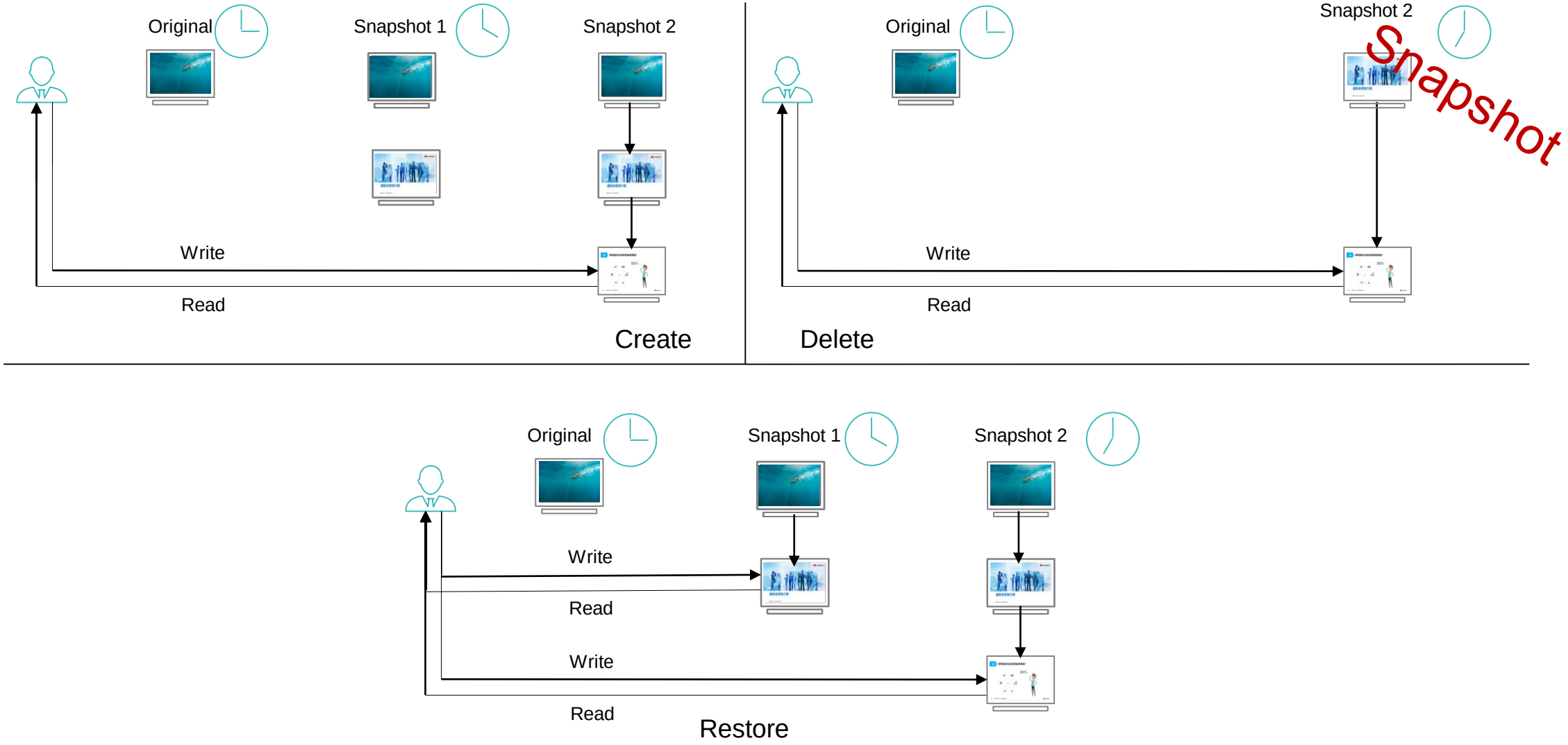


Console control



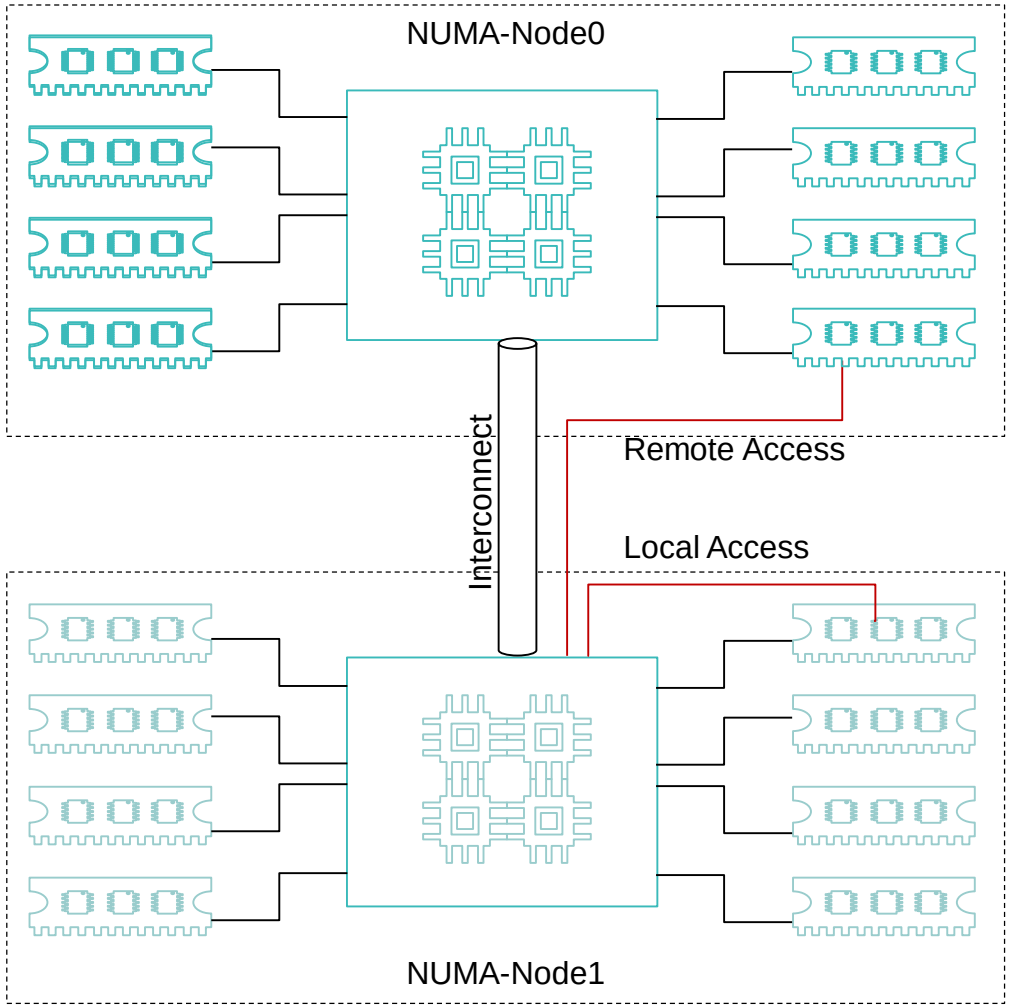
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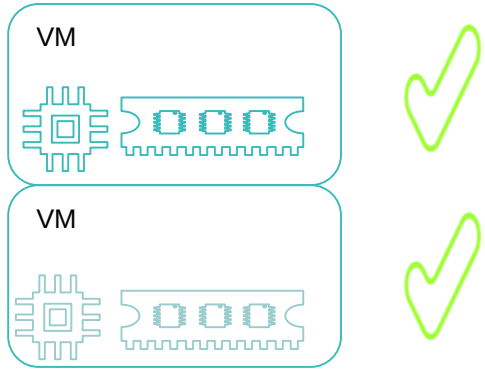
Virtualization (VM) Features (3)

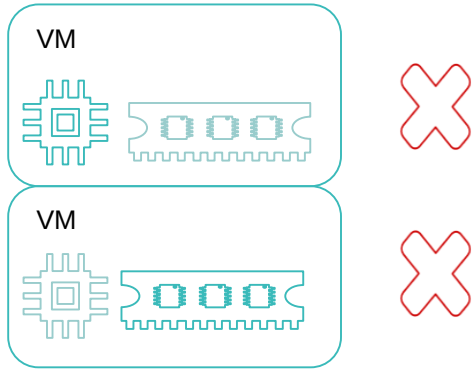


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NUMA







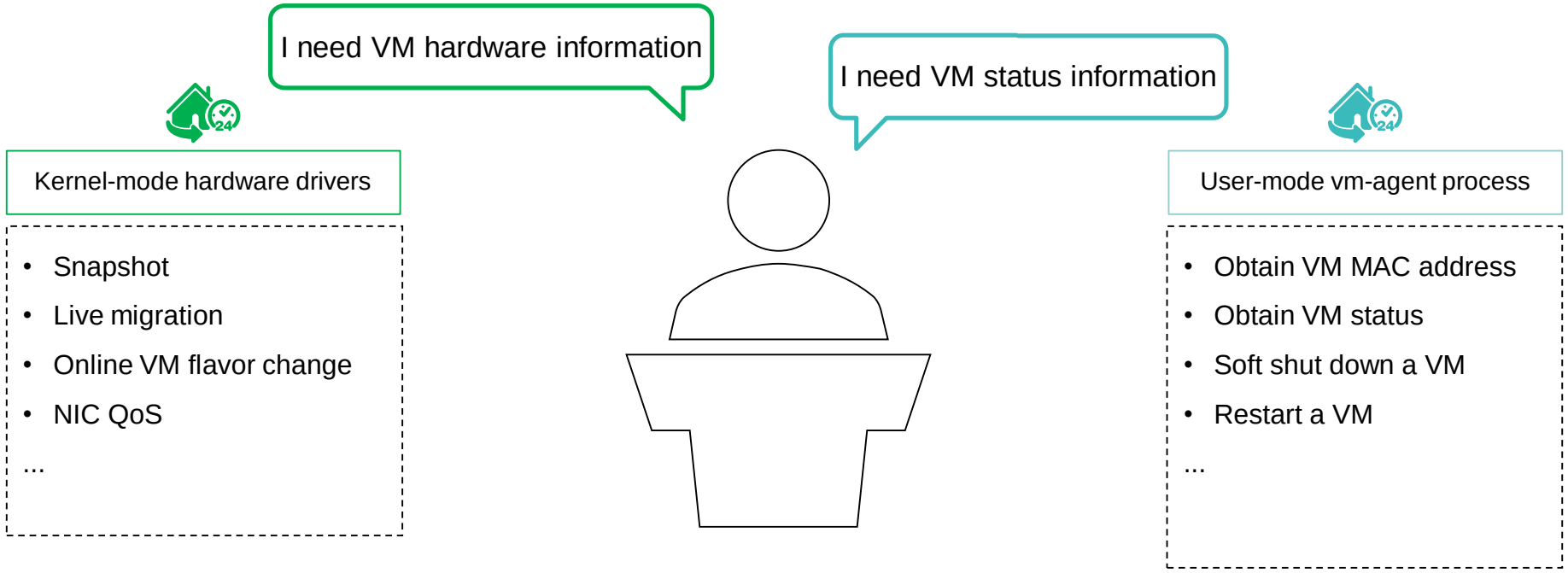
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Prerequisites for Enabling Advanced Features - Tools



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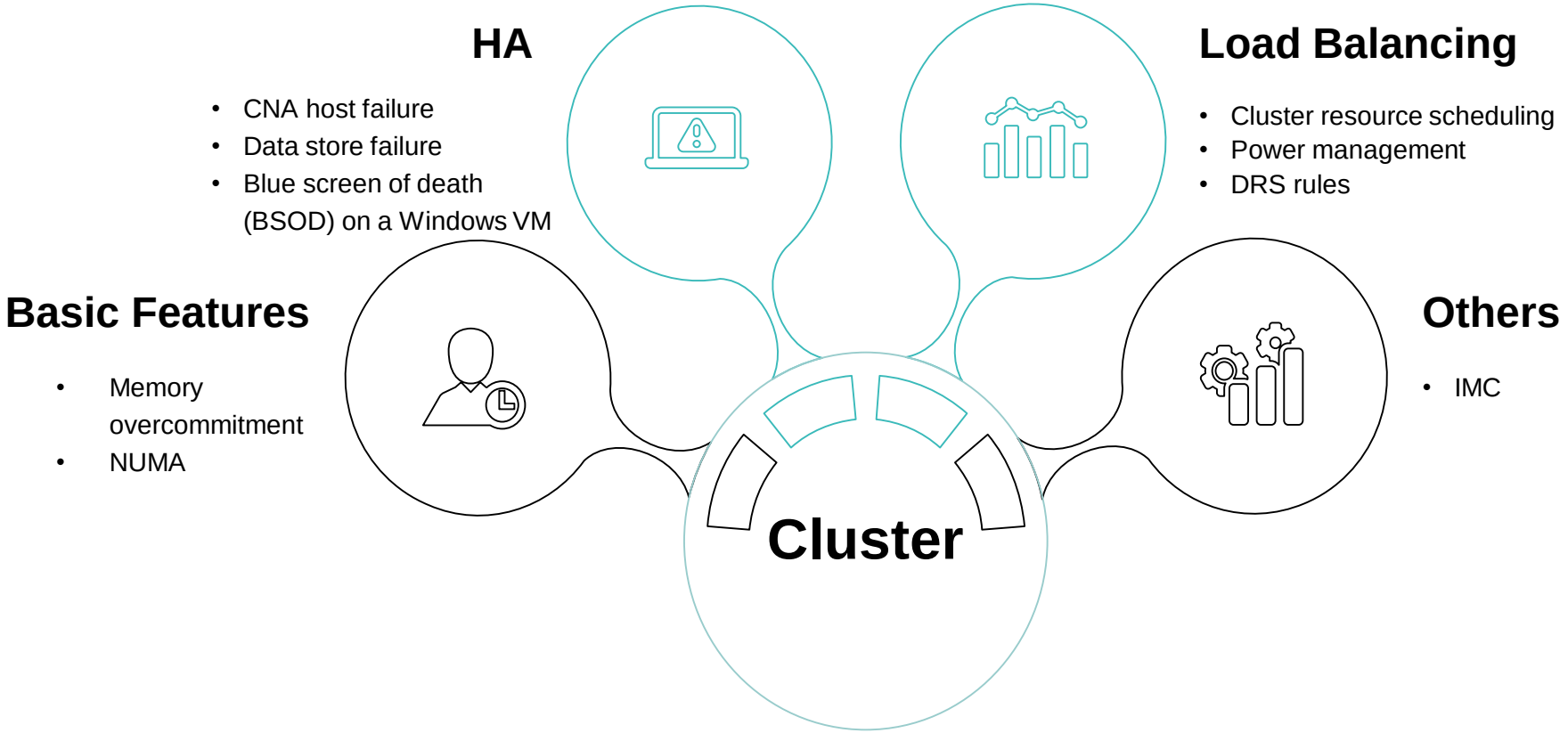
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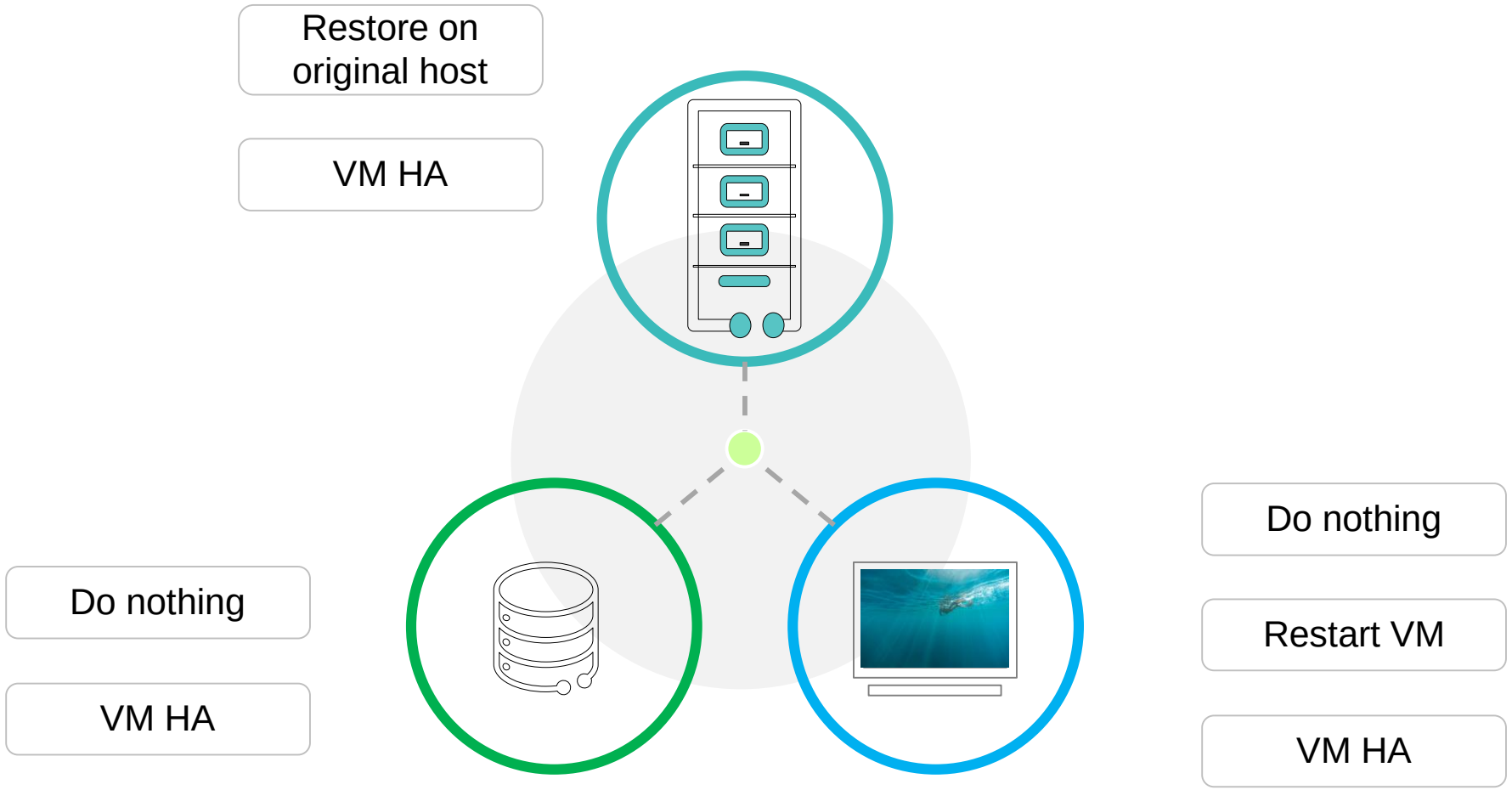
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Cluster Features



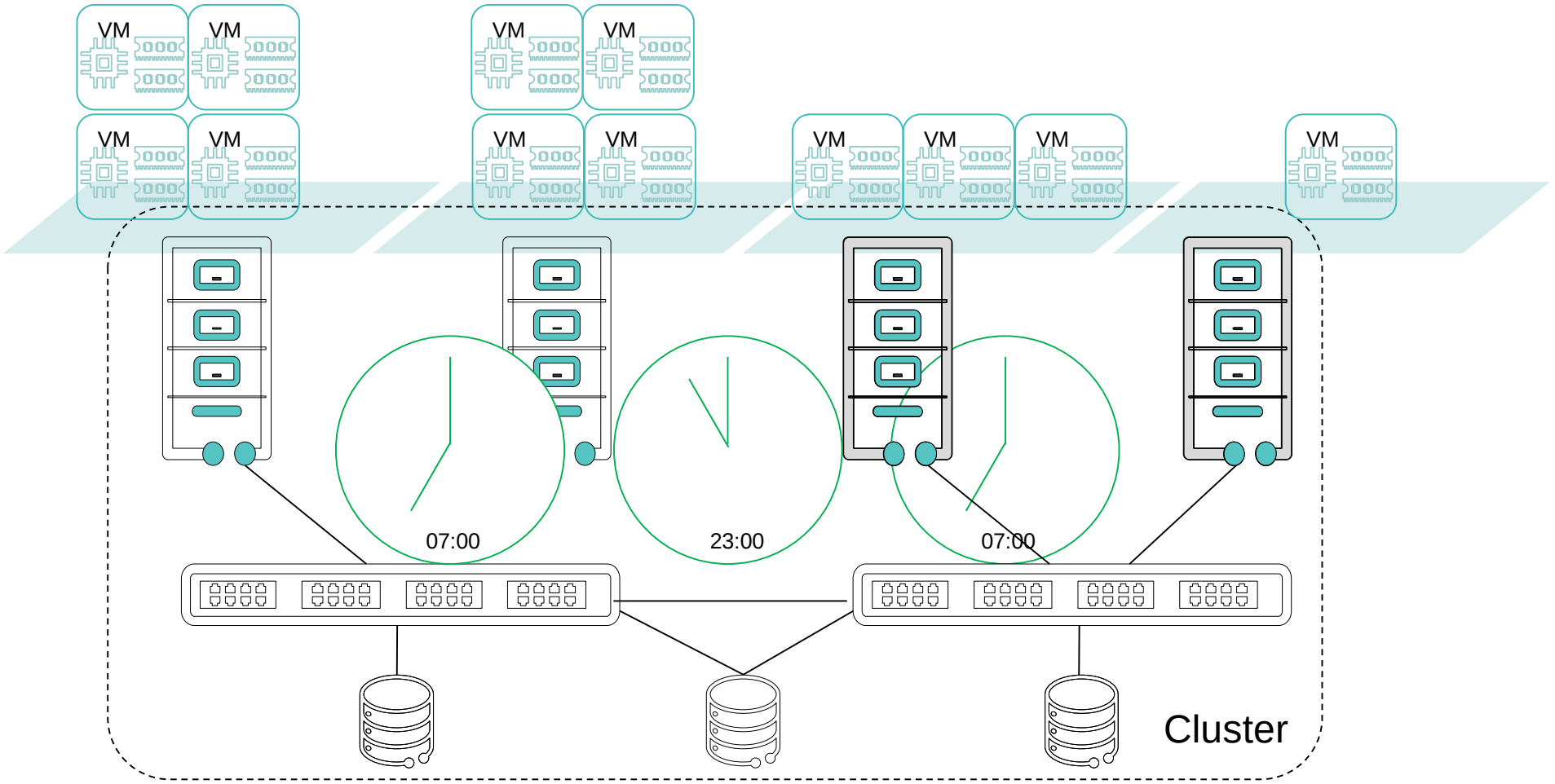
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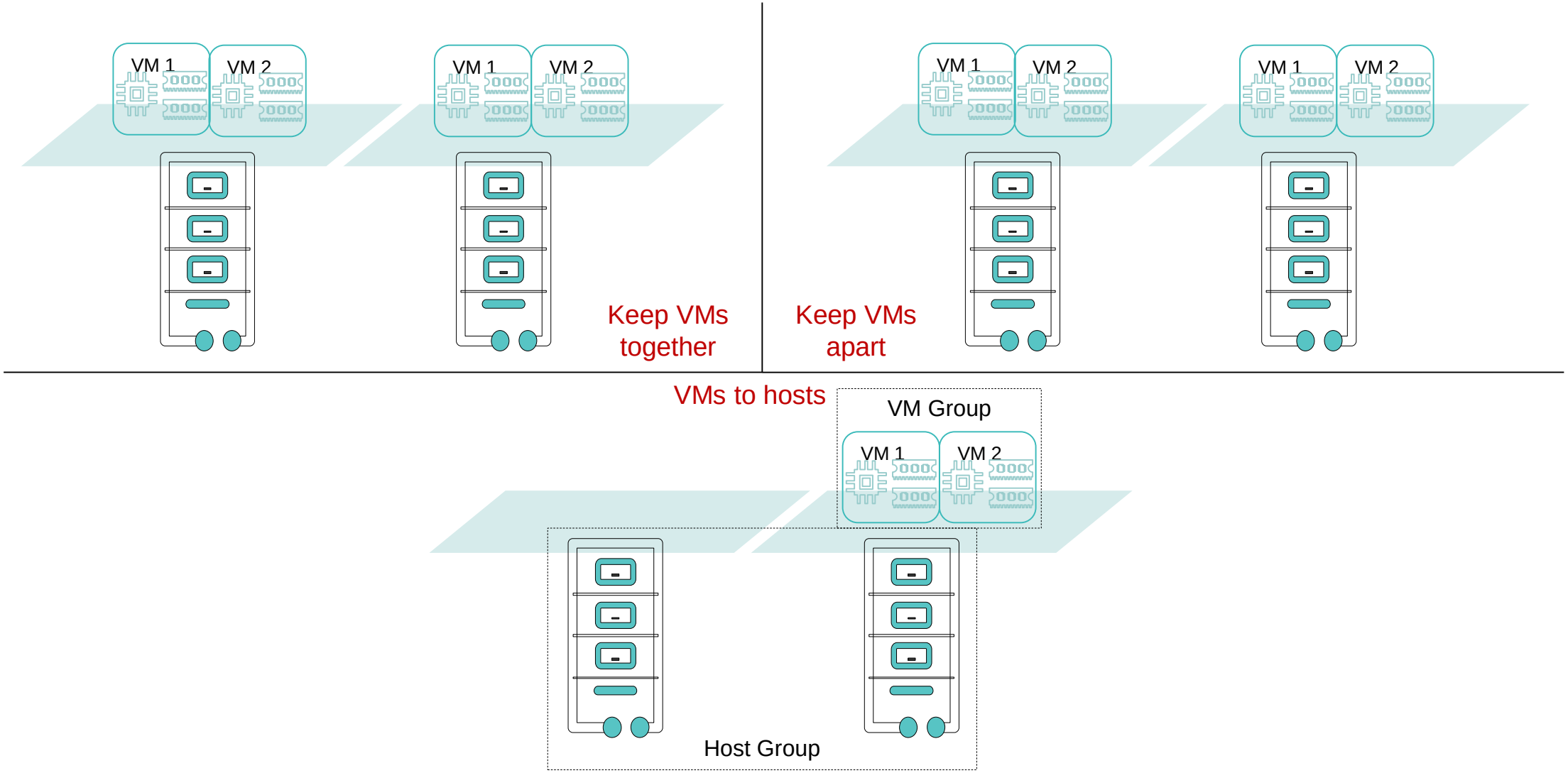
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Power Management



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DRS Rules



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IMC

- On FusionCompute, enabling incompatible migration cluster (IMC) mode for a cluster allows VMs to migrate between hosts that are using CPUs with different performance baselines within this cluster.
- Currently, IMC mode only allows VM live migration between hosts using Intel CPUs.
- IMC mode allows the hosts in a cluster to present the same CPU function set to VMs running on them, despite the fact that these hosts may be using CPUs with different performance baselines.
- For the IMC mode to work for a cluster that already contains hosts and VMs, ensure that the following conditions are met:
  - The CPU generations of the hosts in the cluster are the same as or later than the target IMC mode.
  - The CPU generations of the running or hibernating VMs in the cluster are the same as or earlier than the target IMC mode. If any VM in the cluster does not meet this requirement, stop this VM or migrate it to another cluster.

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VM Features

Resource Management

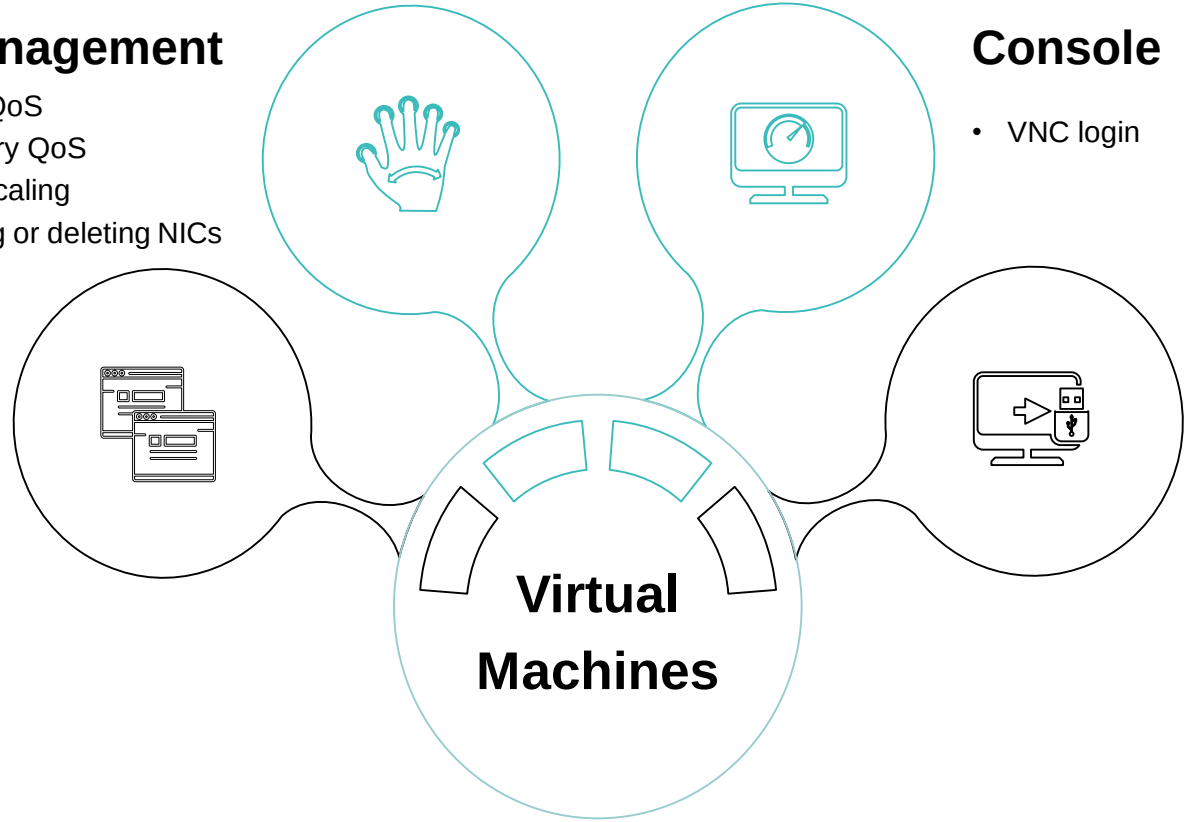
- CPU QoS
- Memory QoS
- Disk scaling
- Adding or deleting NICs

Console

- VNC login

Fast Deployment

- VM deployment using a template
- VM cloning

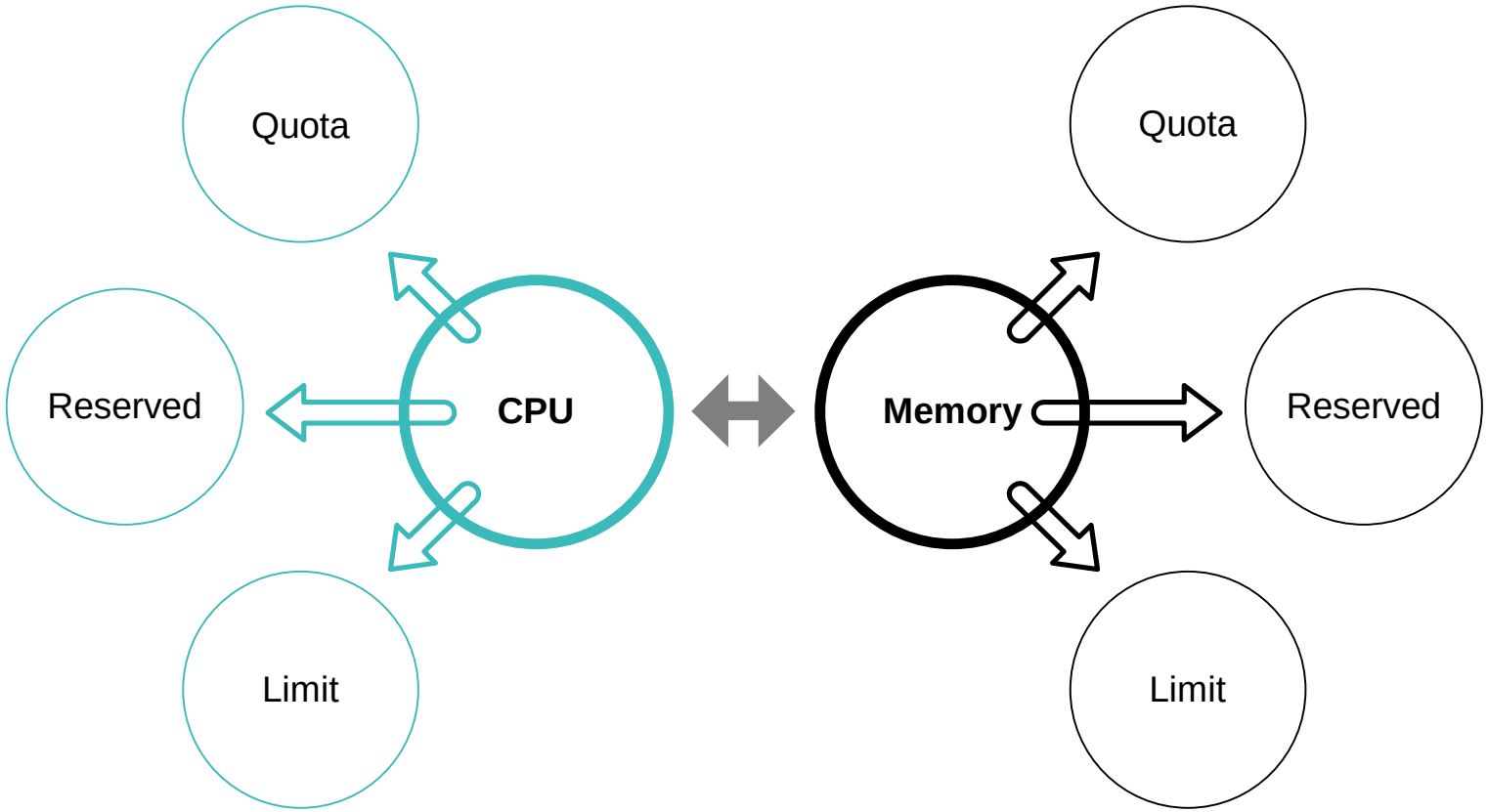


Others

- Snapshot
- Bundling or attaching USB, PCI, or GPU devices, or CD-ROM drive

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CPU and Memory QoS



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Quiz

1. Which of the following is not a snapshot management option?

   A. Create a snapshot

   B. Modify a snapshot

   C. Restore a VM using one of its snapshots

   D. Replicate a snapshot

2. Templates can be used to quickly deploy VMs. At present, templates can be created only by converting or cloning VMs.

   A. TRUE

   B. FALSE

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Summary

At this point you should:

- Understand virtualization features.
- Understand where and when virtualization features may be used.

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Recommendations

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  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Cloud Computing Trends

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Foreword

- Cloud computing has become the foundation of a range of important applications. At the same time, other technologies are also driving the further advancement of cloud computing. This chapter focuses on the relationship between cloud computing and related fields, and introduces you to some of the new technologies related to cloud computing.

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Objectives

- Upon completion of this course, you will:
  - Be able to describe key technologies that underpin cloud computing.
  - Understand the relationship between cloud computing and other fields.
  - Understand some of the most prominent cloud computing trends.

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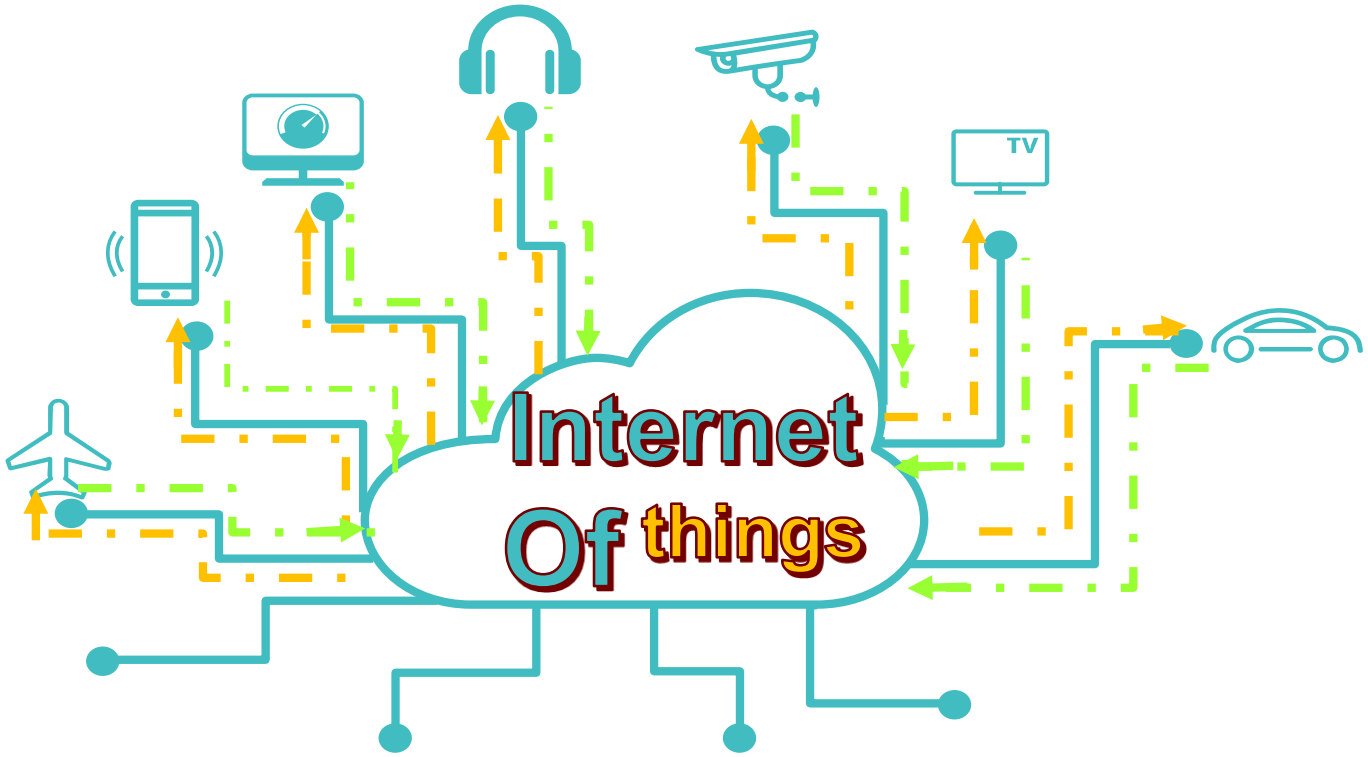
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3. Cloud Computing Trends
4. Other Emerging Technologies

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IoT

- The Internet of Things (IoT) is the concept of connecting everything through the Internet, including all sorts of communication and interaction: people-people, people-things, and things-things.



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Main IoT Technologies



RFID

Sensors

Embedded Systems

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Big Data

- Big data refers to data sets that are too large or complex to be handled by traditional data-processing application software.
- Such data sets are generated by the IoT. Due to a radical increase in total IoT connections, ever more massive data sets will be generated.

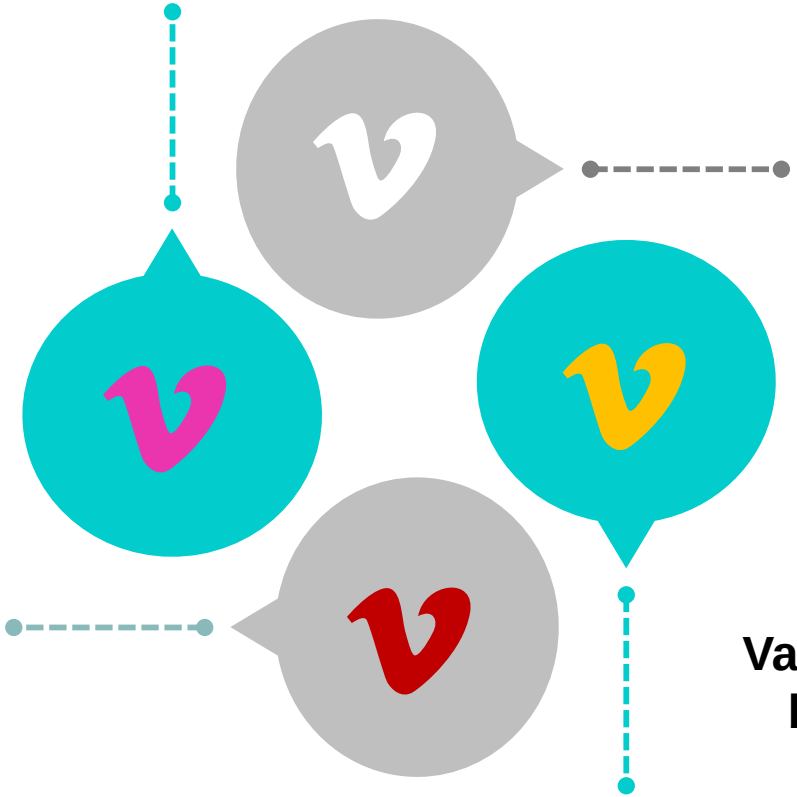


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Characteristics of Big Data

Volume: the amount of data

Variety: the different types of data

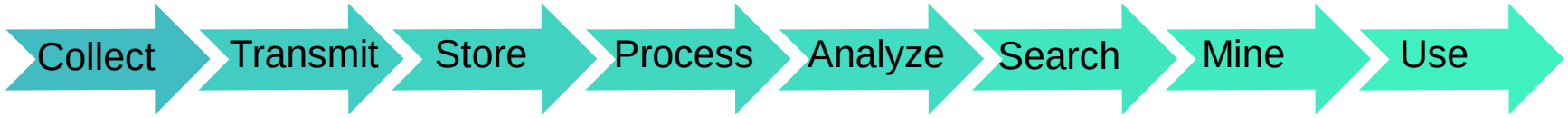


Velocity: the speed at which data is generated

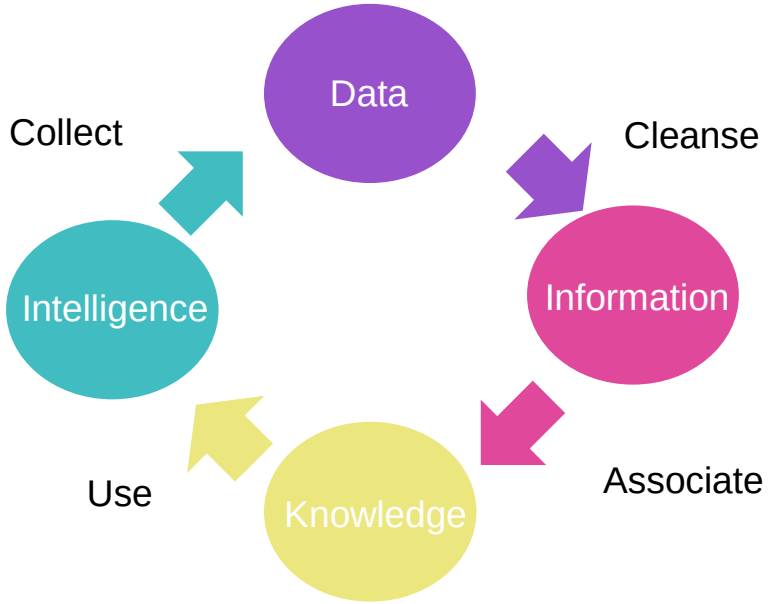
Value: the value that can be extracted from data

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Big Data Processing







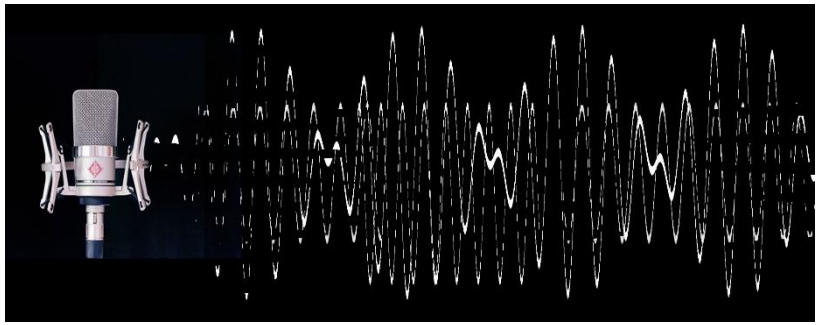
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Artificial Intelligence

- Artificial intelligence (AI) is an area of computer science that focuses on the research into and development of theories, methodology, technology, and application systems for simulating and extending human intelligence with machines.
- Applications of AI include robotics, speech recognition, image recognition, natural language processing.







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Elements of AI

- Big Data

Artificial intelligence is intelligence acquired and summarized by big data analytics.

- Computing Power

Computing power is a prerequisite for applying AI algorithms to extract value from massive data sets.

- Algorithms

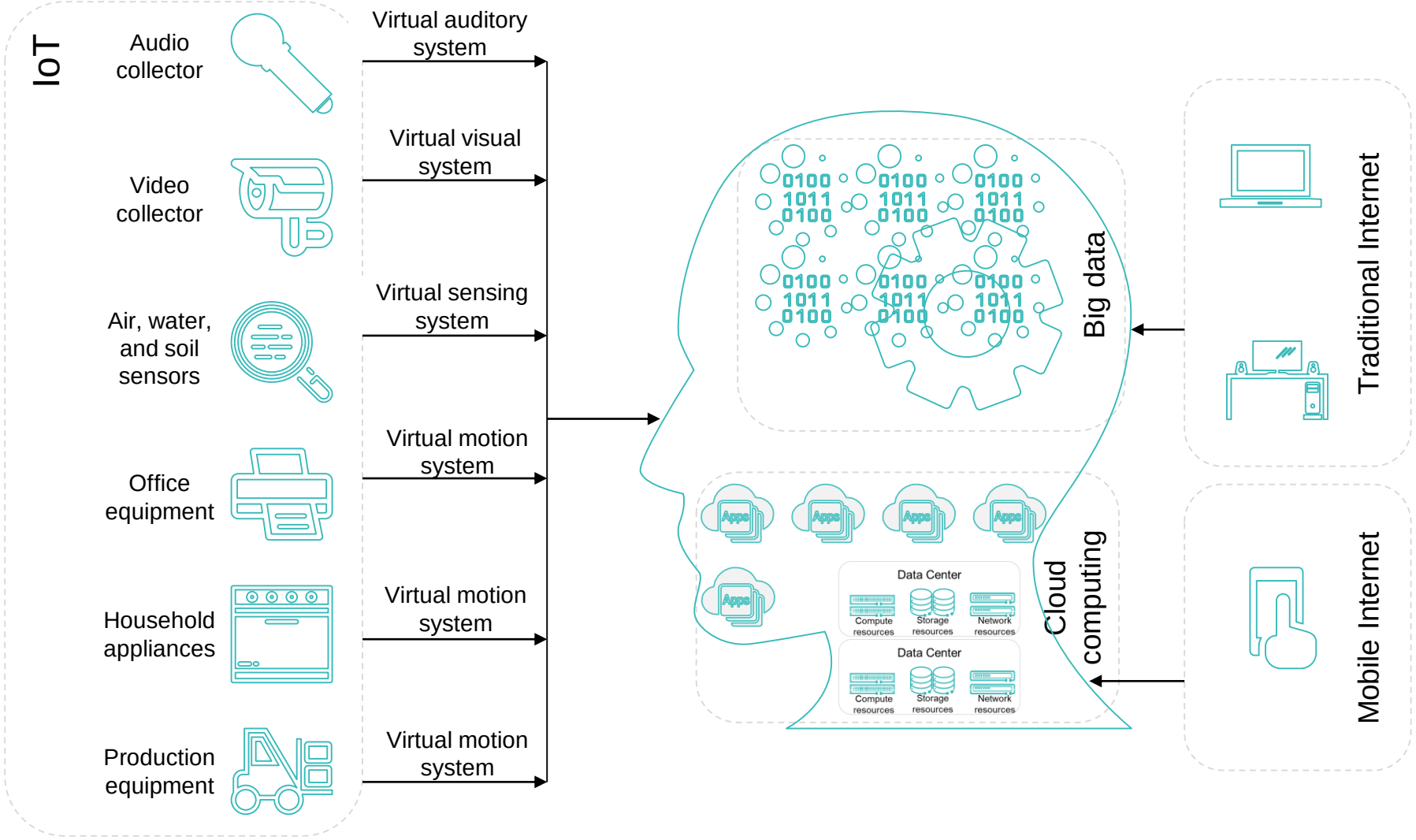
Algorithms are the fundamental way by which AI is realized, and an effective method to extract data intelligence.

- Scenario

Big data, computing power, and algorithms are input that will create meaningful output only if applied to practical business scenarios.

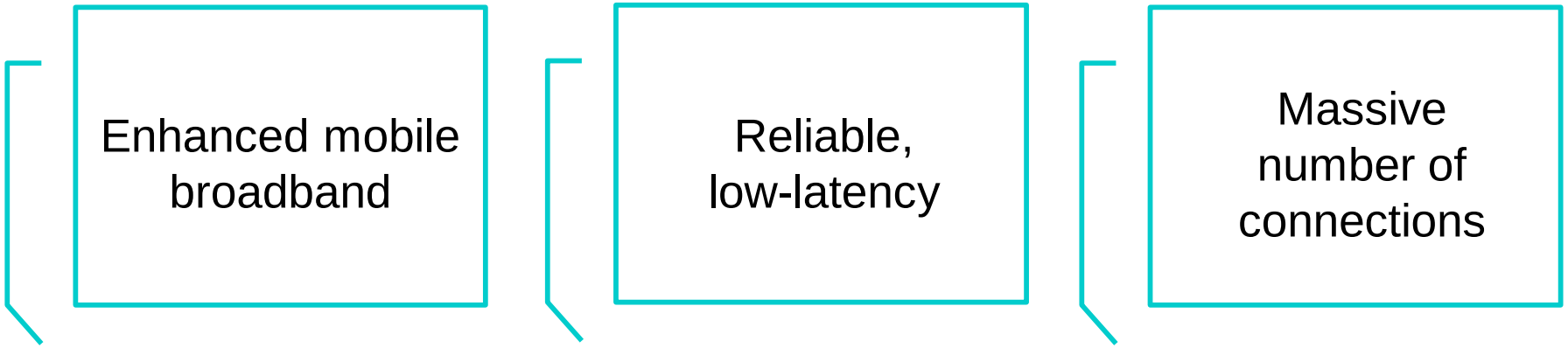
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Cloud Computing, IoT, & Big Data



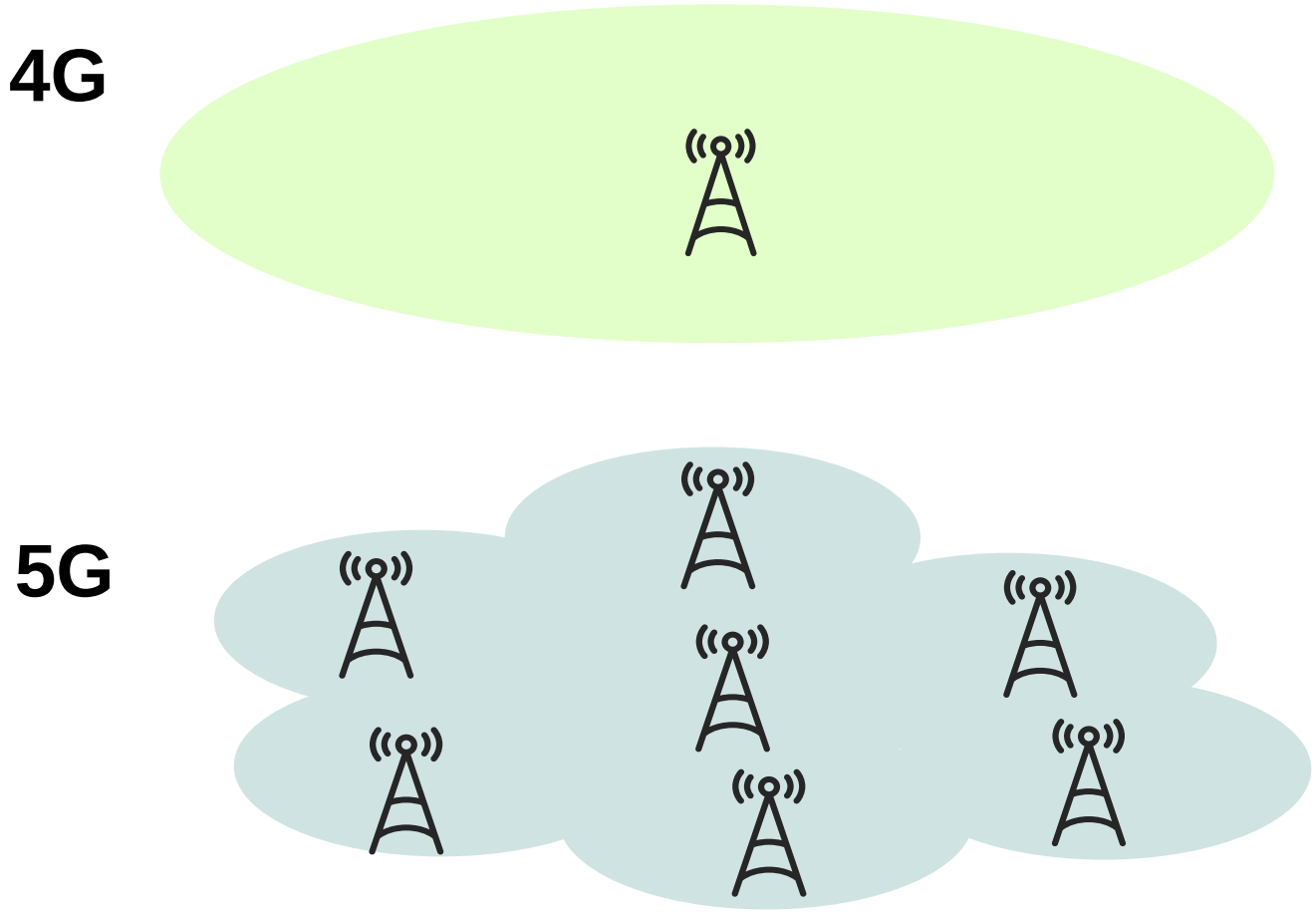
The 5G Era

- 5G refers to the 5th generation of mobile communications technologies and standards.



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How 5G Works: Network Densification



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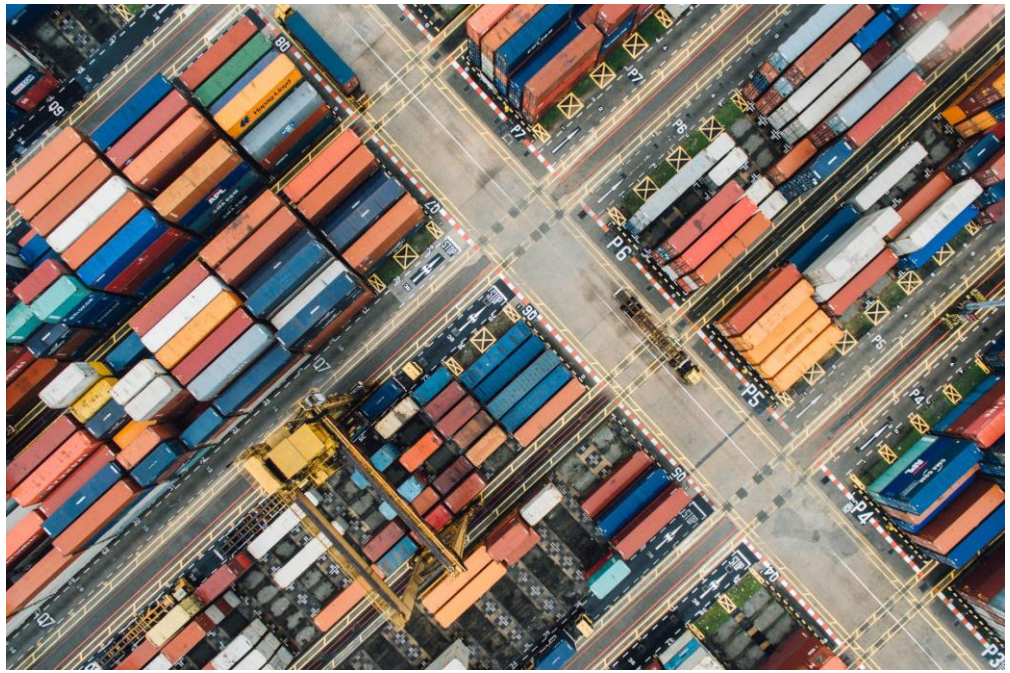
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4. Other Emerging Technologies

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Containers

- Containerization is a lightweight virtualization technology that uses OS-level virtualization. Several containers (isolated user-space instances) can be created on each operating system, each of which is allocated only a subset of the computer's resources.

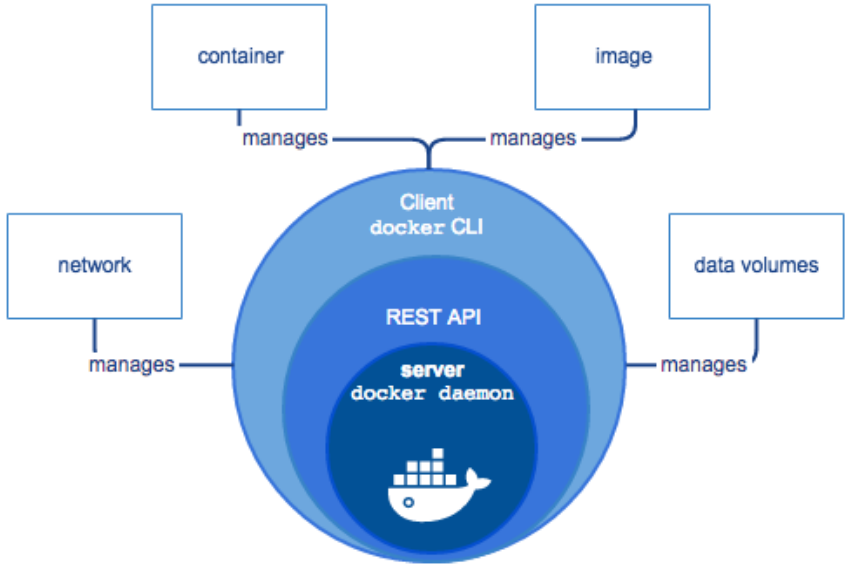


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Docker

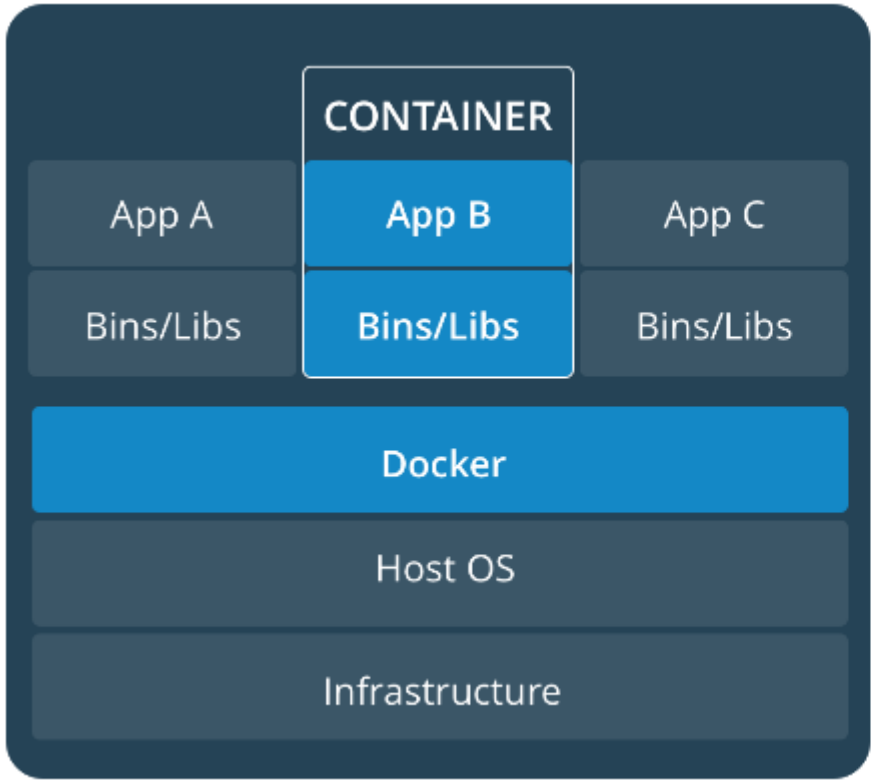
- Docker is an open source application container engine. It allows developers to pack their applications and dependency packages into a portable container, and later run them on any other Linux machine. The sandbox mechanism is used for containers, eliminating any interface between containers.

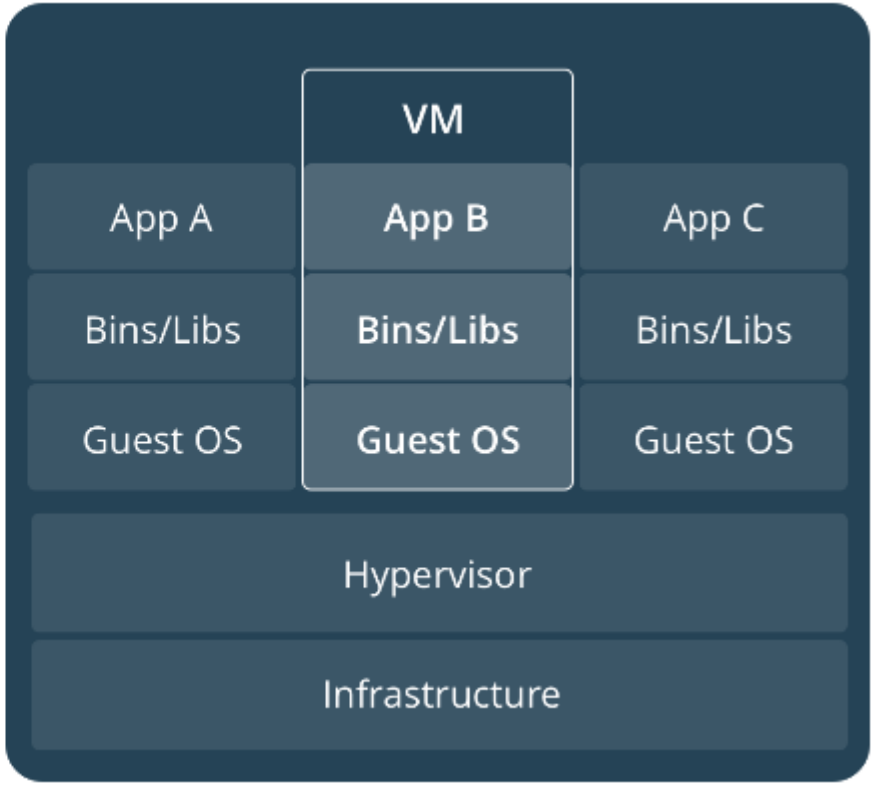




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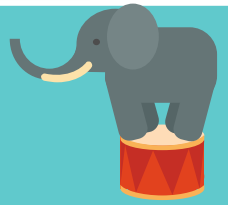
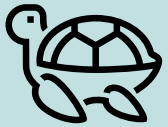
Docker Technology - Differences Between Containers and VMs (1)





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Docker Technology - Differences Between Containers and VMs (2)

Difference	VM	Docker
Size		
Boot speed		
Integration		

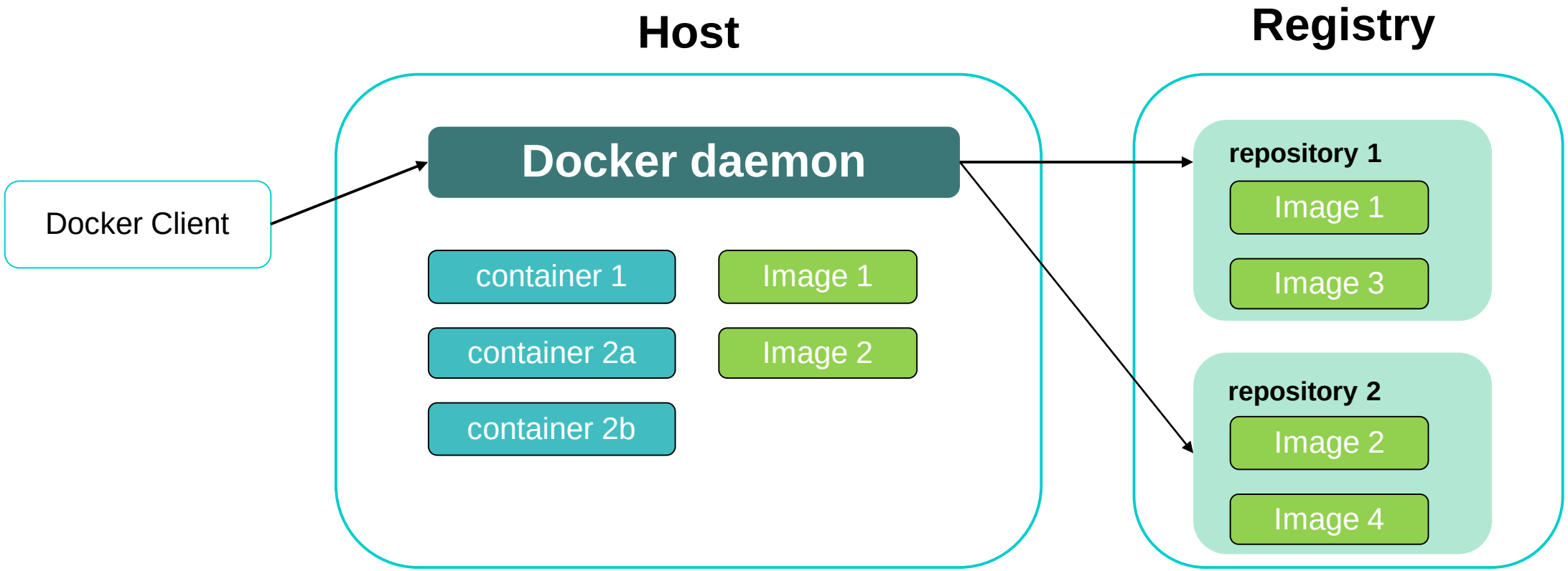
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Docker Elements

- **Image**
  - A Docker image is a read-only template used to create Docker containers. Docker provides a simple mechanism to allow users to create images or update existing ones. Users can also download images created by others and use them without making any changes.
- **Container**
  - Docker uses containers to run applications. A container is a runtime instance of a docker image. A container can be started, started, stopped, and deleted. Each container is an isolated and secure platform.
- **Repository**
  - A repository is a set of Docker images. A Registry is a hosted service containing repositories of images. Each image has a unique tag.

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Docker Architecture



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OpenStack

- Began in 2010 as a joint project of Rackspace Hosting and NASA, OpenStack is a free open-source software platform for cloud computing. OpenStack requires contributions to be released under the Apache 2.0 license.

OpenStack is a free open-source software platform for cloud computing.



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OpenStack Components

Nova **Compute**	**Swift** **Object Storage**	**Cinder** **Block Storage**
Neutron **Network**	**Keystone** **Authentication**	**Ceilometer** **Metering**
Glance **Image**	**Heat** **Orchestration**	**Horizon** **Dashboard (UI Console)**

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Contents

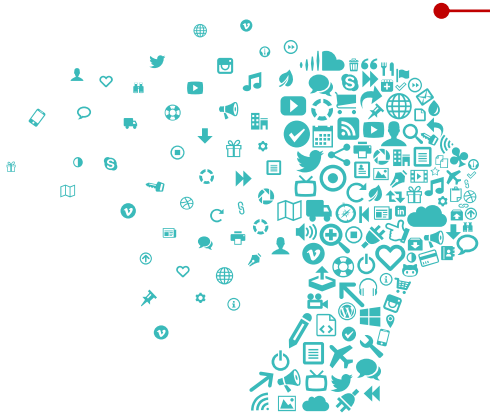
1. Fields Related to Cloud Computing
2. Cloud Computing Technologies
3. **Cloud Computing Trends**
4. Other Emerging Technologies

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Cloud Computing Trends

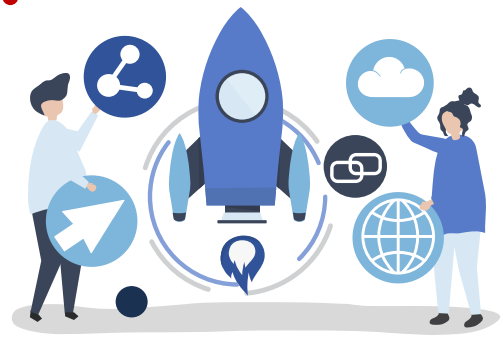
Cloud giants are **strengthening cooperation**, leading to a **more concentrated** market.

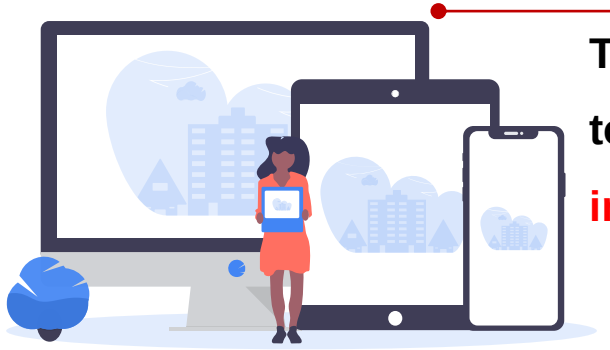




Cloud providers are using **more AI**.

Hybrid cloud and industry cloud are gaining momentum.





There is an emerging trend toward **cloud-device integration**.

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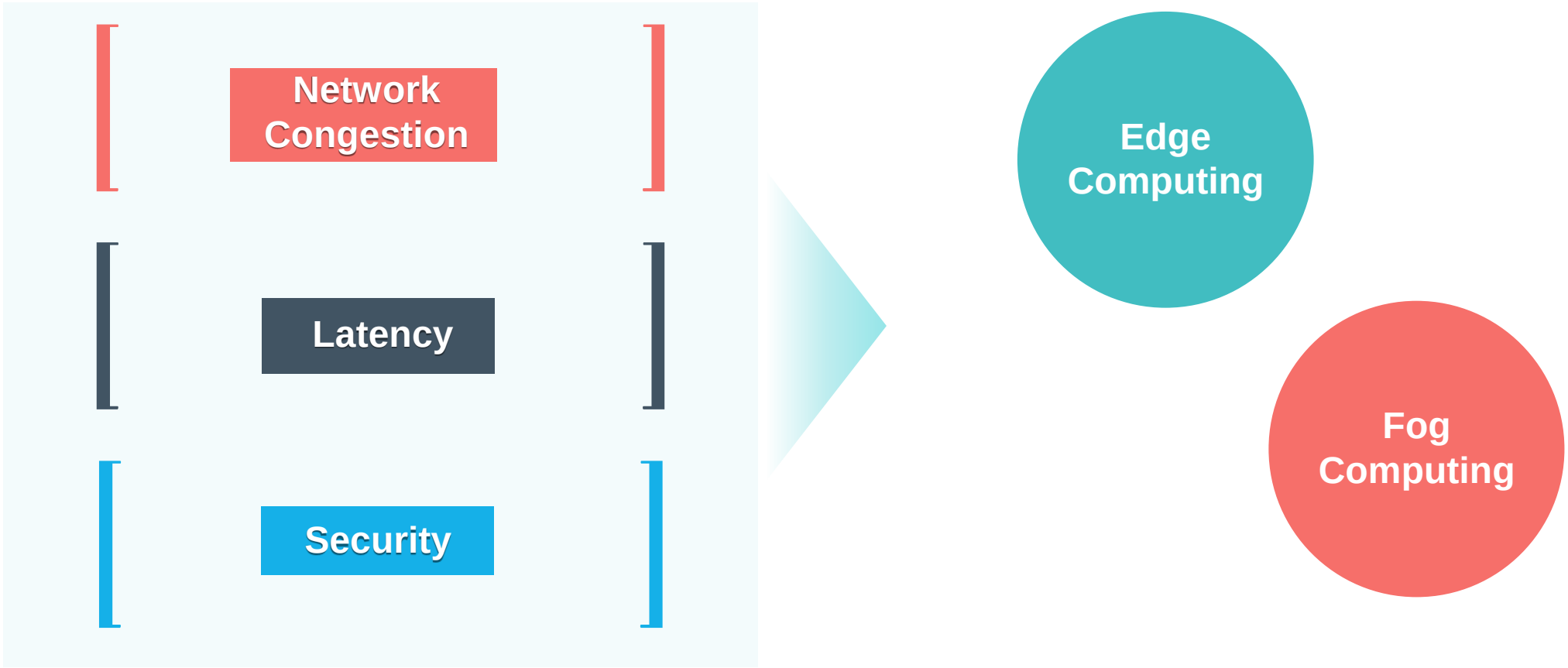
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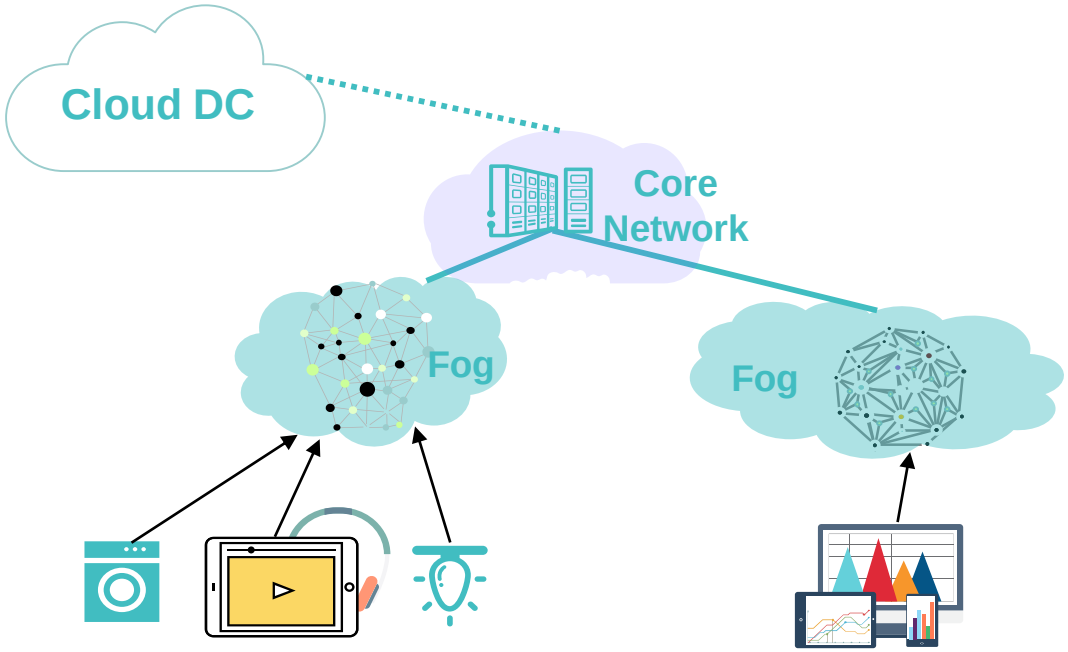
Cloud Computing Extensions



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Fog Computing

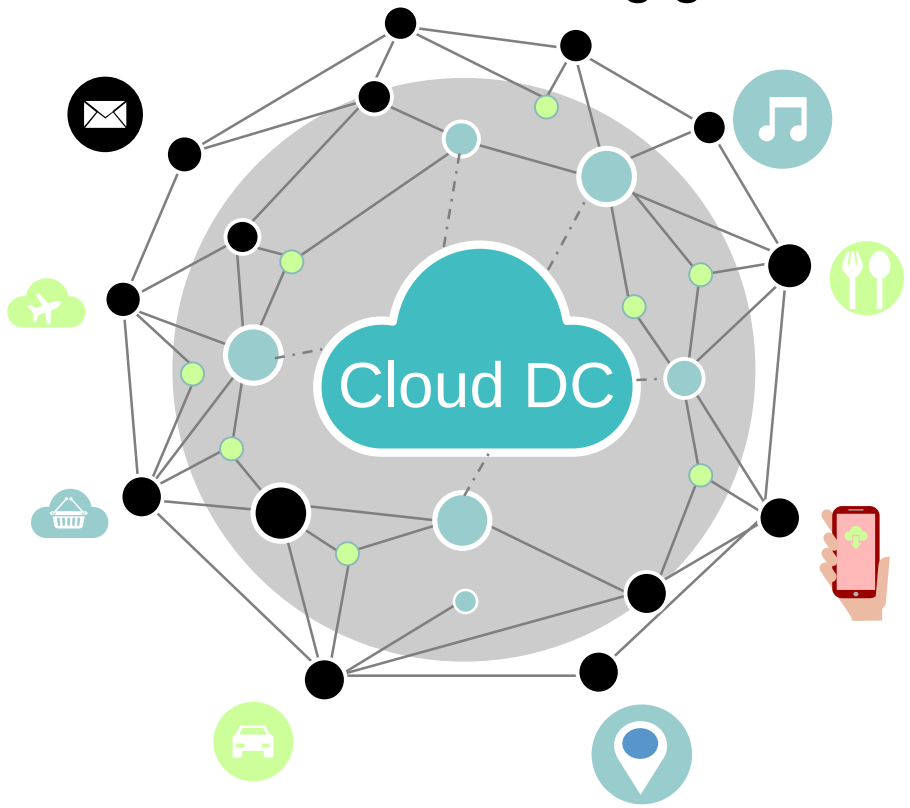
- Fog computing is an extension of cloud computing services to the edge of the network to decrease latency and network congestion. Fog computing uses edge devices to carry out a substantial amount of computation, storage, and communication locally, instead of storing all data in cloud data centers.



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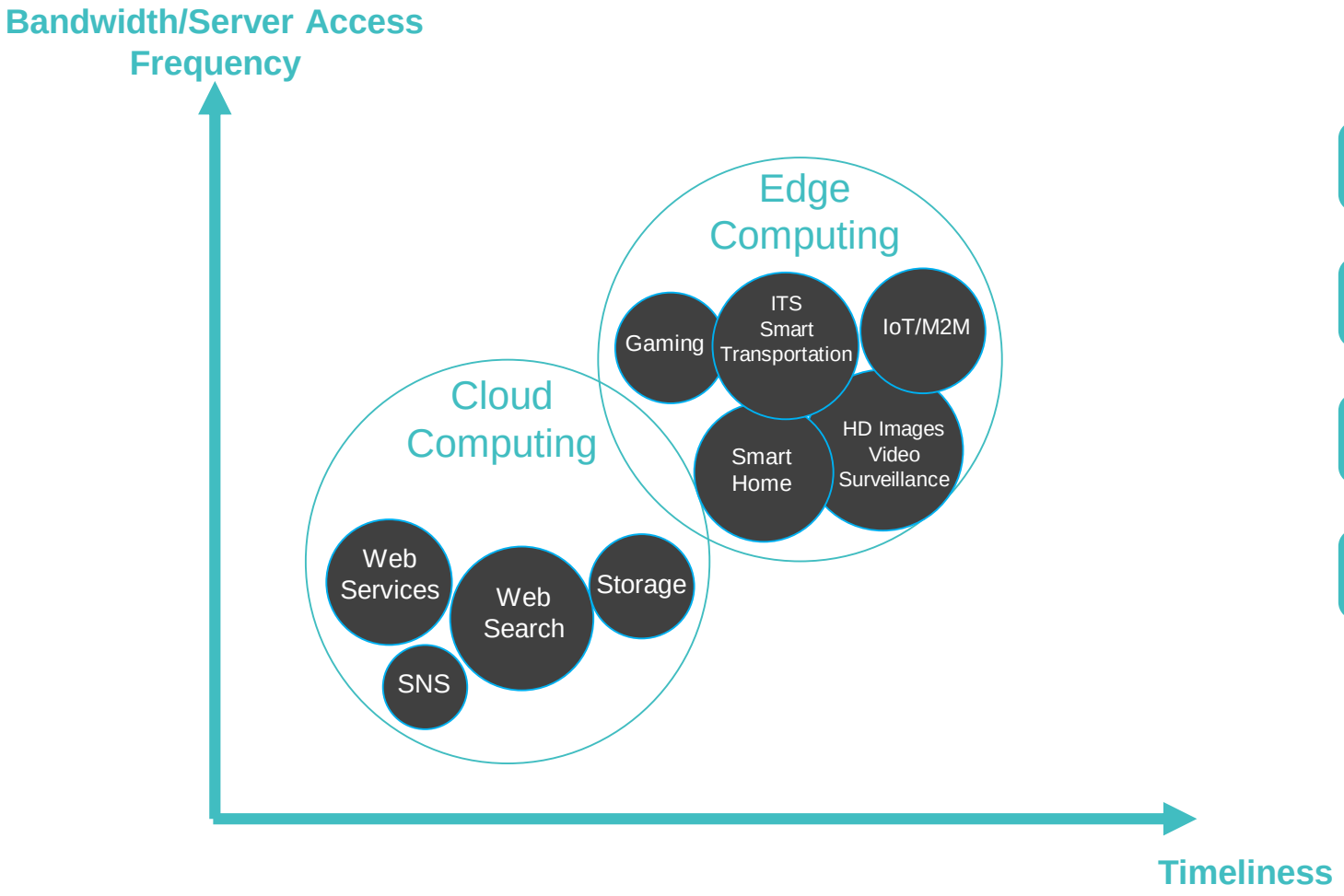
Edge Computing (1)

- Edge computing creates an open platform that integrates core capabilities including network, compute, storage, and applications at the edge of your network, to process data where it is being generated.



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Edge Computing (2)

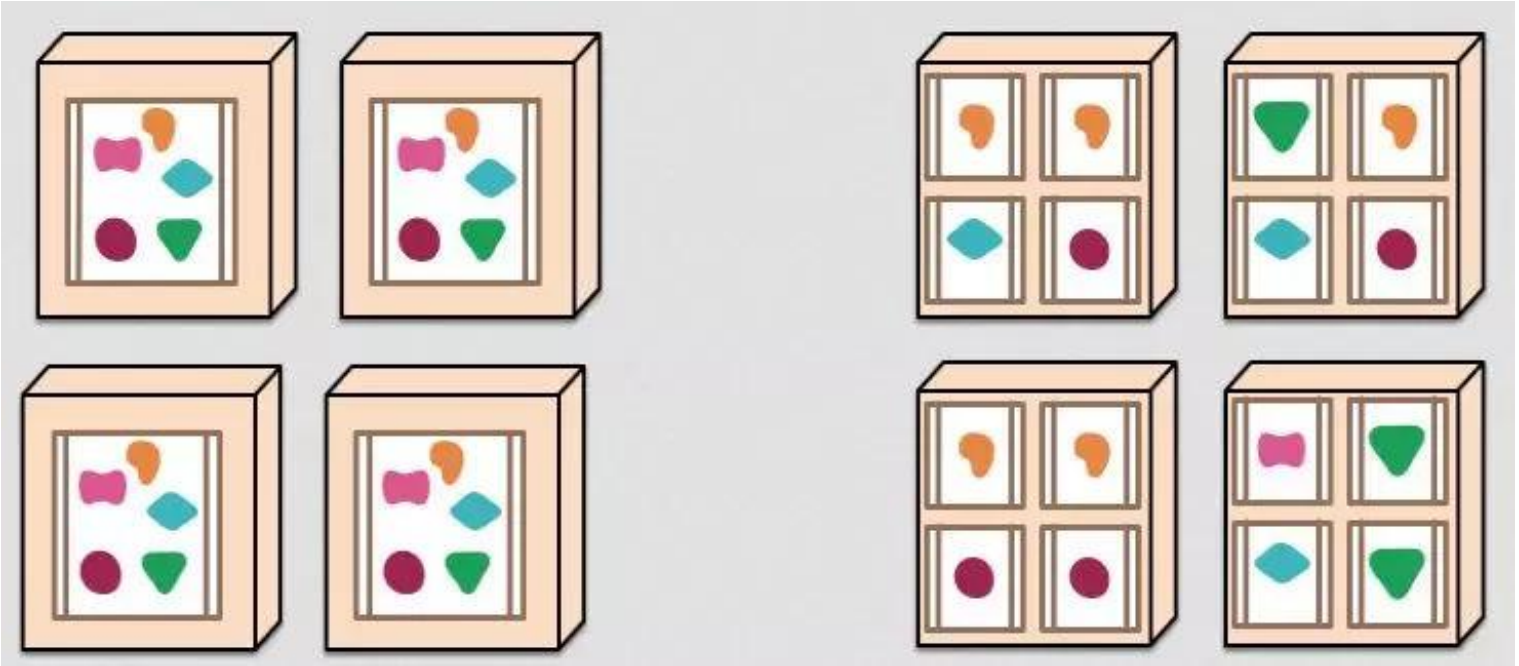


- **Low latency**
- **HD video, bandwidth-intensive**
- **Transmission network efficiency**
- **High-quality user experience**

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Microservices

- Microservices are a software development technique that structures an application as a collection of loosely coupled, fine-grained services. In a microservices architecture, lightweight communication protocols are used, often HTTP resource APIs. The services are built around functionalities and deployed independently and in a fully automated manner. The need for centralized management is minimized. The services can be written in different languages and use different storage types.



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Pros and Cons of Microservices

Pros

- Independent deployment, flexible scalability
- Effective resource isolation
- Optimized development teams

Cons

- Complex development and testing process due to project splitting
- Difficulty of ensuring data consistency between services

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Serverless (1)

- Serverless computing is a model for building and managing microservices-based architectures. It allows you to manage your application deployment on a per-service level, rather than a per-server level.
  - **Function as a Service (FaaS)**: The service provider provides a platform where the developer develops, runs, and manages their applications without the need to build and maintain an infrastructure.
  - **Backend as a Service (BaaS)**: The service provider provides a serverless backend, also known as backend as a service for the mobile application developer. BaaS shifts backend functions including data/file storage, account management, data collection, operation analysis, message push, and social media integration to the cloud, allowing the developer to focus on their apps' frontend.

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Serverless (2)

Previously

Build a framework that runs on a server to respond to multiple events.

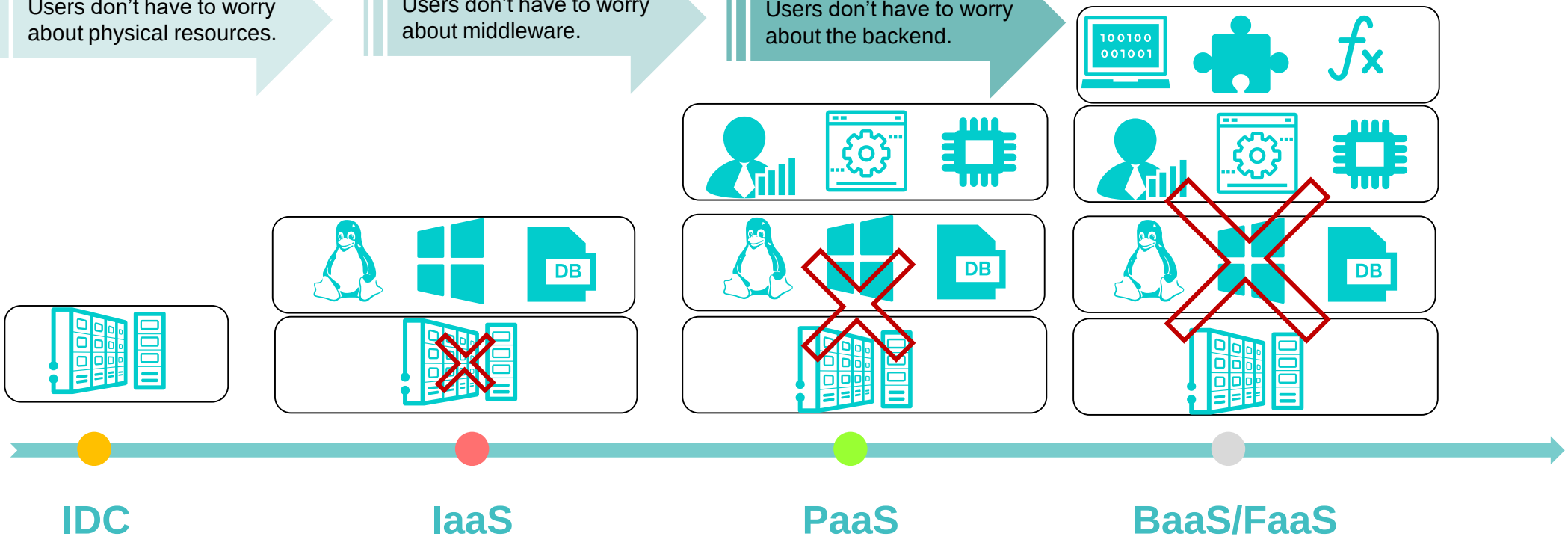
Now

Build or use a separate microservice or function to respond to each event.

Users don't have to worry about physical resources.

Users don't have to worry about middleware.

Users don't have to worry about the backend.



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FaaS Characteristics

- No need for O&M
- Able to focus on the code
- Quick boot, short life cycle
- Infinite elasticity
- Automated

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BaaS Benefits

Higher Efficiency

Reduces costs associated with mobile application development environments and improves efficiency.

Faster TTM

Allows developers to focus on their apps rather than backend functions, and reduces operational costs of apps after launching.

Reduce Resources Needed to Launch Apps

Fewer developers and IT resources are required.

Optimized for Mobile Phones and Tablets

BaaS providers have spent a considerable amount of time and resources on optimizing mobile application data and networks.

Secure and Elastic Infrastructure

BaaS provides bundled infrastructure services, relieving developers of the burden of ensuring elasticity, security, and performance.

A Full Collection of Common APIs

BaaS offers a full collection of commonly used third-party APIs, sparing developers the hassle of collecting these resources.

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Quiz

1. Which of the following is not an element of artificial intelligence? ( )

   A. Big data

   B. Cognitive analysis capability

   C. Computing power

   D. Scenario

2. The only function of the Internet of Things is to collect the data generated by connected objects.

   A. True

   B. False

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Summary

- Fields Related to Cloud Computing
- Cloud Computing Technologies
- Cloud Computing Trends
- Other Emerging Technologies

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Recommendations

- Huawei e-Learning website
  - http://support.huawei.com/learning/Index!toTrainIndex
- Huawei Support case library
  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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Thank You
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Recommendations

- Huawei Learning Website
  - http://learning.huawei.com/en
- Huawei e-Learning
  - https://ilearningx.huawei.com/portal/#/portal/ebg/51
- Huawei Certification
  - http://support.huawei.com/learning/NavigationAction!createNavi?navId=_31&lang=en
- Find Training
  - http://support.huawei.com/learning/NavigationAction!createNavi?navId=_trainingsearch&lang=en

More Information

- Huawei learning APP

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